
Introduction

The goal of this book is to establish an axiomatic framework for higher category theory. This approach is best thought of as a form of *synthetic* category theory: instead of constructing higher categories from more primitive objects, the axioms impose the intended behavior of higher categories directly. This stands in contrast to *analytic* approaches, such as the theory of quasicategories, in which one first constructs higher categories inside a background foundation, typically set theory, and then proves that the resulting objects have the desired categorical properties.

Mathematical motivation

Category theory has long been a useful unifying language for abstract mathematics. Higher category theory¹ refines ordinary category theory by allowing all constructions and properties to be considered up to coherent homotopy. In recent decades, higher category theory and homotopy theory have become central tools in algebraic topology, algebraic geometry, representation theory, mathematical physics, and many other fields.

The traditional foundations of the field have rested on specific models, most notably the framework of quasicategories developed by Joyal [Joy08] and Lurie [Lur09; Lur17]. As the subject matured, however, it became clear that most of its results and constructions are *model-independent*: they never touch the combinatorial details of quasicategories, or of any other model. One way to say this is that there is a *language* of category theory, of which the classical set-based notion, a category as a set of objects and a set of morphisms, is one interpretation, and the theory of quasicategories another. The working content of higher category theory lives in the language, not in any single interpretation of it. Following this viewpoint, much recent work is written in a deliberately model-agnostic style that avoids the combinatorial scaffolding of quasicategories.

One primary goal of this book is to provide a detailed account of this model-independent language of category theory, and to identify a list of key categorical axioms one needs to assume to build up the rest of the theory. We believe this allows for a more direct and intuitive engagement with the core concepts of higher category theory, and that it leads to proofs carrying insights that may have been obscured by model-specific details in more traditional treatments of the theory.

Our framework has three further advantages:

- *Parametrized ∞ -category theory*: Our formalism is designed to serve as the internal language of the ∞ -category of sheaves of ∞ -categories on any ∞ -topos. Thus all results proved in this book are automatically valid in parametrized category theory, as in the work of Barwick et al.

¹More specifically: $(\infty, 1)$ -category theory.

[Bar+16] on genuine equivariant homotopy theory and the work of Martini and Wolf [MW24] on category theory internal to an ∞ -topos.

- *A working language for ∞ -category theory*: The theory specifies the basic notions, operations, and identifications in a language in which one can teach and use ∞ -category theory in the same way one uses naive set theory, making the subject more accessible.
- *Elementary ∞ -category theory*: The axioms are constructive, and hence shed light on the scope of “elementary ∞ -category theory”: Even constructively it is possible to recover substantial parts of standard higher category theory.

An alternative formal approach to higher category theory was developed by Riehl and Verity [RV22]. Their ∞ -cosmoi are certain categories enriched in quasicategories and make systematic use of cotensors by simplicial sets. Their work is model-independent in the sense that a common formalism applies to any ∞ -cosmos satisfying suitable axioms. Our aim is different: the language developed here is model-agnostic, in that it does not refer to a model of higher categories at all.

Philosophical motivation

There is also a philosophical reason to formalize higher category theory: category theory can serve as a foundation for mathematics. This idea goes back at least to Lawvere, and it has been developed in many forms since then; see for instance Landry and Marquis [LM05] for historical context.

The point of view in this book is that categories, in fact $(\infty, 1)$ -categories, should be treated as the building blocks of mathematics, in the role traditionally played by sets or types. One way to study mathematical objects within category theory is through the categories they form; this perspective is already a common practice in fields like algebraic geometry and algebraic topology. We wish to highlight a second way to recover ordinary mathematics as a fragment of the theory of categories: by first restricting the theory to *animae*, those categories in which every morphism is an isomorphism, and then further to *sets*, those animae that admit at most one isomorphism between every two objects. The resulting set theory can then be used to define other mathematical objects in the usual way.

The latter perspective also makes clear that the jump from set-based mathematics to category-based mathematics consists of two conceptual leaps, corresponding to the passage from sets to animae and the passage from animae to categories.

The first leap concerns the notion of *equality*. In set-based mathematics, equality is treated as a property: two elements of a set are either equal or they are not, and there is no further question of *how* they are equal. In anima-based mathematics, equality is instead treated as an intrinsic structure. To illustrate the difference, set-based mathematics treats two abelian groups as the same if they are strictly equal as sets with equal operations, while the anima-based perspective on such equalities would amount to group isomorphisms. The structured perspective on equality is not new; for example:

- Makkai [Mak98] proposed the **Principle of Isomorphism**: All grammatically correct properties of objects of a category are to be invariant under isomorphism.

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- Voevodsky’s *homotopy type theory (HoTT)* [Uni13] builds in a flexible notion of equality which has the key feature that it is compatible with the **Univalence Axiom**, which roughly states that two isomorphic types (groups, topological spaces, etcetera) are equal to each other.

We take it as our main guiding principle throughout the book:

Fundamental Principle of Homotopy Theory: When expressing that two entities are equal, we must always specify *how* they are equal by providing an isomorphism between them. All valid mathematical constructions should preserve these isomorphisms.

Taking this principle seriously, we are forced to replace sets as our basic notion of ‘collections of mathematical objects’: equality in sets is too rigid. The new fundamental object in this paradigm is called an *anima*, which may be described as a collection of objects such that for each two objects there is an anima of isomorphisms between them. This gives animae the structure of a groupoid. By considering equality of morphisms, we also have *isomorphisms between isomorphisms*, and repeating this process we see that animae naturally behave like ∞ -groupoids. Other common names in the literature are *spaces* or *homotopy types*; HoTT for example uses the latter framing. In this book, however, the word *space* is reserved for an actual topological space.

The second leap is to replace animae by categories as the primitive building blocks of the theory. In practice, a class of mathematical objects often naturally comes with more general relations between objects than mere isomorphisms; think of homomorphisms between groups, continuous maps between topological spaces, functors between categories. The concept of a category encapsulates this by encoding morphisms between objects and behavior of their compositions.

The classical perspective on categories is analytic: a category consists of a *set* of objects and a *set* of morphisms, satisfying various axioms. This perspective clashes with Makkai’s principle of isomorphism, as it allows one to distinguish two isomorphic objects: categories should have an *anima* of objects and a *anima* of morphisms. In light of this, it may be tempting to try to define categories directly within HoTT. However, we argue that it is better to make a second leap, and directly take categories to be the primitive objects of the theory, with sets and animae arising as special cases.

Just like the fundamental principle of homotopy theory forces animae to behave like ∞ -groupoids, it also forces categories to behave like $(\infty, 1)$ -categories: equality between morphisms must be encoded by some notion of 2-isomorphisms, equality between those by 3-isomorphisms, and so on. In this way, the language of category theory natively speaks about $(\infty, 1)$ -categories, and strict 1-categories are merely a fragment of this theory.

For broader discussions of categorical and structural foundations of mathematics, we refer to Tsementzis [Tse17], Marquis [Mar13], Rodin [Rod14], and Awodey [Awo14].

Synthetic category theory eats itself

One of the distinctive features of synthetic category theory is that it can interpret itself. We take this as a key design criterion: There should be a universe *Cat* whose objects are small synthetic categories, and all the axioms of the theory we develop should ultimately be satisfied in the internal language of *Cat*. In this sense, it is analogous to the role of Grothendieck universes in set theory:

sets large enough to contain the usual set-theoretic constructions, which can therefore serve as a model of set theory inside set theory. A systematic verification of this design criterion is deferred to Volume II.

The axiom that makes this precise is *directed univalence*, which relates morphisms in $\mathbf{Cat}$ to functors between synthetic categories. The main consequence of this axiom is the familiar Grothendieck correspondence: for every small synthetic category C , functors

$$C \rightarrow \mathbf{Cat}$$

correspond to small cocartesian fibrations over C . To illustrate why this relates the external theory of synthetic categories to the internal theory of $\mathbf{Cat}$, it is illustrative to consider the terminal case $C = *$. In this case, the Grothendieck correspondence says that objects of $\mathbf{Cat}$ correspond to small cocartesian fibrations over the terminal category, hence to small synthetic categories. Applying the principle to more general instances of C then provides the translation between the internal language of $\mathbf{Cat}$ and the external language of categories that we are setting up.

Restricting directed univalence to *anima* recovers Voevodsky’s univalence axiom: isomorphisms between *anima* correspond to equalities in the subuniverse $\mathbf{An} \subseteq \mathbf{Cat}$ of *anima*. But the directed form is stronger: it also says that maps between *anima* correspond to morphisms in $\mathbf{An}$. This also marks a crucial difference with homotopy type theory: HoTT has a universe of types, and univalence explains equality in that universe, but this does not yet give a self-interpretation of HoTT inside HoTT. The problem is to internalize the syntax of type theory together with its higher coherence in plain homotopy type theory. Synthetic category theory is designed so that the theory eats itself: the universe $\mathbf{Cat}$ should serve not only as one object of the theory, but also as a semantic universe in which the theory can be interpreted. This model theorem will be developed in Volume II.

Relation to type theory

In this book, we present the theory of synthetic categories in ordinary mathematical language, specifying its basic notions, operations, and coherence identifications. The main objective of this book is to provide a model-agnostic presentation of synthetic category theory, and build up standard category theory from a sizable list of foundational axioms. No knowledge of type theory is required to read this book.

Nevertheless, our work is heavily inspired by type-theoretic considerations, and we expect it to be possible to write down rigorous type-theoretic syntax and derivation rules that allow the theory to be formalized in proof assistants. We are exploring several approaches:

- A presentation of the theory à la Martin-Löf is in preparation in collaboration with Michele Riva;
- A presentation of the theory within the type theory CaTT of ω -categories [FM17; BFM24] is in preparation in collaboration with Ivan Kobe.

For readers familiar with dependent type theory, we wish to flag two differences with synthetic category theory.

The first difference concerns the *Curry–Howard correspondence*: the idea that types may be interpreted as mathematical statements, and terms of a type as proofs of those statements. Constructions

of types, like the disjoint union $X \sqcup Y$ or the product $X \times Y$, then admit logical interpretations, like the logical disjunction $P \vee Q$ and logical conjunction $P \wedge Q$. Under this correspondence, propositions appear as the special case of (-1) -truncated types: those types which admit at most one term.

In synthetic category theory, it is no longer the case that arbitrary types correspond to mathematical statements. General types are now interpreted as categories, which allow for non-invertible maps between objects. The types corresponding to statements are the primitive *animae*, which the groupoid recognition axiom identifies with the categories in which every morphism is an isomorphism.

The second difference concerns *type dependency*. In ordinary dependent type theory, for each context Γ , the types in context Γ form a category, and any substitution

$$f: \Delta \rightarrow \Gamma$$

induces a pullback operation on types. This pullback operation preserves the logical structure of the theory: products, sums, identity types, and the other type-forming operations are stable under substitution. In this sense, substitution along an arbitrary map of contexts preserves the logic.

In synthetic category theory, this principle breaks in a controlled way. It continues to behave as expected when the contexts involved are *animae*. But if Γ is a general synthetic category, then a type in context Γ should be thought of as a category over Γ , that is, as a functor

$$p: X \rightarrow \Gamma.$$

The resulting theory of categories over Γ is not, in general, again a semantic interpretation of synthetic category theory.

However, there is a better category-theoretic replacement: among functors $X \rightarrow \Gamma$, one should restrict to cocartesian fibrations over Γ . The resulting category *does* form a semantic interpretation of synthetic category theory. This theory is not equivalent to the naive theory of all categories over Γ , except when Γ is an *anima*. A precise form of this result is treated in Volume II.

Still, even when we restrict to cocartesian fibrations, pullback along an arbitrary functor $f: \Delta \rightarrow \Gamma$ does not generally preserve all categorical structure: unlike in type theory, substitution does not fully preserve the logic.

We wish to regard this phenomenon as a feature rather than a bug: It is one of the ways in which synthetic category theory is more expressive than ordinary dependent type theory. The comparison between the naive theory of categories over Γ and the theory of cocartesian fibrations over Γ is itself structured: it is governed by adjunctions and by canonical comparison maps, many of which are not invertible. Such non-invertible comparison maps are not pathological in category theory: they often form an integral part of the mathematical discourse of the subject.

Consistency

The consistency of the theory presented in this book is controlled by the usual models of higher category theory. Cisinski’s work on higher categories [Cis19], together with the work of Cisinski and Nguyen on the universal cocartesian fibration [CN22], shows that Joyal’s theory of quasicategories gives a semantic interpretation of synthetic category theory. Thus the theory is at least as consistent as the usual set-theoretic foundations used to construct quasicategories.

Let us note that the proof of directed univalence, namely the fact that there is a reasonable ∞ -category of ∞ -categories, subsumes Voevodsky’s proof of the univalence axiom for Kan complexes [KL21].

As mentioned before, the work of Buchholtz, Gratzer and Weinberger [GWB26] on simplicial type theory is expected to provide a constructive semantic interpretation of synthetic category theory within the framework of Riehl and Shulman [RS17]. Conversely, the present book proves among other things that the synthetic category of synthetic categories is equivalent to the synthetic category of univalent Segal objects in *anima*, which can be used as the basis for an internal version of simplicial homotopy type theory. This indicates that synthetic category theory and simplicial type theory are mutually consistent, constructively.

Organization of the book

The present version of the book is organized as follows:

- (1) In Chapter 1, we introduce the language of *synthetic category theory*. Categories, *anima*, functors and isomorphism *anima* between functors are taken as primitive, and all our axioms are expressed in terms of them. We introduce products, coproducts, pullbacks and functor categories via universal properties, and use mapping *anima* to define the collection of objects of a category.
- (2) In Chapter 2, we study the internal structure of categories. We introduce morphisms and commutative diagrams, and formulate the Segal axiom and the Rezk axiom to ensure that the objects we call synthetic categories do indeed behave like categories. We then relate primitive *anima* to the structural notion of a *groupoid* and study groupoid cores. *Anima* play a fundamental role in our approach: an object of a synthetic category C is a functor $x: \Gamma \rightarrow C$ whose source is allowed to be an arbitrary *anima* Γ .
- (3) In Chapter 3, we introduce various fundamental categorical constructions in the setting of synthetic category theory: full subcategories, localizations, geometric realizations, relative functor categories, joins and slices.
- (4) In Chapter 4 (outside selection), we develop the language of categories over a context, terms as sections, reindexing, and context extension. The theory developed in this book remains valid for categories in an *anima* context. Some results even work in arbitrary categorical contexts; we distinguish those by underlining them.
- (5) In Chapter 5 (outside selection), we introduce adjunctions, initial and terminal objects, limits and colimits, left and right fibrations, and cartesian and cocartesian fibrations, among other things. We also introduce the Functoriality of universals axiom, which states that in order to provide a left adjoint section to a cartesian fibration $p: E \rightarrow C$ it is enough to show that all the fibers of p admit initial objects.
- (6) One of the most fundamental theorems in category theory is the fact that a functor is an equivalence if and only if it is fully faithful and strongly surjective. We prove this result in Chapter 6 (outside selection).
- (7) The main goal of Chapter 7 (outside selection) is to axiomatize a synthetic category $\mathbf{Cat}$ of small synthetic categories. It comes equipped with a cocartesian fibration $\pi_{\text{univ}}: \mathbf{Cat}_\bullet \rightarrow \mathbf{Cat}$ satisfying the property that for every small synthetic category C there is an equivalence between functors from C to $\mathbf{Cat}$ and cocartesian fibrations over C with small fibers. The equivalence

is given by pulling back the universal small cocartesian fibration π_{univ} along the given functor $C \rightarrow \text{Cat}$.

- (8) Chapter 8 (outside selection) studies initial and terminal functors, proper and smooth functors, weak homotopy equivalences, and base change.
- (9) Chapter 11 (outside selection) develops opposite categories, twisted arrow categories, the Yoneda lemma, representable approximation, locally small categories, and descent.
- (10) Chapter 9 (outside selection) develops sets, natural numbers, simplices, covers, truncation.
- (11) Chapter 10 (outside selection) introduces covers and truncation, thus opening the possibility of dealing constructively with mere existence statements. We also discuss sets and posets.
- (12) Chapter 12 (outside selection) free completions by small colimits and descent. In particular, Quillen's theorem B is proven.
- (13) Chapter 13 (outside selection) studies products in localizations, localizations and adjunctions, and the calculus of fractions.
- (14) Chapter 14 (outside selection) explains how univalent Segal anima and the Rezk nerve functor yield a new description of the universe of categories.
- (15) Chapter 15 (outside selection) develops filtered categories and filtered colimits of anima, together with the theory of finite categories.
- (16) Chapter 16 (outside selection) studies left or right exact localizations as well as presentable categories with respect to a fixed universe (later in the book, this will specialize to finite posets or finite sets, for instance).
- (17) Chapter 17 (outside selection) studies compact generation, tensor product of finitely cocomplete categories, and Ind-objects.
- (18) Chapter 18 (outside selection) develops finite-colimit completions, stable categories, spectra, and Verdier quotients.
- (19) Chapter 19 (outside selection) develops additive and Waldhausen categories, completions by sifted colimits, the S -construction, additivity, the universal property of K -theory (and of the Waldhausen S -construction in general), and the localization theorem in algebraic K -theory.
- (20) Chapter 20 (outside selection) introduces a new accessibility axiom and develops the theory of presentable categories in full generality - existence of left Bousfield localizations, presentation as left Bousfield localizations of categories of presheaves, adjoint functor theorem.

Overview of the axioms

The axioms are introduced progressively, since several require preliminary categorical language before they can be stated. This chapter gives a concise overview of the actual axiomatic package and the principal constructions used in its formulation. Precise statements and dependencies appear in the chapters that follow.

The language of synthetic category theory

Synthetic category theory rests upon a language of categories, functors and identifications, together with specified operations and coherences. We will now provide a rough overview of this language, leaving the details to Chapter 1.

Axiom A. There are categories and, among them, *anima*e. Between categories there are functors, with identity functors and composition. For parallel functors $f, g : C \rightarrow D$, there is an *anima* of identifications $(f = g)$, with prewhiskering, postwhiskering, and inversion functors. An object $c : C$ is a functor $* \rightarrow C$, and an identification $\alpha : f = g$ is a functor $* \rightarrow (f = g)$. Composition of identifications is a functor, with unit, associativity, and inverse identifications. Composition of functors has unitors and an associator, satisfying the triangle and pentagon identities and the specified whiskering and interchange laws.

*Anima*e serve as the homotopy-coherent collections of the theory, playing the role of sets in ordinary mathematics. The distinguishing feature is the notion of equality in an *anima*: given two identifications $p, q : x =_X y$, we may talk about identifications between p and q , and this process may be iterated. This also applies to identifications between functors.

A functor $f : C \rightarrow D$ may be displayed in term notation as $x : C \vdash f(x) : D$. Substitution displays composition of functors; identifications of expressions are identifications of the whole functors.

A functor $f : C \rightarrow D$ is said to be an *equivalence* if there exists another functor $g : D \rightarrow C$ and identifications $\text{id}_C = gf$ and $\text{id}_D = fg$.

There are various ways to build new categories out of old ones. In each case, we will specify the behavior of these constructions by prescribing how to define functors into/out of them and how to define identifications between functors into/out of them. We record only the coherence data and relations used explicitly in the development.

Axiom B. We demand the existence of the following categories:

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- (1) A terminal anima $*$: for every C there is a functor $p_C : C \rightarrow *$, and for $f, g : C \rightarrow *$ the anima $(f = g)$ is contractible.
 - (2) A product $C \times D$ with two projections. Functors into it are formed by pairing, and for $h, k : T \rightarrow C \times D$ projection induces an equivalence
$$(h = k) \xrightarrow{\sim} (\text{pr}_C h = \text{pr}_C k) \times (\text{pr}_D h = \text{pr}_D k).$$
 - (3) A mapping anima $\text{Map}(C, D)$ with evaluation $\text{Map}(C, D) \times C \rightarrow D$. Functors $X \times C \rightarrow D$ with anima parameter X can be curried to $X \rightarrow \text{Map}(C, D)$, and uncurrying induces equivalences of anima of identifications.
 - (4) An initial category $\emptyset$, with $\text{Map}(\emptyset, T)$ contractible for every category T .
 - (5) A coproduct $C \sqcup D$ with inclusions, such that restriction gives an equivalence $\text{Map}(C \sqcup D, T) \xrightarrow{\sim} \text{Map}(C, T) \times \text{Map}(D, T)$ for every T .
 - (6) A pullback $C \times_E D$ with projections and a matching identification. Maps to the pullback are constructed by mapping into C and D and exhibiting an identification; anima of identifications of functors into the pullback are the corresponding pullbacks of anima of identifications.
 - (7) A functor category $\text{Fun}(C, D)$ with evaluation inducing equivalences $\text{Map}(T, \text{Fun}(C, D)) \xrightarrow{\sim} \text{Map}(T \times C, D)$ for every category T . This provides currying and uncurrying for arbitrary categorical parameters.

Products of anima are anima, as are pullbacks of anima over anima. We further demand strictness of the initial category and universality of coproducts, in the sense of Axioms B.4' and B.5'.

We define the collection of objects in a category C to be $C^\simeq = \text{Map}(*, C)$. We define an *absolute object* of a category C to be a functor $x : * \rightarrow C$.

The internal structure of categories

We now turn to morphisms and their composition, developed in Chapter 2. To speak of *morphisms*, we need another axiom:

Axiom C. There is a category $[1]$, called the *walking morphism*, which comes equipped with an initial object 0 and a terminal object 1 . These induce an equivalence $\langle 0, 1 \rangle : * \sqcup * \xrightarrow{\sim} [1]^\simeq$.

We may then define a *morphism* in C to be a functor $f : [1] \rightarrow C$. We refer to the objects $f(0)$ and $f(1)$ of C as the *source* and *target* of f , respectively, and often write $f : x \rightarrow y$ to indicate that f comes equipped with identifications $x = f(0)$ and $y = f(1)$. Similarly, a *natural transformation* of functors $C \rightarrow D$ is defined as a functor $C \times [1] \rightarrow D$, or equivalently a functor $C \rightarrow \text{Fun}([1], D)$.

The *walking commutative triangle* is the category $[2] := \text{Fun}([1], [1])$. It comes with objects 0 , 1 and 2 , and with functors $d_0, d_1, d_2 : [1] \rightarrow [2]$ and $s_0, s_1 : [2] \rightarrow [1]$ satisfying the simplicial identities. A *commutative triangle* in C is a functor $[2] \rightarrow C$, and a *commutative square* is a functor $[1] \times [1] \rightarrow C$. Every commutative square determines two commutative triangles by precomposition with the maps $j_0 := (s_0, s_1) : [2] \rightarrow [1] \times [1]$ and $j_1 := (s_1, s_0) : [2] \rightarrow [1] \times [1]$, and we demand that the original square can be uniquely recovered from this data:

Axiom D (Commutative square axiom). For every category C , the functor

$$(j_0^*, j_1^*): \text{Fun}([1] \times [1], C) \rightarrow \text{Fun}([2], C) \times_{d_1, \text{Fun}([1], C), d_1} \text{Fun}([2], C)$$

is an equivalence.

Every object $x: * \rightarrow C$ gives rise to an *identity morphism* $\text{id}_x: x \rightarrow x$, defined as the composite $[1] \xrightarrow{P([1])} * \xrightarrow{x} C$. The definition of *composition* of morphisms in C requires another axiom:

Axiom E (Segal axiom). For every category C , the restriction functor

$$(d_2^*, d_0^*): \text{Fun}([2], C) \rightarrow \text{Fun}([1], C) \times_C \text{Fun}([1], C)$$

is an equivalence of categories.

Given morphisms $f: x \rightarrow y$ and $g: y \rightarrow z$, the axiom provides a unique commutative triangle $\sigma: [2] \rightarrow C$ satisfying $d_2^*(\sigma) = f$ and $d_0^*(\sigma) = g$. The resulting morphism $g \circ f := d_1^*(\sigma): [1] \rightarrow C$ is called the *composite of f and g* . One can prove that composition of morphisms is automatically unital and associative.

Having both identity morphisms and composition, we may now introduce the notion of *isomorphism* in C : a morphism $f: x \rightarrow y$ in C is an isomorphism if it admits a left inverse $g: y \rightarrow x$ (satisfying $gf = \text{id}_x$) as well as a right inverse $h: y \rightarrow x$ (satisfying $fh = \text{id}_y$). Using diagram categories, we may in fact construct for every C a category $\text{Iso}(C)$ whose objects are precisely the isomorphisms in C (equipped with the data of left and right inverses). The canonical example of an isomorphism in C is the identity morphism $\text{id}_x: x \rightarrow x$, in which case the left and right inverses are simply id_x itself. This construction provides a functor $i: C \rightarrow \text{Iso}(C)$, which we demand to be an equivalence:

Axiom F (Rezk axiom). Every isomorphism in a category C is isomorphic to an identity morphism. More precisely, the functor $i: C \rightarrow \text{Iso}(C)$ is an equivalence.

Axioms E and F are inspired by Rezk's notion of *complete Segal spaces*.

We now introduce the following recognition principle for animae.

Axiom G (Groupoid recognition axiom). A category is an anima if and only if it is a groupoid.

Constructions of categories

Equipped with a basic language of categories, functors and identifications, we can start formulating our remaining axioms of category theory. We first recall the *groupoid cores* of categories. Using the above notion of isomorphisms in a category C , we say that C is a *groupoid* if all of its morphisms are isomorphisms.

For every category C , there is an anima $C^\simeq$ called the *groupoid core of C* , equipped with a functor $\iota_C: C^\simeq \rightarrow C$ simply defined as

$$C^\simeq = \text{Map}(*, C).$$

Every functor $f: X \rightarrow C$ from an anima X factors through ι_C , and for functors $g, h: X \rightarrow C^\simeq$ every identification $\iota_C \circ g = \iota_C \circ h$ may be lifted to an identification $g = h$.

For categories C and D , the primitive mapping anima $\text{Map}(C, D)$ is related to the functor category by a preferred equivalence

$$\text{Map}(C, D) \simeq \text{Fun}(C, D)^\simeq.$$

We also demand the existence of subcategories, localizations, geometric realizations and joins, discussed in detail in Chapter 3:

Axiom H (Subcategory axiom). Consider a category C equipped with a *collection of morphisms* $m: M \hookrightarrow \text{Map}([1], C)$ of C which is *closed under composition* (see Definition 3.1.6 for a precise definition of these terms). Then there exists a category $\langle M \rangle_C$ equipped with a functor $i_M: \langle M \rangle_C \rightarrow C$ satisfying:

- The induced functor $(i_M)_*: \text{Map}([1], \langle M \rangle_C) \rightarrow \text{Map}([1], C)$ factors through M ;
- The functor i_M is *universal* with respect to the previous property: every other functor $f: D \rightarrow C$ whose induced functor $f_*: \text{Map}([1], D) \rightarrow \text{Map}([1], C)$ factors through M admits a unique factorization through $\langle M \rangle_C$.

Axiom I (Localization axiom). For every category C and every collection of morphisms $W \hookrightarrow \text{Map}([1], C)$, there exists a functor $l: C \rightarrow C[W^{-1}]$ exhibiting $C[W^{-1}]$ as a localization of C at the morphisms in W :

- The functor l sends the morphisms in W to isomorphisms in $C[W^{-1}]$;
- The functor l is *universal* with respect to the previous property: every other functor $f: C \rightarrow D$ which sends morphisms in W to isomorphisms in D admits a unique factorization through $C[W^{-1}]$.

In the case where W consists of *all* morphisms in C , we denote the localization by $\|C\|$ and call it the *geometric realization of C* . It is automatically an anima.

Axiom J (Join axioms). For every category Γ and functors $C \rightarrow \Gamma$ and $D \rightarrow \Gamma$, there is a relative join $C \star_\Gamma D \rightarrow \Gamma$. If

$$j_\Gamma := \text{id}_\Gamma \times j: \Gamma \times \partial[1] \longrightarrow \Gamma \times [1]$$

is the weakening of the boundary inclusion, then the height functor $h_{C,D}: C \star_\Gamma D \rightarrow \Gamma \times [1]$ is a dependent product of $C \sqcup D$ over $\Gamma \times \partial[1]$ along j_Γ .

When Γ is an anima, the relative join also satisfies the mapping-out exactness property: it is exhibited by a pushout square

$$\begin{array}{ccc} (C \times_\Gamma D) \sqcup (C \times_\Gamma D) & \longrightarrow & \times(C \times_\Gamma D) \\ \text{pr}_C \sqcup \text{pr}_D \downarrow & & \downarrow \\ C \sqcup D & \longrightarrow & C \star_\Gamma D. \end{array}$$

The absolute join $C \star D$ is the special case $\Gamma = *$.

If we write $[0]$ for the terminal category $*$, we have $[1] \xrightarrow{\sim} [0] \star [0]$, and we may inductively define $[n+1] := [n] \star [0]$.

Cartesian and cocartesian fibrations

An important concept in category theory is that of a cocartesian fibration: a functor $E \rightarrow C$ whose fibers E_c vary covariantly and functorially in the object $c \in C$. In Chapter 5 (outside selection), we introduce a synthetic analogue of this definition: a functor $f: A \rightarrow B$ is called a cocartesian fibration if a certain ‘directed evaluation map’ $\vec{\text{ev}}_0^f: \text{Fun}([1], A) \rightarrow A \vec{\times}_f B$ admits a left adjoint section $\text{lift}_0^f: A \vec{\times}_f B \rightarrow \text{Fun}([1], A)$. Informally, this left adjoint section provides the cocartesian lifts $a \rightarrow \beta_!(a)$ in A for morphisms $\beta: b \rightarrow b'$ in B with specified lift a of b . One may dually define *cartesian fibrations*.

Axiom L (Functoriality of universals axiom). Let $p: E \rightarrow C$ be a cocartesian (resp. cartesian) fibration. If the fiber E_x of p admits a terminal (resp. initial) object for every object x of C , then there is a right (resp. left) adjoint section $s: C \rightarrow E$ of p whose value at x is isomorphic to the chosen terminal (resp. initial) object of E_x .

Axiom M (Exponentiability axiom). Cartesian and cocartesian fibrations are *exponentiable*, meaning that pullback along a (co)cartesian fibration admits a right adjoint, and these right adjoints satisfy a Beck-Chevalley condition.

Given a cocartesian fibration $p: U_\bullet \rightarrow U$, one can formulate what it means for p to satisfy *directed univalence*: roughly speaking, this means that for functors $f_0, f_1: C \rightarrow U$ with pullback fibrations $E_i := f_i^*(U_\bullet) \rightarrow C$, the anima of natural transformations $\alpha: f_0 \rightarrow f_1$ is equivalent to the anima of cocartesian functors $E_0 \rightarrow E_1$ over C . This condition will in particular imply a form of *straightening/unstraightening*. A category U equipped with such a univalent cocartesian fibration p is called a *universe*. We say that U is a *universe of categories* if it is closed under all the categorical operations introduced so far; see Definition 7.4.5 (outside selection) for a complete list.

Axiom N (Directed univalence axiom). For every cocartesian fibration $q: E \rightarrow B$, there exists a universe of categories U such that B is U -small and q has U -small fibers.

We will usually fix a universe of categories $\pi_{\text{univ}}: \text{Cat}_\bullet \rightarrow \text{Cat}$ and work with categories that are small with respect to this universe, i.e. those categories C that can be obtained from an object c of Cat via the following pullback square:

$$\begin{array}{ccc} C & \longrightarrow & \text{Cat}_\bullet \\ \downarrow & \lrcorner & \downarrow \pi_{\text{univ}} \\ * & \xrightarrow{c} & \text{Cat}. \end{array}$$

A consequence of directed univalence is the straightening/unstraightening equivalence

$$\text{Un}: \text{Fun}(C, \text{Cat}) \xrightarrow{\sim} \text{CoCart}(C)$$

for every small category C . Here $\text{CoCart}(C)$ is the subcategory of the slice category $\text{Cat}/_C$ whose objects are small cocartesian fibrations over C and whose morphisms are cocartesian functors. This correspondence motivates the intended interpretation of synthetic category theory in the universe Cat ; its systematic verification belongs to Volume II.

1 The language of synthetic category theory

At the heart of mathematics lies the concept of a set, a collection of mathematical objects. In modern mathematics, sets serve as the building blocks for defining everything from numbers and relations to more complex structures like groups, rings, and topological spaces. Sets are manipulated using basic operations like forming unions, intersections, and complements, and are compared using functions between sets. It is fair to say that the language of set theory forms an essential toolkit for mathematicians.

Although formal axiomatizations of set theory exist, such as those by Zermelo and Fraenkel, they often diverge from the working mathematician’s intuitive understanding of sets. For instance, in formal set theory, every natural number needs to be defined as a set, and in particular one is allowed to speak of the elements of a natural number. Similarly, every function between sets needs to be defined as a set. While this approach is formally valid, it can seem disconnected from everyday mathematical discourse.

In practice, mathematicians often use a more pragmatic approach known as *naive set theory*. In naive set theory, one works with sets based on intuitive understanding, without invoking formal logical or axiomatic implementations. For example, the set of natural numbers is simply the set containing the numbers 0, 1, 2, etcetera, and functions between sets X and Y are simply ‘specifications’ that assign to every element of X an element of Y . The simplicity and accessibility of naive set theory make it relatively easy for students to learn, and indeed in many introductory mathematics textbooks sets are used in the naive way without being rigorously introduced.

A similar trend is occurring in the field of higher category theory. In recent decades, higher category theory has seen rapid growth, in large part thanks to contributions from people like Joyal, Lurie, and others. Their work provides solid foundations for the field, and the language of higher categories is now extending beyond its origins in algebraic topology to fields like algebraic geometry and representation theory. But while the original works of Joyal and Lurie are based on explicit set-theoretical models for higher categories called ‘quasicategories’, more and more research papers are written in a ‘model-independent’ way in which the usage of details about the specific choice of implementations is avoided. We refer to this as the language of *naive category theory*.

The goal of this chapter is to provide a structured form of this language, with explicit rules and axioms for how to manipulate categories and functors. Our approach is *synthetic* in nature rather than *analytic*: constructions of categories, like products or functor categories, are not *defined* in terms of set-level constructions, but are *axiomatized* to exist and satisfy the expected properties. Our rules are deliberately model-agnostic: we expect them to be satisfied in any reasonable foundational

system for (higher) category theory. For example, we wish to cover the following three examples:

- (1) The standard theory of quasicategories;
- (2) Type-theoretic approaches like simplicial type theory [RS17; GWB24];
- (3) The internal theory of the (synthetic) category of small categories $\mathbf{Cat}$.

While the first two examples satisfy various strictness conditions, like strict unitality or associativity for functor composition, the third example forces us to work in a fully ‘homotopical’ setting, where no strictness conditions are imposed.

The chapter is organized as follows. The foundational language is introduced in Section 1.1, and its layer of coherences in Section 1.2. These are used in Section 1.3 to study equivalences of categories. We then introduce a variety of category constructors via their universal properties.

Epistemic status: This chapter started off giving a natural language description of ‘naive category theory’, capturing the model-independent language used in modern papers in higher category theory. Over time, more explicit coherences have been added to the list of axioms when it became clear that some natural language proofs were implicitly using them. We expect the present list to cover most of the required coherences, but it may be the case that we missed some. We hope to eventually settle the complete list of coherences by formalizing everything in a proof assistant.

1.1 The basic vocabulary

We start by introducing the basic vocabulary of synthetic category theory.

Axiom A.1 (Primitives). The following are the primitives of synthetic category theory:

- (1) There are *categories*, which we denote by symbols like C , D , and E . Among categories, there are *animae*.
- (2) For categories C and D , there is a primitive notion of a *functor* $f: C \rightarrow D$. Every category has an *identity functor* $\mathrm{id}_C: C \rightarrow C$. Given $f: C \rightarrow D$ and $g: D \rightarrow E$, there is a *composite functor* $g \circ f: C \rightarrow E$, sometimes also denoted gf .
- (3) There exists an anima $*$, called the *terminal category*.

Categories and functors play the foundational role in synthetic category theory that sets and functions play in naive set theory. The role of the animae among the categories is to encode the idea of a *homotopy-coherent collection*: where objects in classical mathematics assemble into sets, in synthetic category theory they assemble into animae. A category additionally allows objects to be related by morphisms that need not be invertible.

For now, animae form a primitive distinguished class of categories. We regard every category equivalent to an anima as an anima. Once we have introduced morphisms and their invertibility, we

can define groupoids as those categories in which every morphism is invertible. In the next chapter, we will impose the axiom that the animae are precisely the groupoids.

Definition 1.1.1 (Terms). A functor $f : C \rightarrow D$ may be displayed in *term notation* as

$$x : C \vdash f(x) : D.$$

The functor and the expression $f(x)$ are two notations for the same datum. We call $x : C$ its *context*. More generally, we may write $C \vdash t : D$ when the name of the variable is left implicit.

Throughout the chapter, we will indicate how the various rules we introduce may be written in their corresponding term notation. Conversely, any term expression obtained by combining these rules corresponds to a well-defined functor. The notation is inspired by type-theoretic approaches to synthetic category theory, such as [RS17; GWB24].

For example, composition of functors is displayed by substitution: for a second functor $g : D \rightarrow E$, the term notation for the composite $g \circ f$ is the expression $x : C \vdash g(f(x)) : E$. The term notation for the identity functor on C is $x : C \vdash x : C$. We also refer to the latter as the *universal term* of C .

Definition 1.1.2 (Objects). Given a term $x : C$ in context Γ , we say that x is an *object* (in context Γ) if Γ is an anima. If we speak about an *object of C* , we mean an object in an arbitrary anima context. An *absolute object* is a functor $x : * \rightarrow C$, i.e. an object in context $*$.

As already mentioned in the introduction of this book, one major difference between classical mathematics and homotopy theory is the idea that *equality is structure*. In classical mathematics, equality is a yes-no-question: two elements of a set are either equal or not. In homotopy theory, we change our perspective and allow there to be potentially several different identifications. The collection of identifications thus needs to assemble into an anima.

We thus introduce a primitive anima of identifications between every pair of functors. All our definitions and constructions should always respect identifications; we choose to implement this via explicit rules rather than via some deeper foundational principle as is for example done in homotopy type theory.

Axiom A.2 (Identifications). Given parallel functors $f, g : C \rightarrow D$, there is an anima $(f = g)$, called the *anima of identifications of f and g* . An *identification* $\alpha : f = g$ is an absolute object of this anima.

For $h : D \rightarrow E$ and $k : B \rightarrow C$, there are functors

$$h \star - : (f = g) \rightarrow (hf = hg), \quad - \star k : (f = g) \rightarrow (fk = gk),$$

called *postwhiskering* and *prewhiskering*, respectively.

Notation 1.1.3. We also denote identifications $\alpha : f = g$ in term notation as¹

$$x : C \vdash \alpha(x) : f(x) = g(x).$$

¹There is a slight clash between this notation and that of Definition 1.1.1: this does *not* mean a functor $C \rightarrow f(x) = g(x)$, which does not even make sense.

Given an anima A and two absolute objects $a, b : A$, we obtain an anima $a = b$ of identifications between them. In particular, given two identifications $\alpha, \beta : f = g$, we may form the anima $\alpha = \beta$. This construction may be iterated.

Definition 1.1.4. A *commutative square of functors*, often displayed by means of a diagram

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ g \downarrow & & \downarrow h \\ C' & \xrightarrow{f'} & D', \end{array}$$

consists of four functors f, g, h and f' with sources and targets as displayed in the diagram, together with a specified identification $\alpha : h \circ f = f' \circ g$.

We will often refer to the data of just the four functors f, g, h and f' as a *square*, and say that *the square commutes* whenever an identification $\alpha : h \circ f = f' \circ g$ can be provided.

A *commutative triangle of functors*, displayed by means of a diagram

$$\begin{array}{ccc} & D & \\ f \nearrow & & \searrow g \\ C & \xrightarrow{h} & E, \end{array}$$

consists of three functors f, g and h with sources and targets as displayed in the diagram, together with a specified identification $\alpha : h = g \circ f$.

We will similarly display more general diagrams of functors by drawing directed planar graphs with vertices labeled by categories and edges labeled by functors. We consider diagrams whose faces are bounded by two directed paths with the same source and target, and fix a parenthesization for each composite. We say the diagram *commutes* if every face comes equipped with an identification between these two composites.

Definition 1.1.5 (Equivalence of categories). A functor $f : C \rightarrow D$ is said to be an *equivalence* if there exists another functor $g : D \rightarrow C$ equipped with identifications $\text{id}_C = gf$ and $\text{id}_D = fg$. We refer to g as the *inverse* of f . Equivalences will often be introduced by writing $f : C \xrightarrow{\sim} D$. We may write $C \simeq D$ to indicate that C and D are equivalent via an unnamed equivalence.

The next two axioms belong to the universal constructions of group B. We introduce them here because they are needed to formulate the coherence axioms of group A. We begin with the universal property of the terminal category.

Axiom B.1 (Terminal category). For every category T , there is a functor $p_T : T \rightarrow *$. For any functors $f, g : T \rightarrow *$, the functor

$$p_{(f=g)} : (f = g) \longrightarrow *$$

is an equivalence.

In particular, any two functors to $*$ are identified.

Definition 1.1.6 (Contractible category). A category C is called *contractible* if the functor $p_C : C \rightarrow *$ is an equivalence.

Definition 1.1.7 (Constant functor). Let C and D be categories and let x be an absolute object of D . We define the *constant functor* $\text{const}_x : C \rightarrow D$ as the composite $x \circ p_C : C \rightarrow D$:

$$\text{const}_x : C \xrightarrow{p_C} * \xrightarrow{x} D.$$

Remark 1.1.8. Observe that a category C is contractible if and only if there exists an absolute object $x : * \rightarrow C$ equipped with an identification $\text{id}_C = \text{const}_x$. Such an identification $H : \text{id}_C = \text{const}_x$ is called a *contraction of C onto x* .

We will postpone most category constructors until later in the chapter, but we need one crucial one right now already in order to state various coherence relations.

Axiom B.2 (Product of categories). The *product* of categories C and D is a category $C \times D$ equipped with *projection functors* $\text{pr}_1 : C \times D \rightarrow C$ and $\text{pr}_2 : C \times D \rightarrow D$.

Given $f : T \rightarrow C$ and $g : T \rightarrow D$, there is a functor $(f, g) : T \rightarrow C \times D$ equipped with identifications $\beta_{f,g}^1 : \text{pr}_1(f, g) = f$ and $\beta_{f,g}^2 : \text{pr}_2(f, g) = g$. For any two functors $h, k : T \rightarrow C \times D$, the functor

$$(\text{pr}_1 \star -, \text{pr}_2 \star -) : (h = k) \longrightarrow (\text{pr}_1 h = \text{pr}_1 k) \times (\text{pr}_2 h = \text{pr}_2 k)$$

is an equivalence. If C and D are animae, then $C \times D$ is an anima.

We refer to the functor (f, g) as the *pair construction*. In term notation, it is written

$$x : T \vdash (f(x), g(x)) : C \times D,$$

with identifications $\text{pr}_1(f(x), g(x)) = f(x)$ and $\text{pr}_2(f(x), g(x)) = g(x)$. For arbitrary expressions $h(x), k(x) : C \times D$, two identifications $\alpha(x) : \text{pr}_1 h(x) = \text{pr}_1 k(x)$ and $\beta(x) : \text{pr}_2 h(x) = \text{pr}_2 k(x)$ give an identification $(\alpha(x), \beta(x)) : h(x) = k(x)$ by applying an inverse of the displayed equivalence. Its two projections are identified with α and β .

Remark 1.1.9. Given categories C and D , we define a *span* from C to D to be a triple (T, f, g) consisting of a category T together with functors $f : T \rightarrow C$ and $g : T \rightarrow D$. Axiom B.2 may be rephrased as saying that the triple $(C \times D, \text{pr}_1, \text{pr}_2)$ is the *terminal span* from C to D : every span $C \leftarrow T \rightarrow D$ admits a factorization through $C \times D$, and an identification of two functors $T \rightarrow C \times D$ amounts to identifications between their respective components.

Convention 1.1.10. A term of the form $z : C \times D \vdash t : E$ may also be written as

$$x : C, y : D \vdash t(x, y) : E$$

whenever the term t only depends on the variable z through the two projections $\text{pr}_1(z)$ and $\text{pr}_2(z)$. The expression $t(x, y)$ is then obtained from t by replacing $\text{pr}_1(z)$ by x and $\text{pr}_2(z)$ by y . For more than two variables we use a fixed parenthesization of their product.

1.2 Coherences

So far, we have introduced functors and their identifications, but we have essentially no relations between them. We next specify the coherence relations we will use. We first describe composition of identifications and its compatibility with whiskering. We then introduce the unit and associativity identifications for composition of functors, and the relations among them.

For readability, we write most of these coherences in their ‘term notation’, cf. Notation 1.1.3. In each case, the axiom introduces a specified identification of functors, which we will often leave nameless to avoid cluttering of notation.

1.2.1 Composition of identifications

Identifications can be composed: given identifications $\sigma : f = g$ and $\tau : g = h$, we obtain an identification $\tau \circ \sigma : f = h$. We call this *vertical composition* to distinguish it from the horizontal composition introduced below. Every functor has an identity identification, and every identification has an inverse.

Axiom A.3 (Composition of identifications). For every functor f , there is an identity identification $\text{id}_f : f = f$. For parallel functors $f, g, h : C \rightarrow D$, there is a composition functor

$$- \circ - : (g = h) \times (f = g) \longrightarrow (f = h).$$

There is also an *inverse functor*

$$(-)^{-1} : (f = g) \rightarrow (g = f).$$

This composition is unital and associative, and the inverse functor supplies two-sided inverses, through specified identifications of functors on the corresponding animae of identifications. For $\alpha : f = g$, $\beta : g = h$, and $\gamma : h = k$, where $k : C \rightarrow D$, these laws read

$$\begin{aligned} \text{id}_g \circ \alpha &= \alpha, & \alpha \circ \text{id}_f &= \alpha, \\ (\gamma \circ \beta) \circ \alpha &= \gamma \circ (\beta \circ \alpha), \\ \alpha^{-1} \circ \alpha &= \text{id}_f, & \alpha \circ \alpha^{-1} &= \text{id}_g. \end{aligned}$$

The displayed laws are specified identifications of functors on the relevant animae of inputs. For parallel functors $f, g, h, k : C \rightarrow D$, the associativity law compares the two functors

$$(h = k) \times ((g = h) \times (f = g)) \longrightarrow (f = k)$$

given by $(\gamma, (\beta, \alpha)) \mapsto (\gamma \circ \beta) \circ \alpha$ and $(\gamma, (\beta, \alpha)) \mapsto \gamma \circ (\beta \circ \alpha)$. Precomposing these comparisons with a functor into the input anima gives the same laws for the resulting expressions.

We can now also speak of commutative squares, triangles, and more general diagrams of identifications, using the convention of Definition 1.1.4 with vertical composition along their boundaries.

1.2.2 Whiskering and horizontal composition

Whiskering composes an identification with a functor on either side. We first require that it respect identity identifications and vertical composition.

Axiom A.4 (Whiskering and vertical composition). Prewhiskering and postwhiskering preserve identity identifications and vertical composition through specified identifications of functors on the input anima of identifications. For functors $f, g, h: C \rightarrow D$, $k: B \rightarrow C$ and $u: D \rightarrow E$, these laws read

$$\begin{aligned} u \star \text{id}_f &= \text{id}_{u \circ f}, & \text{id}_f \star k &= \text{id}_{f \circ k}, \\ u \star (\tau \circ \sigma) &= (u \star \tau) \circ (u \star \sigma), & (\tau \circ \sigma) \star k &= (\tau \star k) \circ (\sigma \star k), \end{aligned}$$

where $\sigma: f = g$ and $\tau: g = h$. The two vertical-composition comparisons are identifications of functors on $(g = h) \times (f = g)$.

Given identifications $\tau: F = G$ and $\sigma: h = k$, where $F, G: D \rightarrow E$ and $h, k: T \rightarrow D$, there are two ways to obtain an identification $Fh = Gk$: one can first change h to k , or first change F to G . The interchange rule compares these two constructions.

Axiom A.5 (Interchange). The two ways of pasting identifications are identified. More precisely, for $\tau: F = G$ with $F, G: D \rightarrow E$ and $h, k: T \rightarrow D$, there is an identification of functors $(h = k) \rightarrow (Fh = Gk)$

$$(\tau \star k) \circ (F \star -) = (G \star -) \circ (\tau \star h).$$

Here composition with a fixed identification uses it as a constant input. Likewise, for fixed $\sigma: h = k$, the two functors $(F = G) \rightarrow (Fh = Gk)$ given by

$$(- \star k) \circ (F \star \sigma) \quad \text{and} \quad (G \star \sigma) \circ (- \star h)$$

are identified.

Given $\alpha: f = f'$ and $\beta: g = g'$, we write

$$\beta * \alpha := (\beta \star f') \circ (g \star \alpha) : gf = g'f'$$

for their *horizontal composite*. Interchange compares this with $(g' \star \alpha) \circ (\beta \star f)$. In particular, horizontal composition preserves identities and vertical composites. For fixed functors f, f', g, g' , this construction defines a functor

$$- * -: (g = g') \times (f = f') \rightarrow (gf = g'f').$$

1.2.3 Units and associativity of functor composition

Composition of functors is unital and associative through specified identifications, called the *unitors* and *associator*.

Axiom A.6 (Unitors and associator). For functors $f: C \rightarrow D$, $g: D \rightarrow E$ and $h: E \rightarrow F$, there are identifications

$$\lambda_f: \text{id}_D \circ f = f, \quad \rho_f: f \circ \text{id}_C = f, \quad \alpha_{f,g,h}: (h \circ g) \circ f = h \circ (g \circ f).$$

The unitors and associator are compatible with identifications of the functors to which they are applied, in the following sense.

Axiom A.7 (Compatibility with horizontal composition). Horizontal composition of identifications is compatible with the unitality and associativity identifications from Axiom A.6:

- For an identification $\alpha: f = f'$ of functors $C \rightarrow D$ there are commutative squares of identifications

$$\begin{array}{ccc} \text{id}_D \circ f & \xrightarrow{\text{id}_D * \alpha} & \text{id}_D \circ f' \\ \lambda_f \downarrow & & \downarrow \lambda_{f'} \\ f & \xrightarrow{\alpha} & f' \end{array} \quad \text{and} \quad \begin{array}{ccc} f \circ \text{id}_C & \xrightarrow{\alpha * \text{id}_C} & f' \circ \text{id}_C \\ \rho_f \downarrow & & \downarrow \rho_{f'} \\ f & \xrightarrow{\alpha} & f'. \end{array}$$

- For identifications $\alpha: f = f'$, $\beta: g = g'$ and $\gamma: h = h'$, the following square of identifications commutes:

$$\begin{array}{ccc} (h \circ g) \circ f & \xrightarrow{(\gamma * \beta) * \alpha} & (h' \circ g') \circ f' \\ \alpha_{f,g,h} \downarrow & & \downarrow \alpha_{f',g',h'} \\ h \circ (g \circ f) & \xrightarrow{\gamma * (\beta * \alpha)} & h' \circ (g' \circ f'). \end{array}$$

These comparisons are identifications of the corresponding functors on products of the input animae of identifications. Their restrictions along functors into those animae give the same formulas for identifications.

Remark 1.2.1. Taking some of the input identifications to be identities gives compatibility with identifications in each individual functor.

It will sometimes be convenient to abbreviate the conditions from Axiom A.7 as follows:

$$\text{id} * \alpha = \alpha = \alpha * \text{id} \quad \text{and} \quad (\gamma * \beta) * \alpha = \gamma * (\beta * \alpha).$$

More generally, if we are given identifications $\alpha: f = g$ and $\alpha': f' = g'$ of functors $C \rightarrow D$, and we happen to be given identifications $f = f'$ and $g = g'$, we will write $\alpha = \alpha'$ to indicate the existence

of a commutative square of identifications as follows:

$$\begin{array}{ccc} f & \xrightarrow[\equiv]{\alpha} & g \\ \downarrow \equiv & & \downarrow \equiv \\ f' & \xrightarrow[\equiv]{\alpha'} & g'. \end{array}$$

1.2.4 The pentagon and triangle identities

Finally, the unitors and associator must themselves be compatible. The pentagon compares the two ways of reassociating a composite of four functors. The triangle compares removing an identity functor before or after reassociation.

Axiom A.8 (Pentagon and triangle identities). For functors $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$ and $k: D \rightarrow E$, the following pentagon of identifications commutes:

$$\begin{array}{ccccc} & & (k \circ h) \circ (g \circ f) & & \\ & \nearrow \alpha_{f,g,k \circ h} & & \searrow \alpha_{g \circ f,h,k} & \\ ((k \circ h) \circ g) \circ f & & & & k \circ (h \circ (g \circ f)) \\ \downarrow \alpha_{g,h,k} * \text{id}_f & & & & \uparrow \text{id}_k * \alpha_{f,g,h} \\ (k \circ (h \circ g)) \circ f & \xrightarrow{\alpha_{f,h \circ g,k}} & & & k \circ ((h \circ g) \circ f) \end{array}$$

Moreover, for composable functors $f: C \rightarrow D$ and $g: D \rightarrow E$, the following triangle of identifications commutes:

$$\begin{array}{ccc} (g \circ \text{id}_D) \circ f & \xrightarrow{\alpha_{f,\text{id}_D,g}} & g \circ (\text{id}_D \circ f) \\ \searrow \rho_g * \text{id}_f & & \downarrow \text{id}_g * \lambda_f \\ & & g \circ f. \end{array}$$

1.3 Equivalences of categories

We now develop some basic properties of the equivalences introduced in Definition 1.1.5.

Proposition 1.3.1. *Let $f: C \rightarrow D$ be a functor. For f to be an equivalence, it suffices that the following properties hold for every category T :*

- (1) *For $a, b: T \rightarrow C$, any identification $fa = fb$ gives an identification $a = b$.*
- (2) *For any $d: T \rightarrow D$, there is a functor $c: T \rightarrow C$ equipped with an identification $fc = d$.*

Proof. Apply the second property to id_D to obtain $g: D \rightarrow C$ and $fg = \text{id}_D$. The associator, this identification, and the unitors identify $f(gf)$ with $f\text{id}_C$. The first property, with $T = C$, gives $gf = \text{id}_C$. $\square$

Definition 1.3.2 (Section/retraction). Consider a functor $f: C \rightarrow D$. A *section* of f is a functor $s: D \rightarrow C$ equipped with an identification $fs = \text{id}_D$. A *retraction* of f is a functor $r: D \rightarrow C$ equipped with an identification $\text{id}_C = rf$.

Lemma 1.3.3. A functor $f: C \rightarrow D$ is an equivalence if and only if it admits both a section s and a retraction r . In this case there exists an identification $s = r$ and s is an inverse of f .

Proof. If f is an equivalence with inverse $g: D \rightarrow C$, then clearly g is both a section and a retraction of f . Conversely, given a section $s: D \rightarrow C$, $fs = \text{id}_D$ and a retraction $r: D \rightarrow C$, $\text{id}_C = rf$, we obtain an identification $s = r$ by composing the following chain of identifications:

$$s = \text{id}_C \circ s = (r \circ f) \circ s = r \circ (f \circ s) = r \circ \text{id}_D = r.$$

It follows that f is an equivalence with inverse s , since there are identifications $fs = \text{id}_D$ and $sf = rf = \text{id}_C$. $\square$

Corollary 1.3.4. Any two inverses g and h of an equivalence f are identified. $\square$

Lemma 1.3.5. If $f: C \rightarrow D$ and $f': D \rightarrow E$ are equivalences with inverses $g: D \rightarrow C$ and $g': E \rightarrow D$, then also the composite $f'f: C \rightarrow E$ is an equivalence with inverse $gg': E \rightarrow C$.

Proof. We have $gg'f'f = g\text{id}_Df = gf = \text{id}_C$ and $f'f'gg' = f'\text{id}_Dg' = f'g' = \text{id}_E$. $\square$

Lemma 1.3.6. Equivalences of categories satisfy the 2-out-of-3 property: given functors $f: C \rightarrow D$ and $g: D \rightarrow E$, if two of the three functors f , g and gf are equivalences, then so is the third.

Proof. If g and f are equivalences, then so is gf by Lemma 1.3.5. If f and gf are equivalences, then we may rewrite g as the composite $g = g \circ \text{id}_D = g \circ f \circ f^{-1} = gf \circ f^{-1}$, so that g is the composite of two equivalences and hence itself an equivalence. A similar argument shows that f is an equivalence whenever g and gf are equivalences. $\square$

Lemma 1.3.7. Equivalences of categories satisfy the 2-out-of-6 property: given functors $f: C \rightarrow D$, $g: D \rightarrow E$ and $h: E \rightarrow F$ such that gf and hg are equivalences, also the functors f , g , h and hgf are equivalences.

Proof. Consider the functors $f' := f \circ (gf)^{-1}: E \rightarrow D$ and $h' := (hg)^{-1} \circ h: E \rightarrow D$. Then there are identifications

$$g \circ f' = g \circ f \circ (gf)^{-1} = gf \circ (gf)^{-1} = \text{id}_E$$

and

$$h' \circ g = (hg)^{-1} \circ h \circ g = (hg)^{-1} \circ hg = \text{id}_D.$$

Hence it follows from Lemma 1.3.3 that g is an equivalence. In light of Lemma 1.3.6, we then get that also f and h are equivalences, and then so is hgf . $\square$

Definition 1.3.8. We say that a functor $f: C \rightarrow D$ is a *retract* of a functor $f': C' \rightarrow D'$ if there exists a commutative diagram of the form

$$\begin{array}{ccccc} C & \xrightarrow{g} & C' & \xrightarrow{h} & C \\ f \downarrow & & \downarrow f' & & \downarrow f \\ D & \xrightarrow{k} & D' & \xrightarrow{j} & D, \end{array}$$

together with identifications $\alpha: \text{id}_C = h \circ g$ and $\beta: j \circ k = \text{id}_D$, and an identification between $\text{id}_f: f = f$ and the composite identification

$$f = f \circ \text{id}_C \stackrel{\alpha}{=} f \circ h \circ g = j \circ f' \circ g = j \circ k \circ f \stackrel{\beta}{=} \text{id}_D \circ f = f.$$

Lemma 1.3.9. Assume that $f: C \rightarrow D$ is a retract of $f': C' \rightarrow D'$. If f' is an equivalence, then so is f .

Proof. Let $(f')^{-1}: D' \rightarrow C'$ be an inverse of f' . We claim that the composite $w := h \circ (f')^{-1} \circ k: D \rightarrow C$ is inverse to f . Indeed, we have identifications

$$f \circ w = f \circ h \circ (f')^{-1} \circ k = j \circ f' \circ (f')^{-1} \circ k = j \circ k = \text{id}_D$$

and

$$w \circ f = h \circ (f')^{-1} \circ k \circ f = h \circ (f')^{-1} \circ f' \circ g = h \circ g = \text{id}_C.$$

This finishes the proof. $\square$

Lemma 1.3.10. *Prewhiskering or postwhiskering by an equivalence induces equivalences of anima of identifications.*

Proof. Composition with a fixed identification is an equivalence, with inverse given by composition with its inverse. This follows from the vertical unit, associativity, and inverse laws.

Let $F: C \rightarrow D$ have inverse G , and consider two functors $a, b: T \rightarrow C$. We wish to show that the first functor $F \star -$ in the following diagram is an equivalence:

$$(a = b) \xrightarrow{F \star -} (Fa = Fb) \xrightarrow{G \star -} (GFa = GFb) \xrightarrow{F \star -} (FGFa = FGFB).$$

Interchange with $GF = \text{id}_C$ identifies the composite of the first two functors with transport along the identifications $GFa = a$ and $b = GFb$, and so that composite is an equivalence. The composite of the last two is an equivalence for the same reason, using $FG = \text{id}_D$, and so the claim follows from the 2-out-of-6 property. Prewhiskering is treated in the same way. $\square$

Consequently an equivalence lifts a specified identification, together with an identification of its image with that identification. Applying the same result to functors into an anima of identifications also lifts identifications between identifications.

From the universal property of products, we may directly deduce that products are commutative, unital, and associative. For commutativity of products, one argues as follows:

Lemma 1.3.11. For categories C and D , the functor $(\text{pr}_2, \text{pr}_1): C \times D \rightarrow D \times C$ is an equivalence, with inverse given by $(\text{pr}_2, \text{pr}_1): D \times C \rightarrow C \times D$.

Proof. We need to provide identifications $(\text{pr}_2, \text{pr}_1) \circ (\text{pr}_2, \text{pr}_1) = \text{id}_{C \times D}$ and $(\text{pr}_2, \text{pr}_1) \circ (\text{pr}_2, \text{pr}_1) = \text{id}_{D \times C}$. By symmetry, it will suffice to produce the first identification. For this, it will in turn suffice to produce identifications after composing with the functors pr_1 and pr_2 . For the composition with pr_1 we compute that

$$\text{pr}_1 \circ ((\text{pr}_2, \text{pr}_1) \circ (\text{pr}_2, \text{pr}_1)) = (\text{pr}_1 \circ (\text{pr}_2, \text{pr}_1)) \circ (\text{pr}_2, \text{pr}_1) = \text{pr}_2 \circ (\text{pr}_2, \text{pr}_1) = \text{pr}_1,$$

which is indeed identified with $\text{pr}_1 \circ \text{id}_{C \times D}$. The case for composition with pr_2 is analogous. This finishes the proof. $\square$

Unitality may be argued similarly:

Lemma 1.3.12. *For every category C , the projection functor $\text{pr}_1 : C \times * \rightarrow C$ is an equivalence, with inverse given by the functor $(\text{id}_C, p_C) : C \rightarrow C \times *$.*

Proof. The composite $\text{pr}_1 \circ (\text{id}_C, p_C) : C \rightarrow C$ is identified with id_C by the defining property of (id_C, p_C) . To show that the composite $(\text{id}_C, p_C) \circ \text{pr}_1 : C \times * \rightarrow C \times *$ is identified with $\text{id}_{C \times *}$, it will suffice to do so after composing with pr_1 and with pr_2 . The case for pr_2 is clear, since any two functors $C \times * \rightarrow *$ are identified. In the case of pr_1 we compute

$$\text{pr}_1 \circ ((\text{id}_C, p_C) \circ \text{pr}_1) = (\text{pr}_1 \circ (\text{id}_C, p_C)) \circ \text{pr}_1 = \text{id}_C \circ \text{pr}_1 = \text{pr}_1 = \text{pr}_1 \circ \text{id}_{C \times *},$$

where the first identification is associativity, the second is the relation proved in the first paragraph, and the third and fourth are unitality. This finishes the proof. $\square$

Remark 1.3.13. In the previous proofs, we have been very explicit about the usage of associativity and naturality identifications. In the remainder of this book, we will often leave the usage of such identifications implicit in order to make the proofs more concise.

Exercise 1.3.14 (Exercise 1.9.1 (outside selection)). Formulate and prove the associativity of the product of categories.

Construction 1.3.15 (Pairing of identifications). Given functors $f, f' : T \rightarrow C$ and $g, g' : T \rightarrow D$, pairing defines a functor

$$(-, -) : (f = f') \times (g = g') \longrightarrow ((f, g) = (f', g')).$$

For $\varphi : f = f'$ and $\psi : g = g'$, its value (φ, ψ) is equipped with identifications

$$\begin{aligned} \beta_{f', g'}^1 \circ (\text{pr}_1 \star (\varphi, \psi)) &= \varphi \circ \beta_{f, g}^1, \\ \beta_{f', g'}^2 \circ (\text{pr}_2 \star (\varphi, \psi)) &= \psi \circ \beta_{f, g}^2. \end{aligned}$$

There are identifications

$$(\text{id}_f, \text{id}_g) = \text{id}_{(f, g)}, \quad (\varphi' \circ \varphi, \psi' \circ \psi) = (\varphi', \psi') \circ (\varphi, \psi),$$

where $\varphi' : f' = f''$ and $\psi' : g' = g''$.

Proof. Under the identifications $\beta_{f, g}^1 : \text{pr}_1(f, g) = f$ and $\beta_{f, g}^2 : \text{pr}_2(f, g) = g$, φ and ψ correspond to two absolute objects of the two factors of the following product:

$$(\text{pr}_1(f, g) = \text{pr}_1(f', g')) \times (\text{pr}_2(f, g) = \text{pr}_2(f', g')).$$

Since this product is equivalent to the anima $(f, g) = (f', g')$ by Axiom B.2, this provides the identification (φ, ψ) with the two claimed identification.

We leave the identity law and composition law as exercises for the reader. $\square$

Lemma 1.3.16 (Pairing and precomposition). *Let $f: T \rightarrow C$, $g: T \rightarrow D$, and $\sigma: S \rightarrow T$ be functors. Pairing commutes with precomposition: there is an identification*

$$(f, g) \circ \sigma = (f \circ \sigma, g \circ \sigma).$$

This comparison is compatible with identifications of the input functors, with successive precomposition, and with precomposition by an identity functor, using the associativity and unitality identifications.

Proof. After composition with pr_1 and pr_2 , both sides identify with $f \circ \sigma$ and $g \circ \sigma$, respectively. The product axiom therefore supplies the comparison, with these prescribed projections. Its compatibilities can likewise be checked after applying both projections, where they reduce to the coherence laws for composition and identifications. For compatibility with identity functors, one also uses

$$\rho_{ab} = (a \star \rho_b) \circ \alpha_{\text{id}, b, a} \quad (1.1)$$

for composable functors b, a . This follows by prewhiskering with the identity, applying the pentagon for $a, b, \text{id}, \text{id}$ and the triangle identities, and cancelling by Lemma 1.3.10. We leave the remaining checks to the reader. $\square$

Construction 1.3.17. The formation of products of categories is functorial: given $f: C \rightarrow C'$ and $g: D \rightarrow D'$, we obtain a functor

$$f \times g := (f \text{ pr}_1, g \text{ pr}_2) : C \times D \rightarrow C' \times D'.$$

For identifications $\varphi: f = f'$ and $\psi: g = g'$ between functors with these same sources and targets, we define

$$\varphi \times \psi := (\varphi \star \text{pr}_1, \psi \star \text{pr}_2).$$

This gives a functor on the product of the two animae of identifications. It preserves identity identifications and vertical composition by Construction 1.3.15 and the corresponding laws for whiskering.

Exercise 1.3.18 (Exercise 1.9.3 (outside selection)). Show that there are identifications

$$\begin{aligned} \text{id}_C \times \text{id}_D &= \text{id}_{C \times D}, \\ (f' \times g') \circ (f \times g) &= (f' f) \times (g' g) \end{aligned}$$

for $f: C \rightarrow C'$, $f': C' \rightarrow C''$, $g: D \rightarrow D'$, and $g': D' \rightarrow D''$.

1.4 Mapping animae

We now introduce the mapping anima between two categories C and D : an anima whose objects are functors from C to D . The mapping animae will play an important role in the development of the theory: frequently, we may check conditions at the level of mapping animae. We introduce $\text{Map}(C, D)$ via its universal property in terms of currying/uncurrying functors.

Axiom B.3 (Evaluation and currying for mapping animae). For categories C, D , there is an anima $\text{Map}(C, D)$ equipped with an *evaluation functor*

$$\text{ev}_{\text{Map}}: \text{Map}(C, D) \times C \longrightarrow D.$$

If X is an anima and $h: X \times C \rightarrow D$ is a functor, there is a functor $h_c: X \rightarrow \text{Map}(C, D)$ equipped with an identification $(h_c)^u = h$, where

$$a^u := \text{ev}_{\text{Map}} \circ (a \times \text{id}_C): X \times C \longrightarrow D.$$

For any $a, b: X \rightarrow \text{Map}(C, D)$, the functor

$$(-)^u: (a = b) \longrightarrow (a^u = b^u)$$

induced by product functoriality and postwhiskering with evaluation is an equivalence.

We refer to the functor h_c as the *currying* of h , and to a^u as the *uncurrying* of a . Note that the universal property of Map only holds for anima parameters. A more general form of currying with an arbitrary category as parameter will be provided by functor categories in Section 1.7.

Construction 1.4.1 (Functors as points of mapping animae). A functor $f: C \rightarrow D$ determines a point $\ulcorner f \urcorner: * \rightarrow \text{Map}(C, D)$ by currying $f \text{ pr}_2: * \times C \rightarrow D$. Conversely, a point $a: * \rightarrow \text{Map}(C, D)$ determines the functor

$$C \simeq * \times C \xrightarrow{a \times \text{id}_C} \text{Map}(C, D) \times C \xrightarrow{\text{ev}_{\text{Map}}} D.$$

The functor represented by $\ulcorner f \urcorner$ is identified with f . Moreover, for two points a, b , uncurrying followed by prewhiskering with $C \simeq * \times C$ gives an equivalence from $(a = b)$ to the anima of identifications between their represented functors, where we use Lemma 1.3.10. This equivalence also identifies a point with the representative of its represented functor.

From now on we pass freely back and forth between a functor and its representing point using the comparisons just described; in particular, we will usually denote the point $\ulcorner f \urcorner$ simply by f .

Accordingly, a functor $H: \text{Map}(A, B) \rightarrow \text{Map}(C, D)$ sends a functor $f: A \rightarrow B$ to a functor $H(f): C \rightarrow D$, and induces a functor

$$(f = g) \longrightarrow (H(f) = H(g)).$$

These operations respect composition and identities. If H is an equivalence, the induced functor on identification animae is an equivalence as well.

Proposition 1.4.2 (Uncurrying). *For every anima X and categories C, D , uncurrying is represented by an equivalence*

$$(-)^u: \text{Map}(X, \text{Map}(C, D)) \xrightarrow{\sim} \text{Map}(X \times C, D).$$

Proof. Put $P := \text{Map}(X, \text{Map}(C, D))$. Iterated evaluation gives a functor

$$P \times X \times C \longrightarrow \text{Map}(C, D) \times C \xrightarrow{\text{ev}_{\text{Map}}} D.$$

Since P is an anima, currying this functor constructs an actual functor $(-)^u : P \rightarrow \text{Map}(X \times C, D)$. For the inverse, put $Q := \text{Map}(X \times C, D)$, and consider the evaluation functor $Q \times X \times C \rightarrow D$. If we first curry in the variable C , with anima parameter $Q \times X$, and then curry in the variable X , with anima parameter Q , we obtain a functor $Q \rightarrow P$. To verify that both composites are the identity, it suffices by Axiom B.3 to uncurry: once in $X \times C$ for the composite on Q , and successively in X and C for the composite on P . In both cases, the resulting identification follows from the defining identifications for currying. $\square$

Construction 1.4.3. The mapping animae are covariantly functorial in the second variable and contravariantly functorial in the first. For $g : D \rightarrow E$, curry the composite

$$\text{Map}(C, D) \times C \xrightarrow{\text{ev}_{\text{Map}}} D \xrightarrow{g} E$$

to obtain $g_* : \text{Map}(C, D) \rightarrow \text{Map}(C, E)$. For $f : B \rightarrow C$, curry evaluation precomposed with $\text{id}_{\text{Map}(C, D)} \times f$ to obtain $f^* : \text{Map}(C, D) \rightarrow \text{Map}(B, D)$. We call these functors postcomposition with g and precomposition with f , respectively.

An identification $\gamma : g = g'$ induces an identification $\gamma_* : g_* = g'_*$. Indeed, whisker γ with evaluation and use the uncurrying comparisons for g_* and g'_* to obtain an identification between their uncurried functors. Lift this through the equivalence of animae of identifications in the mapping-anima rule. Similarly, for $\varphi : f = f'$, pair φ with the identity on $\text{Map}(C, D)$, postcompose with evaluation, and lift to obtain $\varphi^* : f^* = f'^*$.

Lemma 1.4.4. *For every category C , the anima $\text{Map}(C, *)$ is contractible.*

Proof. We must show that the functor $\text{Map}(C, *) \rightarrow *$ is an equivalence. An inverse is given by the functor $* \rightarrow \text{Map}(C, *)$ obtained by currying the terminal functor $* \times C \rightarrow *$. The composite on $*$ is automatically identified with the identity. For the composite on $\text{Map}(C, *)$, it suffices by the mapping-anima axiom to compare the two uncurried functors $\text{Map}(C, *) \times C \rightarrow *$. These are again identified by the terminal-category axiom. $\square$

Proposition 1.4.5. *For categories C, D, E , the projections induce an equivalence*

$$(\text{pr}_{1,*}, \text{pr}_{2,*}) : \text{Map}(E, C \times D) \xrightarrow{\sim} \text{Map}(E, C) \times \text{Map}(E, D).$$

Proof. We define a functor

$$h : \text{Map}(E, C) \times \text{Map}(E, D) \longrightarrow \text{Map}(E, C \times D)$$

by pairing the two functors

$$\begin{aligned} (\text{Map}(E, C) \times \text{Map}(E, D)) \times E &\xrightarrow{\text{pr}_1 \times \text{id}_E} \text{Map}(E, C) \times E \xrightarrow{\text{ev}_{\text{Map}}} C, \\ (\text{Map}(E, C) \times \text{Map}(E, D)) \times E &\xrightarrow{\text{pr}_2 \times \text{id}_E} \text{Map}(E, D) \times E \xrightarrow{\text{ev}_{\text{Map}}} D, \end{aligned}$$

and currying the resulting functor to $C \times D$. This is possible since $\text{Map}(E, C) \times \text{Map}(E, D)$ is an anima.

We start by showing that the composite $(\text{pr}_{1,*}, \text{pr}_{2,*}) \circ h$ is identified with the identity. It will suffice to show this after projecting to both factors, and by symmetry we may just do this for $\text{Map}(E, C)$.

In other words, we must show that $\text{pr}_{1,*} \circ h$ is identified with pr_1 . It will in turn suffice to show this after uncurrying. But the uncurrying of both functors is identified with

$$(\text{Map}(E, C) \times \text{Map}(E, D)) \times E \xrightarrow{\text{pr}_1 \times \text{id}_E} \text{Map}(E, C) \times E \xrightarrow{\text{ev}_{\text{Map}}} C,$$

by the construction of h and the definition of postcomposition.

We will now show that also the composite $h \circ (\text{pr}_{1,*}, \text{pr}_{2,*})$ is identified with the identity. Uncurrying reduces this to comparing two functors

$$\text{Map}(E, C \times D) \times E \longrightarrow C \times D.$$

It suffices to compare their two projections, and again we may just consider the C -component. The projection of the uncurried identity is $\text{pr}_1 \circ \text{ev}_{\text{Map}}$; the projection of the other functor is $\text{ev}_{\text{Map}} \circ (\text{pr}_{1,*} \times \text{id}_E)$. These are identified by the defining commutative square for postcomposition:

$$\begin{array}{ccc} \text{Map}(E, C \times D) \times E & \xrightarrow{\text{ev}_{\text{Map}}} & C \times D \\ \text{pr}_{1,*} \times \text{id}_E \downarrow & & \downarrow \text{pr}_1 \\ \text{Map}(E, C) \times E & \xrightarrow{\text{ev}_{\text{Map}}} & C. \end{array}$$

The product axiom and then the mapping-anima axiom give the desired identification. $\square$

We next compare uncurrying with substitution in the anima parameter.

Lemma 1.4.6 (Uncurrying and substitution). *Let C, D be categories, X, Y animae, and $f: X \rightarrow \text{Map}(C, D)$ and $\sigma: Y \rightarrow X$ functors. Uncurrying commutes with substitution: there is an identification*

$$(f \circ \sigma)^u = f^u \circ (\sigma \times \text{id}_C).$$

This comparison is compatible with identifications of the input functors, with successive substitution, and with substitution by an identity functor, using the associativity and unitality identifications. Moreover, uncurrying preserves identity identifications and their vertical composition.

Proof. Using the definition of uncurrying, functoriality of products, and associativity, we obtain identifications

$$\begin{aligned} (f \circ \sigma)^u &= \text{ev}_{\text{Map}} \circ ((f \circ \sigma) \times \text{id}_C) \\ &= \text{ev}_{\text{Map}} \circ (f \times \text{id}_C) \circ (\sigma \times \text{id}_C) \\ &= f^u \circ (\sigma \times \text{id}_C). \end{aligned}$$

The compatibilities follow from those for pairing and composition, as in Lemma 1.3.16; the unit comparison also uses equation (1.1). Preservation of identity identifications and vertical composition follows from the corresponding properties of products and whiskering with evaluation. We leave the details to the reader. $\square$

We can now lift composition of functors to an operation on mapping animae.

Construction 1.4.7. For categories C, D and E , consider the double evaluation functor

$$\text{Map}(D, E) \times \text{Map}(C, D) \times C \xrightarrow{\text{id} \times \text{ev}} \text{Map}(D, E) \times D \xrightarrow{\text{ev}} E.$$

Currying in C defines a functor

$$\mu_{C,D,E} : \text{Map}(D, E) \times \text{Map}(C, D) \longrightarrow \text{Map}(C, E),$$

which by construction comes with an identification

$$\text{ev}_{\text{Map}}(\mu(g, f), c) = \text{ev}_{\text{Map}}(g, \text{ev}_{\text{Map}}(f, c)).$$

The projection $* \times C \rightarrow C$ similarly curries to a functor

$$\eta_C : * \longrightarrow \text{Map}(C, C),$$

whose evaluation is identified with that projection.

We may use these functors to obtain composition and evaluation for term expressions. Given terms

$$\Gamma \vdash f : \text{Map}(C, D) \quad \text{and} \quad \Gamma \vdash g : \text{Map}(D, E),$$

we may form

$$\Gamma \vdash g \circ f := \mu_{C,D,E}(g, f) : \text{Map}(C, E).$$

Likewise, given a term $\Gamma \vdash f : \text{Map}(C, D)$ and $\Gamma \vdash x : C$, we define

$$\Gamma \vdash f(x) := \text{ev}_{\text{Map}}(f, x) : D.$$

Proposition 1.4.8 (Coherence of composition). *Composition in mapping animae has specified left and right unitors and an associator. These satisfy the triangle and pentagon identities through specified identifications.*

Proof. Throughout the proof, we make use of the following notation: Given a ‘parameter anima’ P and a functor $f : P \rightarrow \text{Map}(C, D)$, we define

$$\widetilde{f} := (\text{pr}_1, f'') : P \times C \rightarrow P \times D.$$

If $g : P \rightarrow \text{Map}(D, E)$ is any other functor, then there is an identification

$$\widetilde{g} \circ \widetilde{f} = \widetilde{g \circ f};$$

indeed, we may construct this identification componentwise, where it is clear for the first component and follows from the defining property of $\mu_{C,D,E}$ in the second component. The compatibility with the primitive coherence identifications and change of parameter can likewise be checked after both projections, using Lemmas 1.4.6 and 1.3.16 and the coherence axioms for composition.

We now use this to construct the associator. Given a third functor $h : P \rightarrow \text{Map}(E, F)$, we obtain three composable functors

$$P \times C \xrightarrow{\widetilde{f}} P \times D \xrightarrow{\widetilde{g}} P \times E \xrightarrow{\widetilde{h}} P \times F.$$

Using the primitive associator and the previous comparisons, we get identifications

$$(\widetilde{h \circ g}) \circ f = \widetilde{h \circ g} \circ \widetilde{f} = (\widetilde{h} \circ \widetilde{g}) \circ \widetilde{f} = \widetilde{h} \circ (\widetilde{g} \circ \widetilde{f}) = \widetilde{h} \circ \widetilde{g \circ f} = \widetilde{h \circ (g \circ f)}.$$

Passing to the second component of this identification, we obtain an identification between the uncurrying of $(h \circ g) \circ f$ and that of $h \circ (g \circ f)$; we may then use the mapping-anima axiom to lift this to the desired associator.

For the unitors, note that the constant functor $\text{const}_{\text{id}_C} : P \rightarrow \text{Map}(C, C)$ corresponds under (un)currying to the projection functor $\text{pr}_2 : P \times C \rightarrow C$, and so the associated functor $\widetilde{\text{const}_{\text{id}_C}} : P \times C \rightarrow P \times C$ is identified with the identity. We then consider the unitors $\lambda_{\widetilde{f}} : \text{id}_{P \times D} \circ \widetilde{f} = \widetilde{f}$ and $\rho_{\widetilde{f}} : \widetilde{f} \circ \text{id}_{P \times C} = \widetilde{f}$, and with the same arguments as above deduce the desired relations $\text{id}_D \circ f = f$ and $f \circ \text{id}_C = f$.

Observe that both of these constructions are compatible with substitution in the parameter anima P : given a map of animae $\sigma : Q \rightarrow P$, we can construct identifications

$$\alpha_{h,g,f}^P \star \sigma = \alpha_{h\sigma,g\sigma,f\sigma}^Q, \quad \lambda_f^P \star \sigma = \lambda_{f\sigma}^Q, \quad \rho_f^P \star \sigma = \rho_{f\sigma}^Q,$$

up to the substitution comparisons relating e.g. $((h \circ g) \circ f)\sigma$ to $((h\sigma) \circ (g\sigma)) \circ f\sigma$. To see this, note that by Lemma 1.4.6 there is a commutative square

$$\begin{array}{ccc} Q \times C & \xrightarrow{\widetilde{f}\sigma} & Q \times D \\ \sigma \times \text{id}_C \downarrow & & \downarrow \sigma \times \text{id}_D \\ P \times C & \xrightarrow{\widetilde{f}} & P \times D. \end{array}$$

Using this for f , g and h and applying the interchange law and the various primitive coherence identities, we get the relation on associators after uncurrying, which suffices to establish the relation. The analogous argument applies to the unitors.

For the pentagon, assume given a fourth map $k : P \rightarrow \text{Map}(F, G)$, and consider the primitive pentagon identification for the four functors

$$P \times C \xrightarrow{\widetilde{f}} P \times D \xrightarrow{\widetilde{g}} P \times E \xrightarrow{\widetilde{h}} P \times F \xrightarrow{\widetilde{k}} P \times G.$$

For each of the five vertices of the resulting pentagon, we may use iterated identifications of the form $\widetilde{g} \circ \widetilde{f} = \widetilde{g \circ f}$ to put them into the form $\widetilde{u}$ for some u . By the construction of the associator, the substitution compatibility, and interchange, projecting the five edges of the resulting pentagon to the second component identifies them with the uncurryings of the five edges of the desired pentagon. The primitive pentagon identification therefore gives an identification between the uncurryings of the two composites around this pentagon. The mapping-anima axiom then provides the desired pentagon identification.

The triangle identification is obtained by similar means. □

We end the section by establishing various further properties about mapping animae.

Proposition 1.4.9. *For functors $f : A \rightarrow B$ and $g : B \rightarrow C$ and any category E , there is a canonical identification*

$$g_* \circ f_* = (g \circ f)_*$$

of functors $\text{Map}(E, A) \rightarrow \text{Map}(E, C)$.

Proof. After uncurrying, both sides evaluate a functor $E \rightarrow A$ and then compose it with f and g . The primitive associator gives the comparison between these uncurried functors, and we may lift it by Axiom B.3. □

Proposition 1.4.10. *For functors $f: A \rightarrow B$ and $g: B \rightarrow C$ and any category E , there is a canonical identification*

$$f^* \circ g^* = (g \circ f)^*$$

of functors $\text{Map}(C, E) \rightarrow \text{Map}(A, E)$.

Proof. This is proved in the same way, using the primitive associator after uncurrying. $\square$

Proposition 1.4.11. *For a functor $f: C \rightarrow D$, the following are equivalent:*

- (i) *The functor f is an equivalence.*
- (ii) *For every category E , the functor $f_*: \text{Map}(E, C) \rightarrow \text{Map}(E, D)$ is an equivalence.*
- (iii) *For every category E , the functor $f^*: \text{Map}(D, E) \rightarrow \text{Map}(C, E)$ is an equivalence.*

If C and D are animae, it suffices in (ii) or (iii) to let E range over animae.

Proof. We prove (i) $\Leftrightarrow$ (ii); the other case is dual. An inverse of f induces an inverse of every f_* . Conversely, pick an inverse of $f_*: \text{Map}(D, C) \rightarrow \text{Map}(D, D)$, and let $g: D \rightarrow C$ be the evaluation of this inverse at id_D . The unit of the equivalence and the beta comparison for postcomposition give an identification $f \circ g = \text{id}_D$. Now, since $f_*: \text{Map}(C, C) \rightarrow \text{Map}(C, D)$ is an equivalence, it induces an equivalence on animae of identifications. It follows that the chain of identifications

$$f \circ (g \circ f) = (f \circ g) \circ f = \text{id}_D \circ f = f = f \circ \text{id}_C$$

lifts to an identification $g \circ f = \text{id}_C$. Here we use Construction 1.4.1 and the beta comparison to identify the action of f_* on functors with postcomposition by f . If C and D are animae, both choices of E in this argument are animae, showing the final claim. $\square$

1.4.1 Collections of objects

We now define the *groupoid core* of a category, which we may think of as its collection of objects.

Definition 1.4.12. For any category C , we define the *groupoid core* of C to be

$$C^\simeq := \text{Map}(*, C).$$

It comes equipped with the canonical functor

$$\iota_C: C^\simeq \simeq C^\simeq \times * \xrightarrow{\text{ev}_{\text{Map}}} C.$$

The functor ι_C displays the objects of C with the anima $C^\simeq$ as parameter. It is universal among functors from animae in the following sense.

Corollary 1.4.13 (Universal property of the groupoid core). *For any anima X and any category D , postcomposition induces an equivalence*

$$(\iota_D)_*: \text{Map}(X, D^\simeq) \xrightarrow{\sim} \text{Map}(X, D).$$

Proof. By Proposition 1.4.2 and the equivalence $X \times * \simeq X$, there are equivalences

$$\text{Map}(X, D^\simeq) = \text{Map}(X, \text{Map}(*, D)) \simeq \text{Map}(X \times *, D) \simeq \text{Map}(X, D).$$

By the definition of ι_D , their composite is postcomposition with ι_D . $\square$

Remark 1.4.14. The anima $C^\simeq$ also describes all functors of the form $x: \Gamma \rightarrow C$ where Γ is an anima. By the previous result, it suffices to understand the universal such functor, i.e. $\iota_C: C^\simeq \rightarrow C$.

Proposition 1.4.15. *For every anima A , the canonical functor $\iota_A: A^\simeq \rightarrow A$ is an equivalence. Its inverse is obtained by currying the projection $A \times * \rightarrow A$.*

Proof. For every anima X , postcomposition with ι_A induces an equivalence by Corollary 1.4.13. Since both $A^\simeq$ and A are animae, Proposition 1.4.11 shows that ι_A is an equivalence. The defining identification for currying identifies its composite with the stated inverse with id_A . $\square$

Construction 1.4.16. A functor $f: C \rightarrow D$ induces a functor

$$f^\simeq := f_*: C^\simeq = \text{Map}(*, C) \longrightarrow \text{Map}(*, D) = D^\simeq.$$

The defining identification for postcomposition makes the square

$$\begin{array}{ccc} C^\simeq & \xrightarrow{f^\simeq} & D^\simeq \\ \iota_C \downarrow & & \downarrow \iota_D \\ C & \xrightarrow{f} & D \end{array}$$

commute. There are identifications $(\text{id}_C)^\simeq = \text{id}_{C^\simeq}$ and $g^\simeq \circ f^\simeq = (g \circ f)^\simeq$, obtained by uncurrying and using the unit and associativity identifications for functors; the latter is the case $E = *$ of Proposition 1.4.9. An identification $\alpha: f = f'$ similarly induces an identification $\alpha^\simeq: f^\simeq = (f')^\simeq$, by whiskering α with evaluation and then currying.

Lemma 1.4.17. *For categories C and D , the functor $(\text{pr}_C^\simeq, \text{pr}_D^\simeq): (C \times D)^\simeq \rightarrow C^\simeq \times D^\simeq$ is an equivalence.*

Proof. This is a special case of Proposition 1.4.5. $\square$

1.5 Initial categories and coproducts

Mapping animae allow us to express universal properties for functors out of a category. We use them to introduce the initial category and coproducts.

Axiom B.4 (Initial category). There is a category $\emptyset$, called the *initial category*, such that $\text{Map}(\emptyset, T)$ is contractible for every category T .

Thus there is a functor $\emptyset \rightarrow T$ for every T , and for any two such functors their anima of identifications is contractible. Indeed, their representing points lie in the contractible anima $\text{Map}(\emptyset, T)$, and Lemma 1.3.10 and Construction 1.4.1 give the assertion about animae of identifications.

We further ask the initial category to be *strictly* initial:

Axiom B.4' (Strictness of the initial category). Given a category C , any functor $C \rightarrow \emptyset$ is an equivalence.

Lemma 1.5.1. *For a category C , the projection map $C \times \emptyset \rightarrow \emptyset$ is an equivalence.*

Proof. This is immediate from Axiom B.4'. □

Axiom B.5 (Coproduct of categories). The *coproduct* of categories C and D is a category $C \sqcup D$ equipped with *inclusion functors* $i_C : C \rightarrow C \sqcup D$ and $i_D : D \rightarrow C \sqcup D$, such that for every category T the restriction functor

$$(i_C^*, i_D^*) : \text{Map}(C \sqcup D, T) \longrightarrow \text{Map}(C, T) \times \text{Map}(D, T)$$

is an equivalence. We also call $C \sqcup D$ the *disjoint union* of C and D .

Construction 1.5.2 (Copairing). For $f : C \rightarrow T$ and $g : D \rightarrow T$, applying an inverse to the displayed equivalence (i_C^*, i_D^*) to the pair (f, g) provides a functor

$$\langle f, g \rangle : C \sqcup D \rightarrow T$$

with identifications $\langle f, g \rangle i_C = f$ and $\langle f, g \rangle i_D = g$. For any $h, k : C \sqcup D \rightarrow T$, restriction also induces an equivalence

$$(h = k) \xrightarrow{\sim} (hi_C = ki_C) \times (hi_D = ki_D).$$

To see this, use Lemma 1.3.10 for the equivalence in the coproduct axiom, then the product rule and the comparison of Construction 1.4.1. In particular, a pair of identifications $\alpha : hi_C = ki_C$ and $\beta : hi_D = ki_D$ results in an identification $\langle \alpha, \beta \rangle : h = k$, with the prescribed restrictions to C and D .

In term notation, the axiom allows us to write expressions of the form

$$z : C \sqcup D \vdash \langle f, g \rangle(z) : T,$$

whenever given two expressions $x : C \vdash f(x) : T$ and $y : D \vdash g(y) : T$; the relations can be phrased as $\langle f, g \rangle(i_C(x)) = f(x)$ and $\langle f, g \rangle(i_D(y)) = g(y)$.

Once we have distributivity available (see Section 1.6.3) we can also use term notation for expressions with more additional parameters.

Exercise 1.5.3 (Exercise 1.9.2 (outside selection)). Show that the coproduct of categories is associative, commutative and unital:

$$(C \sqcup D) \sqcup E \xrightarrow{\sim} C \sqcup (D \sqcup E), \quad C \sqcup D \xrightarrow{\sim} D \sqcup C, \quad \emptyset \sqcup C \xrightarrow{\sim} C \xrightarrow{\sim} C \sqcup \emptyset.$$

Construction 1.5.4. The formation of coproducts of categories is functorial: given two functors $f: C \rightarrow C'$ and $g: D \rightarrow D'$, we obtain a new functor

$$f \sqcup g := \langle i_{C'} \circ f, i_{D'} \circ g \rangle: C \sqcup D \rightarrow C' \sqcup D'.$$

We leave it to the reader to verify that there are identifications

$$\text{id}_C \sqcup \text{id}_D = \text{id}_{C \sqcup D} \quad \text{and} \quad (f' \sqcup g') \circ (f \sqcup g) = (f' \circ f) \sqcup (g' \circ g).$$

1.6 Pullbacks of categories

We will now axiomatize the *pullback* of two functors $f: C \rightarrow E$ and $g: D \rightarrow E$ of categories. This will be similar to the axiom for the product $C \times D$, but it is more involved due to the fact that we need to record compatibilities with the structure maps to E .

Definition 1.6.1. Consider two functors $f: C \rightarrow E$ and $g: D \rightarrow E$. Given a category T , we define a *cone diagram on T* to be a triple $c = (t_C, t_D, \tau)$ consisting of functors $t_C: T \rightarrow C$ and $t_D: T \rightarrow D$ together with an identification $\tau: f \circ t_C = g \circ t_D$. In other words, a cone diagram consists of a commutative square of the form

$$\begin{array}{ccc} T & \xrightarrow{t_C} & C \\ t_D \downarrow & & \downarrow f \\ D & \xrightarrow{g} & E. \end{array}$$

We call T the *cone point* of the diagram. Given another cone diagram $c' = (t'_C, t'_D, \tau')$, an *identification of cone diagrams* $c = c'$ is a triple $\Phi: (\varphi_C, \varphi_D, H)$, where $\varphi_C: t_C = t'_C$ and $\varphi_D: t_D = t'_D$ are identifications, and H is an identification

$$\tau' \circ (f \star \varphi_C) = (g \star \varphi_D) \circ \tau.$$

In particular, a comparison of cones needs to be compatible with the identifications forming the commutative square. In turn, if $\Psi: (\psi_C, \psi_D, G)$ is another identification of cone diagrams $c = c'$, we say that Φ and Ψ are identified if there are identifications $\varphi_C = \psi_C$ and $\varphi_D = \psi_D$ under which H becomes identified with G .

Given a functor $s: S \rightarrow T$ and a cone diagram $c = (t_C, t_D, \tau)$ on T we obtain another cone diagram $s^*(c) := (t_C \circ s, t_D \circ s, \tau \circ \text{id}_s)$ on S . Similarly, every identification $s_1 = s_2$ of functors $S \rightarrow T$ induces an identification $s_1^*(c) = s_2^*(c)$ of cone diagrams.

Axiom B.6 (Pullback of categories). For functors $f: C \rightarrow E$ and $g: D \rightarrow E$, there is a category $C \times_E D$, called their *pullback* or *fiber product*, with projections $\text{pr}_1: C \times_E D \rightarrow C$ and $\text{pr}_2: C \times_E D \rightarrow D$, and an identification $\tau_{\text{univ}}: f \text{pr}_1 = g \text{pr}_2$. Given $a: T \rightarrow C$, $b: T \rightarrow D$ and $\tau: fa = gb$, there is a functor $(a, b)_\tau: T \rightarrow C \times_E D$ equipped with projection comparisons $c: \text{pr}_1(a, b)_\tau = a$ and $d: \text{pr}_2(a, b)_\tau = b$ and an identification

$$\tau \circ (f \star c) = (g \star d) \circ (\tau_{\text{univ}} \star (a, b)_\tau).$$

If C, D, E are animae, then $C \times_E D$ is an anima.

For any $h, k : T \rightarrow C \times_E D$, the projection-induced functor

$$(h = k) \longrightarrow (\text{pr}_1 h = \text{pr}_1 k) \times_{(f \text{pr}_1 h = g \text{pr}_2 k)} (\text{pr}_2 h = \text{pr}_2 k)$$

is an equivalence; here the two functors defining the pullback on the right are

$$\alpha \longmapsto (\tau_{\text{univ}} \star k) \circ (f \star \alpha), \quad \beta \longmapsto (g \star \beta) \circ (\tau_{\text{univ}} \star h).$$

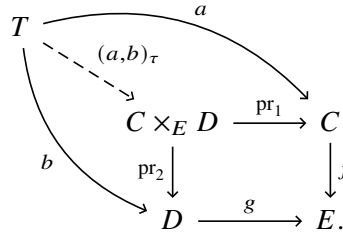
Interchange with τ_{univ} identifies their composites with the two projections from $(h = k)$ and thereby defines the displayed functor.

The final clause uses the pullback construction, now applied to three animae of identifications. In term notation the factorization is written

$$x : T \vdash (a(x), b(x))_{\tau(x)} : C \times_E D.$$

The subscript may be omitted when it is clear from context, or left implicit to simplify notation. An identification between two such expressions is written $(\alpha(x), \beta(x))_{\eta(x)}$, where η is the identification between the two composites in the middle anima of identifications of the axiom.

The resulting map may be displayed as follows:



Definition 1.6.2 (Pullback square). A commutative square

$$\begin{array}{ccc} P & \xrightarrow{p} & C \\ q \downarrow & & \downarrow f \\ D & \xrightarrow{g} & E \end{array}$$

with matching identification $\sigma : fp = gq$ is a *pullback square*, or is *cartesian*, if its induced functor $(p, q)_\sigma : P \rightarrow C \times_E D$ is an equivalence. We indicate this by putting the symbol $\sqsubset$ in the square.

A pullback square has the same universal property as the chosen pullback: every cone factors through it with an identification of cone diagrams, and its animae of identifications are given by the displayed pullbacks of animae of identifications. Indeed, transfer the two clauses of the axiom through the equivalence $P \rightarrow C \times_E D$, using Lemma 1.3.10. Conversely, these clauses construct inverse functors between P and the chosen pullback.

Note that any functor between two pullbacks of the same cospan that respects the projections and the specified identifications making the squares is an equivalence. Indeed, the induced functors to the chosen pullback are equivalences and form a commutative triangle with this functor, so the claim follows by two-out-of-three.

Proposition 1.6.3. *A commutative square of categories is a pullback square if and only if applying $\text{Map}(T, -)$ gives a pullback square for every category T .*

Proof. We first show that applying $\text{Map}(T, -)$ to the chosen pullback $C \times_E D$ results in a pullback square. Postcomposition with the two projections and whiskering with τ_{univ} gives a comparison functor

$$\text{Map}(T, C \times_E D) \longrightarrow \text{Map}(T, C) \times_{\text{Map}(T, E)} \text{Map}(T, D).$$

For an inverse, let P be the anima on the right. Uncurrying the projections to $\text{Map}(T, C)$ and $\text{Map}(T, D)$ gives maps from $P \times T$ to C and D . Moreover, the identification in the commutative square for $C \times_E D$ results in an identification of two functors $P \times T \rightarrow E$. The pullback axiom thus supplies a functor from $P \times T$ to $C \times_E D$, with the projection comparisons and their compatibility with the squares. Currying gives a candidate inverse functor $P \rightarrow \text{Map}(T, C \times_E D)$.

To compare the composite on the right to the identity, we first do so on its two projections, where it follows by currying the identifications we just constructed. To show that the resulting two commutative squares are identified, we may again do so after uncurrying, where it follows from unwinding definitions.

For the composite on the left, we need to show its uncurrying agrees with the evaluation map. This can in turn be tested after both projections and comparing the commutativity identifications.

The claim now also follows for an arbitrary pullback square by considering the equivalence from its cone point to $C \times_E D$. $\square$

Lemma 1.6.4. *For functors $C \rightarrow E$ and $D \rightarrow E$, the functor $(\text{pr}_C^\sim, \text{pr}_D^\sim): (C \times_E D)^\sim \rightarrow C^\sim \times_{E^\sim} D^\sim$ is an equivalence.*

Proof. This is a special case of Proposition 1.6.3. $\square$

Exercise 1.6.5 (Exercise 1.9.4 (outside selection)). Show that the fiber product is commutative, associative and unital: for functors $C \rightarrow \Gamma$, $D \rightarrow \Gamma$ and $B \rightarrow \Gamma$, there are preferred equivalences

$$C \times_\Gamma D \xrightarrow{\sim} D \times_\Gamma C, \quad B \times_\Gamma (C \times_\Gamma D) \xrightarrow{\sim} (B \times_\Gamma C) \times_\Gamma D, \quad C \times_\Gamma \Gamma \xrightarrow{\sim} C.$$

Exercise 1.6.6 (Exercise 1.9.5 (outside selection)). Consider categories C and D , and let $p_C: C \rightarrow *$ and $p_D: D \rightarrow *$ be their functors to the terminal category. Show that the functor $(\text{pr}_1, \text{pr}_2): C \times_* D \rightarrow C \times D$ is an equivalence.

Construction 1.6.7. The construction of pullbacks of categories is functorial: given a commutative diagram

$$\begin{array}{ccccc} C & \xrightarrow{f} & \Gamma & \xleftarrow{g} & D \\ \varphi \downarrow & & \downarrow \chi & & \downarrow \psi \\ C' & \xrightarrow{f'} & \Gamma' & \xleftarrow{g'} & D' \end{array}$$

there is an induced functor

$$\varphi \times_\chi \psi := (\varphi \circ \text{pr}_1, \psi \circ \text{pr}_2): C \times_\Gamma D \rightarrow C' \times_{\Gamma'} D'.$$

Exercise 1.6.8. In the above situation, show that if each of the functors φ , ψ and χ is an equivalence, then so is $\varphi \times_\chi \psi$.

Corollary 1.6.9. *Equivalences are closed under pullback: if $g: D \rightarrow E$ is an equivalence and $f: C \rightarrow E$ is an arbitrary functor, then also $C \times_E D \rightarrow C$ is an equivalence.*

Proof. This is a special case of Exercise 1.6.8 by taking $C' = C$, $\Gamma = \Gamma' = E$, $D' = E$, $\varphi = \text{id}_C$, $\chi = \text{id}_E$ and $\psi = g$: this exhibits the projection $C \times_E D \rightarrow C$ as the composite of the equivalence $\varphi \times_\chi \psi: C \times_E D \rightarrow C \times_E E$ with the unitality equivalence $C \times_E E \xrightarrow{\sim} C$ from Exercise 1.6.5. $\square$

1.6.1 Fibers of functors

Pullbacks of categories lead to the notion of a *fiber* of a functor:

Definition 1.6.10 (Fiber). Let $f: C \rightarrow D$ be a functor and let $x: * \rightarrow D$ be an absolute object. We define the *fiber* of f over x as the pullback

$$\begin{array}{ccc} C_x & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow f \\ * & \xrightarrow{x} & D. \end{array}$$

One particular case is the fiber of a map $b: * \rightarrow C$ over some object $a: * \rightarrow C$. The absolute terms of this fiber can be identified with identifications $\alpha: a = b$. We impose as an axiom that the same holds in families:

Axiom B.7 (Representability of identifications). For every category C and absolute objects $a, b \in C$, there is an identification

$$\varepsilon: \text{const}_a = \text{const}_b$$

of functors $(a = b) \rightarrow C$ such that the square

$$\begin{array}{ccc} (a = b) & \xrightarrow{p_{(a=b)}} & * \\ p_{(a=b)} \downarrow & \lrcorner & \downarrow b \\ * & \xrightarrow{a} & C \end{array}$$

is a pullback.

For a functor $u: T \rightarrow (a = b)$, the corresponding identification between the constant functors $T \rightarrow C$ is obtained by prewhiskering ε with u , using the terminal-category comparisons to identify its source and target.

1.6.2 Properties of pullback squares

Lemma 1.6.11 (Pasting lemma for pullback squares). *Consider a commutative diagram*

$$\begin{array}{ccccc} C_1 & \xrightarrow{g_1} & C_2 & \xrightarrow{g_2} & C_3 \\ f_1 \downarrow & & \downarrow f_2 & & \downarrow f_3 \\ D_1 & \xrightarrow{h_1} & D_2 & \xrightarrow{h_2} & D_3. \end{array}$$

- (1) *There is a preferred equivalence $D_1 \times_{D_3} C_3 \xrightarrow{\sim} D_1 \times_{D_2} (D_2 \times_{D_3} C_3)$.*
- (2) *If the right-hand square is a pullback square, then the left-hand square is a pullback square if and only if the outer rectangle is a pullback square.*

Proof. For part (1), we define the functor as $(\text{pr}_1, (h_1 \circ \text{pr}_1, \text{pr}_2))$. We claim that an inverse is given by the functor

$$(\text{pr}_1, \text{pr}_2 \circ \text{pr}_2): D_1 \times_{D_2} (D_2 \times_{D_3} C_3) \rightarrow D_1 \times_{D_3} C_3.$$

To show that the two composites are identified with the respective identities, it suffices to construct these identifications after projecting to the two components and verifying that these identifications agree after composing with the functors to D_3 . But after unwinding the definitions, this holds tautologically by the very construction of the two functors in both directions.

Part (2) is an immediate consequence of the 2-out-of-3 property from Lemma 1.3.6 applied to the following commutative square:

$$\begin{array}{ccc} C_1 & \xrightarrow{\quad} & D_1 \times_{D_2} C_2 \\ \downarrow & & \downarrow \cong \\ D_1 \times_{D_3} C_3 & \xrightarrow{\cong} & D_1 \times_{D_2} (D_2 \times_{D_3} C_3), \end{array}$$

where the right vertical map is an equivalence by applying Exercise 1.6.8 to the equivalence $C_2 \xrightarrow{\sim} D_2 \times_{D_3} C_3$. $\square$

Lemma 1.6.12. *Consider a commutative square*

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ g \downarrow & & \downarrow h \\ E & \xrightarrow{k} & F \end{array}$$

and assume that h is an equivalence. Then g is an equivalence if and only if the square is a pullback square.

Proof. Since h is an equivalence, also the projection map $\text{pr}_2: D \times_F E \rightarrow E$ is an equivalence by Corollary 1.6.9. Since the composite of the map $(f, g): C \rightarrow D \times_F E$ with pr_2 is identified with g , the claim follows from the 2-out-of-3 property of equivalences, Lemma 1.3.6. $\square$

Lemma 1.6.13. Consider functors $f: C \rightarrow E$ and $g: D \rightarrow E$. Then the commutative square

$$\begin{array}{ccc} C \times_E D & \longrightarrow & E \\ (\text{pr}_1, \text{pr}_2) \downarrow & & \downarrow \Delta \\ C \times D & \xrightarrow{f \times g} & E \times E \end{array}$$

is a pullback square.

Proof. This follows immediately from the pasting lemma for pullback squares, Lemma 1.6.11, applied to the following commutative diagram:

$$\begin{array}{ccccc} C \times_E D & \xrightarrow{\text{pr}_2} & D & \xrightarrow{g} & E \\ (\text{pr}_1, \text{pr}_2) \downarrow & & \downarrow (g, \text{id}_D) & & \downarrow (\text{id}_E, \text{id}_E) \\ C \times D & \xrightarrow{f \times \text{id}_D} & E \times D & \xrightarrow{\text{id}_E \times g} & E \times E \\ \text{pr}_1 \downarrow & & \downarrow \text{pr}_1 & & \\ C & \xrightarrow{f} & E. & & \end{array} \quad \square$$

1.6.3 Universality of coproducts

The strictness of the initial category immediately implies its universality:

Lemma 1.6.14. For every functor $g: D \rightarrow E$, the commutative square

$$\begin{array}{ccc} \emptyset & \longrightarrow & \emptyset \\ \downarrow & & \downarrow \\ D & \xrightarrow{g} & E \end{array}$$

is a pullback square.

Proof. We need to show that the induced map $\emptyset \rightarrow \emptyset \times_E D$ is an equivalence. But this follows directly from Axiom B.4' by applying 2-out-of-3 to the maps $\emptyset \rightarrow \emptyset \times_E D \xrightarrow{\text{pr}_1} \emptyset$. $\square$

We now postulate a analogous universality property on coproducts of categories:

Axiom B.5' (Universality of coproducts). Consider categories Γ_0 and Γ_1 , define $\Gamma := \Gamma_0 \sqcup \Gamma_1$, and let $i_0: \Gamma_0 \rightarrow \Gamma$ and $i_1: \Gamma_1 \rightarrow \Gamma$ denote the two inclusions.

(1) For functors $f: C \rightarrow \Gamma_0$ and $g: D \rightarrow \Gamma_1$, the commutative squares

$$\begin{array}{ccc} C & \xrightarrow{i_C} & C \sqcup D \\ f \downarrow & & \downarrow f \sqcup g \\ \Gamma_0 & \xrightarrow{i_0} & \Gamma_0 \sqcup \Gamma_1 \end{array} \quad \text{and} \quad \begin{array}{ccc} D & \xrightarrow{i_D} & C \sqcup D \\ g \downarrow & & \downarrow f \sqcup g \\ \Gamma_1 & \xrightarrow{i_1} & \Gamma_0 \sqcup \Gamma_1 \end{array}$$

are pullback squares.

(2) For a functor $h: E \rightarrow \Gamma$, the functor

$$\langle \text{pr}_1, \text{pr}_1 \rangle: (E \times_{\Gamma} \Gamma_0) \sqcup (E \times_{\Gamma} \Gamma_1) \rightarrow E$$

is an equivalence.

Corollary 1.6.15 (Disjointness of coproducts). *Coproducts are disjoint: for categories C and D , the commutative square*

$$\begin{array}{ccc} \emptyset & \longrightarrow & C \\ \downarrow & & \downarrow i_C \\ D & \xrightarrow{i_D} & C \sqcup D \end{array}$$

is a pullback square. □

Lemma 1.6.16. *For functors $f: C \rightarrow \Gamma$, $g: D \rightarrow \Gamma$ and $\varphi: \Gamma' \rightarrow \Gamma$, the commutative square*

$$\begin{array}{ccc} (C \times_{\Gamma} \Gamma') \sqcup (D \times_{\Gamma} \Gamma') & \longrightarrow & C \sqcup D \\ \downarrow & & \downarrow \langle f, g \rangle \\ \Gamma' & \xrightarrow{\varphi} & \Gamma \end{array}$$

is a pullback square.

Proof. We need to show that the induced functor

$$(C \times_{\Gamma} \Gamma') \sqcup (D \times_{\Gamma} \Gamma') \rightarrow (C \sqcup D) \times_{\Gamma} \Gamma'$$

is an equivalence. But this is a special case of Axiom B.5' applied to the functor $f = \text{pr}_1: (C \sqcup D) \times_{\Gamma} \Gamma' \rightarrow C \sqcup D$: by the pasting law of pullback squares there are equivalences

$$C \times_{C \sqcup D} (C \sqcup D) \times_{\Gamma} \Gamma' \simeq C \times_{\Gamma} \Gamma' \quad \text{and} \quad D \times_{C \sqcup D} (C \sqcup D) \times_{\Gamma} \Gamma' \simeq D \times_{\Gamma} \Gamma'.$$

This finishes the proof. □

Corollary 1.6.17 (Distributivity). *For categories C , D and E , the functor*

$$\langle E \times i_C, E \times i_D \rangle: E \times C \sqcup E \times D \rightarrow E \times (C \sqcup D)$$

is an equivalence.

Proof. This is an instance of the previous lemma by taking $\Gamma = *$, $\Gamma' = E$, $f = p_C: C \rightarrow *$, $g = p_D: D \rightarrow *$, and $\varphi = p_E: E \rightarrow *$. □

Consequently, expressions with an additional parameter $t: T$ may be defined by cases: combine the functors $T \times C \rightarrow E$ and $T \times D \rightarrow E$ by copairing, then precompose with the inverse of distributivity. Their restriction comparisons are the comparisons of the resulting expression on the two summands.

Lemma 1.6.18. *The following two conditions are equivalent:*

(1) *The category $* \sqcup *$ is contractible;*

(2) Every category is contractible.

Proof. It is clear that (2) implies (1). Conversely, assume that $* \sqcup *$ is contractible, i.e. that the map $* \sqcup * \rightarrow *$ is an equivalence. Since this map admits a section given by the left inclusion $l: * = * \sqcup \emptyset \hookrightarrow * \sqcup *$, it follows that l is an equivalence. In particular, its fiber over the right point must be contractible. Since this fiber is the empty category $\emptyset$ by Corollary 1.6.15, we see that the map $\emptyset \rightarrow *$ is an equivalence, and it follows from taking the product with any category C that also $\emptyset \rightarrow C$ is an equivalence. By 2-out-of-3 this implies that C is contractible. $\square$

1.6.4 Embeddings of categories

We introduce the notion of *embeddings* of categories and prove basic properties about them. We will eventually see that embeddings precisely correspond to *subcategories*, see Proposition 6.4.1 (outside selection).

Definition 1.6.19 (Embedding). A functor $f: C \rightarrow D$ is called an *embedding* if the diagonal $\Delta_f := (\text{id}_C, \text{id}_C): C \rightarrow C \times_D C$ is an equivalence, or equivalently if the commutative diagram

$$\begin{array}{ccc} C & \xlongequal{\quad} & C \\ \parallel & & \downarrow f \\ C & \xrightarrow{f} & D \end{array}$$

is a pullback square.

Proposition 1.6.20. A functor $f: C \rightarrow D$ is an embedding if and only if for every category E the induced map $f_*: \text{Map}(E, C) \rightarrow \text{Map}(E, D)$ is an embedding.

Proof. This follows immediately from Proposition 1.6.3 by considering the commutative diagram

$$\begin{array}{ccc} C & \xlongequal{\quad} & C \\ \parallel & & \downarrow f \\ C & \xrightarrow{f} & D. \end{array}$$

$\square$

Corollary 1.6.21. Given an embedding $f: C \rightarrow D$, the functor $f^\simeq: C^\simeq \rightarrow D^\simeq$ is also an embedding.

Proof. In light of the definition of embeddings, this is immediate from Lemma 1.6.4. $\square$

Lemma 1.6.22. A functor $f: C \rightarrow D$ is an embedding if and only if the commutative square

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ (\text{id}_C, \text{id}_C) \downarrow & & \downarrow (\text{id}_D, \text{id}_D) \\ C \times C & \xrightarrow{f \times f} & D \times D \end{array}$$

is a pullback square.

Proof. By Lemma 1.6.13 the pullback of this diagram is $C \times_D C$. Since the induced map $C \rightarrow C \times_D C$ is precisely the diagonal $\Delta_f: C \rightarrow C \times_D C$ of f , the claim follows. $\square$

Lemma 1.6.23. *Consider a pullback square*

$$\begin{array}{ccc} C' & \xrightarrow{g} & C \\ u \downarrow & & \downarrow v \\ D' & \xrightarrow{f} & D \end{array}$$

of categories. If v is an embedding, then also u is an embedding.

Proof. This is left as an exercise for the reader, see Exercise 1.9.10 (outside selection) $\square$

Lemma 1.6.24. *Let $f: C \rightarrow D$ be an embedding which admits a section $s: D \rightarrow C$. Then f is an equivalence.*

Proof. As $fs = \text{id}_D$, it remains to show that $sf = \text{id}_C$. Consider the following commutative diagram:

$$\begin{array}{ccccc} C & \xrightarrow{\quad h \quad} & C & \xlongequal{\quad} & C \\ f \downarrow & & \parallel & \lrcorner & \downarrow f \\ D & \xrightarrow{s} & C & \xrightarrow{f} & D. \end{array}$$

Since the right square is a pullback square by assumption on f , there exists a map $h: C \rightarrow C$ making the diagram commute. It follows that $sf = h = \text{id}_C$, finishing the proof. $\square$

We will now prove some more intricate properties about embeddings, which may be skipped on first reading and referred back to whenever needed.

Lemma 1.6.25. *Let $f: C \rightarrow D$ and $g: D \rightarrow E$ be functors and assume that g is an embedding. Then the square*

$$\begin{array}{ccc} C & \xlongequal{\quad} & C \\ f \downarrow & & \downarrow gf \\ D & \xrightarrow{g} & E \end{array}$$

is a pullback square.

Proof. This is immediate from the pasting law of pullback squares applied to the following commutative diagram:

$$\begin{array}{ccc} C & \xlongequal{\quad} & C \\ f \downarrow & & \downarrow f \\ D & \xlongequal{\quad} & D \\ \parallel & & \downarrow g \\ D & \xrightarrow{g} & E. \end{array}$$

Indeed, the bottom square is a pullback square by assumption on g , while the top square is clearly a pullback square. $\square$

Lemma 1.6.26. *Consider a commutative triangle*

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ & \searrow gf & \swarrow g \\ & E & \end{array}$$

of functors. If g is an embedding, then f is an embedding if and only if gf is an embedding.

Proof. Consider the following commutative diagram:

$$\begin{array}{ccccc} C & \xlongequal{\quad} & C & \xlongequal{\quad} & C \\ \parallel & & \downarrow f & & \downarrow gf \\ C & \xrightarrow{f} & D & \xrightarrow{g} & E. \end{array}$$

Since g is an embedding, the right square is a pullback square by Lemma 1.6.25. It follows from the pasting law of pullback squares, Lemma 1.6.11, that the left square is cartesian if and only if the outer rectangle is cartesian, proving the claim. $\square$

Corollary 1.6.27. *Consider a commutative diagram*

$$\begin{array}{ccc} A & \xrightarrow{u} & C \\ f \downarrow & & \downarrow g \\ B & \xrightarrow{v} & D \end{array}$$

of functors such that v is an embedding. Then u is an embedding if and only if the induced map $(f, u): A \rightarrow B \times_D C$ is an embedding.

Proof. The map $u: A \rightarrow C$ factors as the composite $A \xrightarrow{(f, u)} B \times_D C \xrightarrow{\text{pr}_2} C$. The map pr_2 is an embedding by Lemma 1.6.23, and hence the statement follows from Lemma 1.6.26. $\square$

Lemma 1.6.28. *Consider a commutative diagram*

$$\begin{array}{ccc} C' & \xrightarrow{u} & C \\ f' \downarrow & & \downarrow f \\ D' & \xrightarrow{v} & D \\ g' \downarrow & & \downarrow g \\ E' & \xrightarrow{w} & E. \end{array}$$

If v and w are embeddings and the composite square is a pullback square, then also the top square is a pullback square.

Proof. Consider the following commutative diagram:

$$\begin{array}{ccccc} C' & \xlongequal{\quad} & C' & \xrightarrow{u} & C \\ f' \downarrow & & \downarrow (g'f', fu) & & \downarrow f \\ D' & \xrightarrow{(g', v)} & E' \times_E D & \longrightarrow & D \\ & & \downarrow & & \downarrow \\ & & E' & \xrightarrow{w} & E. \end{array}$$

Since w is an embedding, the map $E' \times_E D \rightarrow D$ is an embedding by Lemma 1.6.23. Since $v: D' \rightarrow D$ is also an embedding, it thus follows from Lemma 1.6.26 that the map $(g', v): D' \rightarrow E' \times_E D$ is an embedding, and thus by Lemma 1.6.25 the top left square is a pullback square. Since the right bottom square and right outer rectangle are pullback squares, so is the top right square by the pasting law Lemma 1.6.11. Another instance of Lemma 1.6.11 shows that the outer rectangle is a pullback square, as desired. $\square$

1.7 Functor categories

Given two categories C and D , we are interested in forming the *functor category* $\text{Fun}(C, D)$ whose objects are precisely the functors from C to D .

Given any functor $e: F \times C \rightarrow D$ and any category T , we can form a functor

$$u_T: \text{Map}(T, F) \rightarrow \text{Map}(T \times C, D)$$

by currying the composite

$$\text{Map}(T, F) \times T \times C \xrightarrow{\text{ev}_{\text{Map}} \times \text{id}_C} F \times C \xrightarrow{e} D.$$

It satisfies the property that it evaluations on an absolute object of $\text{Map}(T, F)$, corresponding to a functor $g: T \rightarrow F$, as

$$u_T(g) = e \circ (g \times \text{id}_C).$$

Definition 1.7.1. A *functor category* from C to D is a category F equipped with a functor $e: F \times C \rightarrow D$ such that, for any category T , the induced map

$$u_T: \text{Map}(T, F) \xrightarrow{\sim} \text{Map}(T \times C, D)$$

is an equivalence. We then call F the category of functors from C to D and e the *evaluation functor*.

Proposition 1.7.2. If F and F' both are categories of functors from C to D with evaluation functors e and e' respectively, then there is an equivalence $\varphi: F' \xrightarrow{\sim} F$ as well as $\sigma: e \circ (\varphi \times \text{id}_C) = e'$.

Proof. Let v be an inverse of the equivalence $u_{F'}: \text{Map}(F', F) \xrightarrow{\sim} \text{Map}(F' \times C, D)$ and define $\varphi := v(e'): F' \rightarrow F$. By Construction 1.4.1, the identification $u_{F'} \circ v = \text{id}$ and the beta comparison for $u_{F'}$ provide

$$\sigma: e \circ (\varphi \times \text{id}_C) = e'.$$

Exchanging the roles of (F, e) and (F', e') , we similarly obtain a functor $\psi: F \rightarrow F'$ together with $e' \circ (\psi \times \text{id}_C) = e$. Combining the two, the uncurried functors associated to $\varphi \circ \psi$ and id_F are identified:

$$e \circ ((\varphi \circ \psi) \times \text{id}_C) = e' \circ (\psi \times \text{id}_C) = e = e \circ (\text{id}_F \times \text{id}_C).$$

Since u_F is an equivalence, it reflects this identification and gives $\varphi \circ \psi = \text{id}_F$. The same argument shows $\psi \circ \varphi = \text{id}_{F'}$, so φ is an equivalence with inverse ψ . $\square$

Axiom B.8 (Functor category). For any two categories C and D , there is a category $\text{Fun}(C, D)$ equipped with an *evaluation functor* $\text{ev}: \text{Fun}(C, D) \times C \rightarrow D$ that exhibits it as a functor category from C to D .

Definition 1.7.3. Given another category T , choose an inverse v_T of the equivalence u_T . For a functor $f: T \times C \rightarrow D$, define its *currying* by applying v_T :

$$f_c := v_T(f): T \longrightarrow \text{Fun}(C, D).$$

It comes equipped with a commutative diagram as follows:

$$\begin{array}{ccc} T \times C & \xrightarrow{f_c \times \text{id}_C} & \text{Fun}(C, D) \times C \\ & \searrow f & \downarrow \text{ev} \\ & & D. \end{array}$$

Conversely, given a functor $g: T \rightarrow \text{Fun}(C, D)$, we call the composite

$$T \times C \xrightarrow{g \times \text{id}_C} \text{Fun}(C, D) \times C \xrightarrow{\text{ev}} D$$

the *uncurrying* of g and denote it by g^u . The discussion before shows it is identified with $u_T(g)$. Similarly, an identification $\beta: g = h$ induces an identification $\beta^u: g^u = h^u$. At the level of mapping animae, uncurrying is the equivalence

$$(-)^u: \text{Map}(T, \text{Fun}(C, D)) \xrightarrow{\sim} \text{Map}(T \times C, D)$$

with inverse defined by currying.

For parallel functors $g, h: T \rightarrow \text{Fun}(C, D)$, the induced functor

$$(g = h) \longrightarrow (g^u = h^u)$$

is an equivalence. Indeed, represent g, h by points of $\text{Map}(T, \text{Fun}(C, D))$, apply Lemma 1.3.10 to u_T , and use Construction 1.4.1 at both ends.

Notation 1.7.4. In term notation, an expression $x: T, y: C \vdash f(x, y): D$ may therefore be curried to $x: T \vdash (y \mapsto f(x, y)): \text{Fun}(C, D)$. Conversely, evaluation of $x: T \vdash F(x): \text{Fun}(C, D)$ and $x: T \vdash c(x): C$ is formed by pairing, giving $x: T \vdash F(x)(c(x)): D$. These conventions also apply to identifications of the whole functors, with the beta and eta comparisons above.

Remark 1.7.5. By considering the terminal category, we observe that absolute objects $* \rightarrow \text{Fun}(C, D)$ correspond to functors $C \simeq * \times C \rightarrow D$, justifying the notation and terminology of $\text{Fun}(C, D)$.

Construction 1.7.6 (Evaluation functor). Given categories C and D , and $x: * \rightarrow C$, the *evaluation functor at x* is defined as the composite

$$\text{ev}_x: \text{Fun}(C, D) \simeq \text{Fun}(C, D) \times * \xrightarrow{\text{id} \times x} \text{Fun}(C, D) \times C \xrightarrow{\text{ev}} D.$$

The functoriality of the product naturally provides functoriality for the functor categories:

Construction 1.7.7 (Functoriality of functor categories). Given a functor $g: D \rightarrow D'$, we define the functor $g \circ -: \text{Fun}(C, D) \rightarrow \text{Fun}(C, D')$ as the currying of the composite

$$\text{Fun}(C, D) \times C \xrightarrow{\text{ev}} D \xrightarrow{g} D'.$$

By assumption we obtain a commutative square as follows:

$$\begin{array}{ccc} \text{Fun}(C, D) \times C & \xrightarrow{\text{ev}} & D \\ (g \circ -) \times \text{id}_C \downarrow & & \downarrow g \\ \text{Fun}(C, D') \times C & \xrightarrow{\text{ev}} & D'. \end{array}$$

Similarly, given a functor $f: C' \rightarrow C$, we define the functor $- \circ f: \text{Fun}(C, D) \rightarrow \text{Fun}(C', D)$ as the currying of the composite

$$\text{Fun}(C, D) \times C' \xrightarrow{\text{id} \times f} \text{Fun}(C, D) \times C \xrightarrow{\text{ev}} D.$$

It satisfies the property that it makes the following square commute:

$$\begin{array}{ccc} \text{Fun}(C, D) \times C' & \xrightarrow{\text{id} \times f} & \text{Fun}(C, D) \times C \\ (- \circ f) \times \text{id}_{C'} \downarrow & & \downarrow \text{ev} \\ \text{Fun}(C', D) \times C' & \xrightarrow{\text{ev}} & D. \end{array}$$

In a completely analogous way, every identification $\beta: g = g'$ of functors $D \rightarrow D'$ induces an identification $(\beta \circ -): (g \circ -) = (g' \circ -)$ of functors $\text{Fun}(C, D) \rightarrow \text{Fun}(C, D')$, and similarly for the construction $- \circ f$.

Exercise 1.7.8 (Exercise 1.9.6 (outside selection)). Formulate and prove that the assignments $g \mapsto (g \circ -)$ and $f \mapsto (- \circ f)$ are functorial, in the sense that they respect identity functors and composition of functors.

1.7.1 Properties of functor categories

The universal property of functor categories allows us to deduce their compatibilities with the terminal category, the initial category, products, coproducts and pullbacks from the corresponding properties of mapping animae. We will use Proposition 1.4.11 to test the comparison functors after applying $\text{Map}(T, -)$.

Lemma 1.7.9. *For every category C the functor category $\text{Fun}(C, *)$ is contractible.*

Proof. For every category T , the universal property of the functor category gives

$$\text{Map}(T, \text{Fun}(C, *)) \simeq \text{Map}(T \times C, *) \simeq *,$$

where the last equivalence is Lemma 1.4.4. Thus $\text{Fun}(C, *) \rightarrow *$ induces an equivalence on mapping animae for every T , and is an equivalence by Proposition 1.4.11. $\square$

Lemma 1.7.10. *Let C , D and E be categories. Then the functor*

$$(\text{pr}_1 \circ -, \text{pr}_2 \circ -): \text{Fun}(C, D \times E) \rightarrow \text{Fun}(C, D) \times \text{Fun}(C, E)$$

is an equivalence.

Proof. For every category T , the universal property of functor categories and Proposition 1.4.5 give equivalences

$$\begin{aligned} \text{Map}(T, \text{Fun}(C, D \times E)) &\simeq \text{Map}(T \times C, D \times E) \\ &\simeq \text{Map}(T \times C, D) \times \text{Map}(T \times C, E) \\ &\simeq \text{Map}(T, \text{Fun}(C, D)) \times \text{Map}(T, \text{Fun}(C, E)) \\ &\simeq \text{Map}(T, \text{Fun}(C, D) \times \text{Fun}(C, E)). \end{aligned}$$

This composite is induced by the functor in the statement: after uncurrying, its two components are postcomposition with pr_1 and pr_2 , by Lemma 1.7.7. The result follows from Proposition 1.4.11. $\square$

Lemma 1.7.11. *Let T be a category and let $f: C \rightarrow E$ and $g: D \rightarrow E$ be functors. Then the functor*

$$(\text{pr}_1 \circ -, \text{pr}_2 \circ -): \text{Fun}(T, C \times_E D) \rightarrow \text{Fun}(T, C) \times_{\text{Fun}(T, E)} \text{Fun}(T, D)$$

induced by the identification $(f \circ -) \circ (\text{pr}_1 \circ -) = (g \circ -) \circ (\text{pr}_2 \circ -)$ is an equivalence.

Proof. By Proposition 1.6.3, it suffices to show that the pullback square defining $C \times_E D$ remains a pullback square after applying $\text{Map}(S, \text{Fun}(T, -))$ for every category S . Under uncurrying, its four corners are identified with

$$\text{Map}(S \times T, C \times_E D), \quad \text{Map}(S \times T, C), \quad \text{Map}(S \times T, D), \quad \text{Map}(S \times T, E).$$

By Lemma 1.7.7, the four functors become postcomposition with the original projections, f , and g . These comparisons respect the specified commutativity identification, as follows by uncurrying its defining postcomposition identification. We conclude that this square agrees with the one obtained by applying $\text{Map}(S \times T, -)$ to the original pullback square, including its commutativity data. So the resulting square is a pullback by Proposition 1.6.3. $\square$

Lemma 1.7.12. *For categories C , D and E , there is a preferred equivalence*

$$\text{Fun}(C, \text{Fun}(D, E)) \xrightarrow{\sim} \text{Fun}(C \times D, E).$$

Proof. We define the preferred functor as the curried functor of the following composite:

$$\text{Fun}(C, \text{Fun}(D, E)) \times C \times D \xrightarrow{\text{ev} \times \text{id}_D} \text{Fun}(D, E) \times D \xrightarrow{\text{ev}} E.$$

To construct an inverse, it will by currying suffice to construct a functor $\text{Fun}(C \times D, E) \times C \rightarrow \text{Fun}(D, E)$, which we may take to be the curried functor of the following composite

$$\text{Fun}(C \times D, E) \times C \times D \xrightarrow{\text{ev}} E.$$

By construction, these two functors are compatible with the chain of equivalences

$$\begin{aligned}
\text{Map}(T, \text{Fun}(C, \text{Fun}(D, E))) &\simeq \text{Map}(T \times C, \text{Fun}(D, E)) \\
&\simeq \text{Map}((T \times C) \times D, E) \\
&\simeq \text{Map}(T \times (C \times D), E) \\
&\simeq \text{Map}(T, \text{Fun}(C \times D, E)).
\end{aligned}$$

This shows that applying $\text{Map}(T, -)$ to the preferred functor is an equivalence for all T , hence that the preferred functor is an equivalence. $\square$

Lemma 1.7.13. *For a category C , the functor $\text{ev}_*: \text{Fun}(*, C) \rightarrow C$ is an equivalence.*

Proof. We leave it to the reader to verify that an inverse is given by currying the equivalence $C \times * \xrightarrow{\sim} C$. $\square$

Lemma 1.7.14. *For a category C , the functor category $\text{Fun}(\emptyset, C)$ is contractible, i.e. $\text{Fun}(\emptyset, C) \xrightarrow{\sim} *$.*

Proof. For every category T , there are equivalences

$$\text{Map}(T, \text{Fun}(\emptyset, C)) \simeq \text{Map}(T \times \emptyset, C) \simeq \text{Map}(\emptyset, C) \simeq *,$$

using Lemma 1.5.1 and the initial-category axiom. Hence $\text{Fun}(\emptyset, C) \rightarrow *$ induces an equivalence on mapping animae for every T , and Proposition 1.4.11 applies. $\square$

Lemma 1.7.15. *For categories C, D and E , restriction along the inclusions $i_C: C \hookrightarrow C \sqcup D$ and $i_D: D \hookrightarrow C \sqcup D$ induces an equivalence*

$$(i_C^*, i_D^*): \text{Fun}(C \sqcup D, E) \xrightarrow{\sim} \text{Fun}(C, E) \times \text{Fun}(D, E).$$

Proof. For every category T , there are equivalences

$$\begin{aligned}
\text{Map}(T, \text{Fun}(C \sqcup D, E)) &\simeq \text{Map}(T \times (C \sqcup D), E) \\
&\simeq \text{Map}((T \times C) \sqcup (T \times D), E) \\
&\simeq \text{Map}(T \times C, E) \times \text{Map}(T \times D, E) \\
&\simeq \text{Map}(T, \text{Fun}(C, E)) \times \text{Map}(T, \text{Fun}(D, E)) \\
&\simeq \text{Map}(T, \text{Fun}(C, E) \times \text{Fun}(D, E)),
\end{aligned}$$

using distributivity, the coproduct axiom, and Proposition 1.4.5. This composite is induced by (i_C^*, i_D^*) : after uncurrying, its components are restriction along $T \times i_C$ and $T \times i_D$, which correspond under Corollary 1.6.17 to the two coproduct inclusions. The result follows from Proposition 1.4.11. $\square$

1.7.2 Mapping animae and functor categories

We now compare the mapping anima with the functor category. Currying mapping-anima evaluation defines a canonical functor

$$\iota_{C,D} := (\text{ev}_{\text{Map}})_c : \text{Map}(C, D) \longrightarrow \text{Fun}(C, D).$$

The following result identifies $\text{Map}(C, D)$ with the collection of objects of $\text{Fun}(C, D)$.

Proposition 1.7.16. *For any categories C and D , there is a preferred equivalence*

$$\text{Map}(C, D) \xrightarrow{\sim} \text{Fun}(C, D)^{\simeq}.$$

Under this equivalence, the canonical functor $\iota_{\text{Fun}(C,D)} : \text{Fun}(C, D)^{\simeq} \rightarrow \text{Fun}(C, D)$ is identified with $\iota_{C,D}$.

Proof. The universal property of functor categories gives equivalences

$$\text{Fun}(C, D)^{\simeq} = \text{Map}(*, \text{Fun}(C, D)) \simeq \text{Map}(* \times C, D) \simeq \text{Map}(C, D).$$

The inverse of their composite is the desired equivalence. To check compatibility with the canonical functors, it suffices to uncurry. Both resulting functors $\text{Map}(C, D) \times C \rightarrow D$ are identified with mapping-anima evaluation. $\square$

Proposition 1.7.17. *For any anima X , postcomposition with $\iota_{C,D}$ induces an equivalence*

$$(\iota_{C,D})_* : \text{Map}(X, \text{Map}(C, D)) \xrightarrow{\sim} \text{Map}(X, \text{Fun}(C, D)).$$

Proof. By the preceding proposition, this is the universal property of the groupoid core of $\text{Fun}(C, D)$ from Corollary 1.4.13. $\square$

1.8 Pushout squares

Using mapping animae, we may dualize the definition of a pullback square of categories:

Definition 1.8.1 (Pushout square). A commutative square of categories

$$\begin{array}{ccc} C' & \xrightarrow{u} & C \\ f' \downarrow & & \downarrow f \\ D' & \xrightarrow{v} & D \end{array}$$

is called a *pushout square* (or *cocartesian*) if, for every category E , the induced square

$$\begin{array}{ccc} \text{Map}(D, E) & \xrightarrow{v^*} & \text{Map}(D', E) \\ f^* \downarrow & & \downarrow (f')^* \\ \text{Map}(C, E) & \xrightarrow{u^*} & \text{Map}(C', E) \end{array}$$

is a pullback square.

Remark 1.8.2. We do not assume in the present metatheory that arbitrary pushouts of categories exist. We do not need this as a metatheoretic axiom, although such pushouts exist in all intended models of our theory. We will eventually prove that the synthetic category Cat of small synthetic categories admits small colimits, which in particular provides pushouts of small categories.

Proposition 1.8.3. *A commutative square of categories*

$$\begin{array}{ccc} C' & \xrightarrow{u} & C \\ f' \downarrow & & \downarrow f \\ D' & \xrightarrow{v} & D \end{array}$$

is a pushout square if and only if, for every category E , the induced square

$$\begin{array}{ccc} \text{Fun}(D, E) & \xrightarrow{v^*} & \text{Fun}(D', E) \\ f^* \downarrow & & \downarrow (f')^* \\ \text{Fun}(C, E) & \xrightarrow{u^*} & \text{Fun}(C', E) \end{array}$$

is a pullback square.

Proof. For any category T , the commutative square

$$\begin{array}{ccc} \text{Map}(T, \text{Fun}(D, E)) & \xrightarrow{v^*} & \text{Map}(T, \text{Fun}(D', E)) \\ f^* \downarrow & & \downarrow (f')^* \\ \text{Map}(T, \text{Fun}(C, E)) & \xrightarrow{u^*} & \text{Map}(T, \text{Fun}(C', E)) \end{array}$$

is equivalent to the commutative square

$$\begin{array}{ccc} \text{Map}(T \times D, E) & \xrightarrow{v^*} & \text{Map}(T \times D', E) \\ f^* \downarrow & & \downarrow (f')^* \\ \text{Map}(T \times C, E) & \xrightarrow{u^*} & \text{Map}(T \times C', E) \end{array}$$

which in turn is equivalent to the square

$$\begin{array}{ccc} \text{Map}(D, \text{Fun}(T, E)) & \xrightarrow{v^*} & \text{Map}(D', \text{Fun}(T, E)) \\ f^* \downarrow & & \downarrow (f')^* \\ \text{Map}(C, \text{Fun}(T, E)) & \xrightarrow{u^*} & \text{Map}(C', \text{Fun}(T, E)). \end{array}$$

The latter is a pullback square for all T and E if and only if it is so for all E in the special case where $T = *$ since there is a canonical equivalence $E \xrightarrow{\sim} \text{Fun}(*, E)$. On the other hand, the first square is a pullback square for all T if and only if

$$\begin{array}{ccc} \text{Fun}(D, E) & \xrightarrow{v^*} & \text{Fun}(D', E) \\ f^* \downarrow & & \downarrow (f')^* \\ \text{Fun}(C, E) & \xrightarrow{u^*} & \text{Fun}(C', E) \end{array}$$

is a pullback square. This proves the proposition. $\square$

Lemma 1.8.4 (Pasting lemma for pushout squares). *Consider a commutative diagram*

$$\begin{array}{ccccc}
 C_1 & \xrightarrow{g_1} & C_2 & \xrightarrow{g_2} & C_3 \\
 f_1 \downarrow & & \downarrow f_2 & & \downarrow f_3 \\
 D_1 & \xrightarrow{h_1} & D_2 & \xrightarrow{h_2} & D_3.
 \end{array}$$

If the left-hand square is a pushout square, then the right-hand square is a pushout square if and only if the outer rectangle is a pushout square.

Proof. This follows immediately by applying Lemma 1.6.11 to the commutative diagrams

$$\begin{array}{ccccc}
 \text{Map}(D_3, E) & \xrightarrow{h_2^*} & \text{Map}(D_2, E) & \xrightarrow{h_1^*} & \text{Map}(D_1, E) \\
 f_3^* \downarrow & & \downarrow f_2^* & & \downarrow f_1^* \\
 \text{Map}(C_3, E) & \xrightarrow{g_2^*} & \text{Map}(C_2, E) & \xrightarrow{g_1^*} & \text{Map}(C_1, E).
 \end{array}$$

for each E .

□

2 The internal structure of categories

In the previous chapter, our discussion of categories focused on their external theory: relating them via functors and equivalences, and constructing them via products or functor categories. Something we have mostly ignored so far is the *internal* structure that categories are expected to have: beyond objects, they should also have morphisms, and there should be a unital and associative composition law.

Rather than directly axiomatizing such internal structure, we will instead gain access to it by ‘probing’ a category under consideration with various test categories. We begin with morphisms and commutative diagrams in Section 2.1. The Segal axiom in Section 2.2 then provides composition of morphisms, and the Rezk axiom in Section 2.3 relates isomorphisms to identifications of objects.

Finally, we relate the primitive animae of our theory to the structural notion of a *groupoid*, a category in which every morphism is an isomorphism. The groupoid recognition axiom identifies these two points of entry. This also explains why the collection of objects $C^\simeq$ introduced in the previous chapter is called the groupoid core of C : it remembers the objects of C and their isomorphisms, while forgetting the noninvertible morphisms.

2.1 Morphisms and diagrams

In the case of morphisms, we use the following test category.

Axiom C.1 (Walking morphism). There is a category $[1]$ equipped with objects $0, 1 : * \rightarrow [1]$. Moreover, the functors $0, 1 : * \rightarrow [1]^\simeq$ induce an equivalence

$$\langle 0, 1 \rangle : * \sqcup * \xrightarrow{\sim} [1]^\simeq.$$

In particular, $* \sqcup *$ is an anima, since it is equivalent to the anima $[1]^\simeq$.

By substitution along $p_\Gamma : \Gamma \rightarrow *$, the two displayed objects also give terms

$$\Gamma \vdash 0, 1 : [1]$$

in every categorical context Γ .

The category $[1]$ is called the *walking morphism* and should be thought of as the poset $\{0 \leq 1\}$.

Let us introduce some standard terminology regarding functors out of $[1]$.

Definition 2.1.1 (Morphisms). A *morphism* in a category C is defined as a functor $f : [1] \rightarrow C$, or equivalently as an object of the arrow category $\text{Fun}([1], C)$.

Every morphism $f : [1] \rightarrow C$ has a *source/domain* $f(0) : * \rightarrow C$ and a *target/codomain* $f(1) : * \rightarrow C$, obtained by precomposition with $0, 1 : * \rightarrow [1]$. If f is a morphism in C , and x and y are objects of C , we will often write $f : x \rightarrow y$ to indicate that f comes equipped with identifications $x = f(0)$ and $y = f(1)$, which we will mostly leave implicit in the notation. In this case we say that f is a *morphism from x to y* .

We write $\text{ev}_0, \text{ev}_1 : \text{Fun}([1], C) \rightarrow C$ for the source and target functors, given by evaluation at 0 and 1, cf. Construction 1.7.6.

Definition 2.1.2 (Natural transformations). If C and D are categories, we define a *natural transformation of functors* $C \rightarrow D$ to be a morphism in $\text{Fun}(C, D)$. By (un)currying, this may equivalently be encoded as a functor $[1] \times C \rightarrow D$.

Definition 2.1.3 (Arrow category/hom anima). For a category C , we define its *arrow category* as

$$\text{Ar}(C) := \text{Fun}([1], C).$$

Given absolute objects x and y of C , we define the *hom anima* $C(x, y)$ via the following pullback square:

$$\begin{array}{ccc} C(x, y) & \longrightarrow & \text{Ar}(C) \\ \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ * & \xrightarrow{(x, y)} & C \times C. \end{array}$$

At this point, the pullback defines a category. We call it the hom anima in anticipation of Corollary 5.3.5 (outside selection), where it will be shown to be an anima. Observe that the absolute objects of $C(x, y)$ are triples (f, α, β) , where $f : x' \rightarrow y'$ is a morphism in C and $\alpha : x = x'$ and $\beta : y = y'$ are identifications in C . We will sometimes use the alternative notation $\text{Hom}_C(x, y)$ for $C(x, y)$.

Convention 2.1.4 (Morphism expressions). For functors $f, g : C \rightarrow D$, the notation

$$x : C \vdash \alpha(x) : f(x) \rightarrow g(x)$$

means a functor $H : C \times [1] \rightarrow D$ together with identifications $H(-, 0) = f$ and $H(-, 1) = g$. Currying gives its equivalent description as a functor $C \rightarrow \text{Ar}(D)$ with the same endpoint comparisons. In the absolute case, we use $[1] \rightarrow D$ directly.

An identification between two morphism expressions with the same displayed source and target is an identification of their interval diagrams together with compatibility with the two endpoint comparisons. For absolute morphisms between fixed objects $a, b : D$, it is an identification in $D(a, b)$. In context $x : C$, with the same fixed endpoints, it is an identification between functors $C \rightarrow D(a, b)$. For general f and g that vary with x , it is simply a functor into D with the two endpoint compatibilities.

Construction 2.1.5 (From identifications to morphism expressions). A primitive identification $\alpha : f = g$ determines a morphism expression from $f(x)$ to $g(x)$. Take the constant interval

$$C \times [1] \xrightarrow{\text{pr}_1} C \xrightarrow{f} D,$$

with its canonical source identification, and with target identification given by α .

Construction 2.1.6. Let $f: C \rightarrow D$ be a functor and consider two absolute objects x and y of C . Then the induced functor $f_*: \text{Ar}(C) \rightarrow \text{Ar}(D)$ induces a functor

$$f: C(x, y) \rightarrow D(f(x), f(y))$$

on fibers.

Example 2.1.7 (Identity morphism). In a common categorical context Γ , a term $\Gamma \vdash x: C$ determines the *identity morphism* $\Gamma \vdash \text{id}_x: x \rightarrow x$, defined by currying the composite

$$\Gamma \times [1] \xrightarrow{\text{pr}_1} \Gamma \xrightarrow{x} C.$$

It comes with a specified identification from its uncurrying to this composite. For every functor $f: C \rightarrow D$, there is a preferred identification $\Gamma \vdash f(\text{id}_x) = \text{id}_{f(x)}$. In the absolute case $\Gamma = *$, this recovers the usual identity morphism $\text{id}_x \in C(x, x)$.

Definition 2.1.8 (Slice categories). Let C be a category and let x be an absolute object of C . We define the *slice categories* $C_{/x}$ and $C_{x/}$ via the following two pullback diagrams:

$$\begin{array}{ccc} C_{/x} & \longrightarrow & \text{Ar}(C) \\ p_x \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ C & \xrightarrow{\text{id}_C \times x} & C \times C \\ \downarrow & \lrcorner & \downarrow \text{pr}_2 \\ * & \xrightarrow{x} & C. \end{array} \quad \text{and} \quad \begin{array}{ccc} C_{x/} & \longrightarrow & \text{Ar}(C) \\ q_x \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ C & \xrightarrow{x \times \text{id}_C} & C \times C \\ \downarrow & \lrcorner & \downarrow \text{pr}_1 \\ * & \xrightarrow{x} & C. \end{array}$$

In particular, an object of the category $C_{/x}$ consists of a pair (y, u) where y is an object of C and $u: y \rightarrow x$ is a morphism in C . Similarly the objects of $C_{x/}$ are pairs (y, v) where now v is a morphism in C of the form $v: x \rightarrow y$.

Definition 2.1.9 (Initial/terminal object). An absolute object x in a category C is called *terminal* if the functor $p_x: C_{/x} \rightarrow C$ is an equivalence. Dually, x is called *initial* if $q_x: C_{x/} \rightarrow C$ is an equivalence.

Remark 2.1.10. Observe that the fiber of p_x over an absolute object y is the hom anima $C(y, x)$, using the pasting law for pullback squares. In particular, if x is terminal then $C(y, x)$ is contractible for every y . Dually, if x is initial then $C(x, y)$ is contractible for all y . The converse will be established later, once additional axioms have been introduced.

Axiom C.2. The absolute object $0: * \rightarrow [1]$ is initial. The absolute object $1: * \rightarrow [1]$ is terminal.

Construction 2.1.11. As a consequence of Axiom C.2, we obtain the following two pullback squares:

$$\begin{array}{ccc} [1] & \xrightarrow{\overline{\text{min}}} & \text{Ar}([1]) \\ \sim \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ [1] & \xrightarrow{(\text{const}_0, \text{id})} & [1] \times [1] \end{array} \quad \text{and} \quad \begin{array}{ccc} [1] & \xrightarrow{\overline{\text{max}}} & \text{Ar}([1]) \\ \sim \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ [1] & \xrightarrow{(\text{id}, \text{const}_1)} & [1] \times [1]. \end{array}$$

Using $\text{Ar}(C) = \text{Fun}([1], C)$, the two top horizontal maps in these diagrams correspond under currying/uncurrying to functors

$$\min: [1] \times [1] \rightarrow [1] \quad \text{and} \quad \max: [1] \times [1] \rightarrow [1].$$

Lemma 2.1.12. *The maps $\max$ and $\min$ satisfy the identifications*

$$\begin{aligned} \max(x, 0) &= x = \max(0, x), & \max(x, 1) &= 1 = \max(1, x), \\ \min(x, 0) &= 0 = \min(0, x), & \min(x, 1) &= x = \min(1, x), \end{aligned}$$

in the sense that the following diagrams commute:

$$\begin{array}{ccc} [1] & \xrightarrow{(\text{id}, 0)} & [1] \times [1] \xleftarrow{(0, \text{id})} [1] \\ & \searrow \text{const}_0 & \downarrow \min \swarrow \text{const}_0 \\ & & [1] \end{array} \quad \begin{array}{ccc} [1] & \xrightarrow{(\text{id}, 1)} & [1] \times [1] \xleftarrow{(1, \text{id})} [1] \\ & \searrow & \downarrow \min \swarrow \\ & & [1] \end{array}$$

$$\begin{array}{ccc} [1] & \xrightarrow{(\text{id}, 0)} & [1] \times [1] \xleftarrow{(0, \text{id})} [1] \\ & \searrow & \downarrow \max \swarrow \\ & & [1] \end{array} \quad \begin{array}{ccc} [1] & \xrightarrow{(\text{id}, 1)} & [1] \times [1] \xleftarrow{(1, \text{id})} [1] \\ & \searrow \text{const}_1 & \downarrow \max \swarrow \text{const}_1 \\ & & [1] \end{array}$$

Proof. The identifications $\min(x, 0) = 0$, $\min(x, 1) = x$, $\max(x, 0) = x$ and $\max(x, 1) = 1$ are immediate from the two pullback squares stated before the lemma. For the other four relations we use that $q_0^{-1}: [1] \xrightarrow{\sim} [1]_{0/}$ sends 0 to $\text{id}_0: 0 \rightarrow 0$ and 1 to $\text{can}: 0 \rightarrow 1$, while $p_1^{-1}: [1] \xrightarrow{\sim} [1]_{/1}$ sends 0 to $\text{can}: 0 \rightarrow 1$ and 1 to $\text{id}_1: 1 \rightarrow 1$. $\square$

2.1.1 Commutative triangles

Apart from the poset $[1] = \{0 \leq 1\}$, which classifies morphisms, there should also be a poset $[2] = \{\bar{0} \leq \bar{1} \leq \bar{2}\}$ classifying *commutative triangles*.

Definition 2.1.13 (Walking commutative triangle). We define the *walking commutative triangle* as

$$[2] := \text{Ar}([1]) = \text{Fun}([1], [1]).$$

We define objects $\bar{0}, \bar{1}, \bar{2}: [2]$ by:

$$\bar{0} := \text{id}_0, \quad \bar{1} := \text{can}, \quad \bar{2} := \text{id}_1,$$

where $\text{can}: [1] \rightarrow [1]$ is the identity of $[1]$, regarded as a morphism $\text{can}: 0 \rightarrow 1$ in $[1]$.

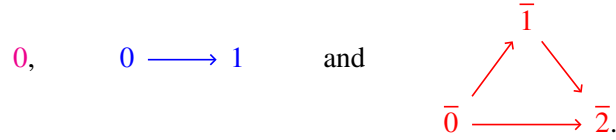
Construction 2.1.14. We construct three *face maps* $d_i: [1] \rightarrow [2]$ for $i = 0, 1, 2$:

- We set $d_0 := \overline{\max}: [1] \rightarrow \text{Fun}([1], [1])$;
- We set $d_1 := p_{[1]}^*: [1] \rightarrow \text{Fun}([1], [1])$;
- We set $d_2 := \overline{\min}: [1] \rightarrow \text{Fun}([1], [1])$.

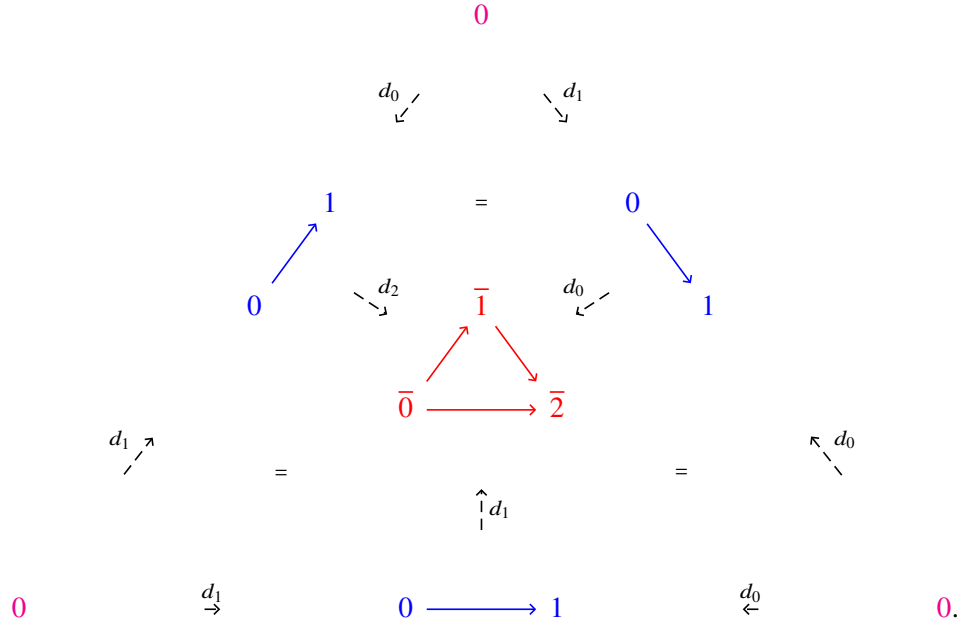
We also construct two *degeneracy maps* $s_i: [2] \rightarrow [1]$ for $i = 0, 1$:

- We set $s_0 := \text{ev}_0 : \text{Fun}([1], [1]) \rightarrow [1]$;
- We set $s_1 := \text{ev}_1 : \text{Fun}([1], [1]) \rightarrow [1]$.

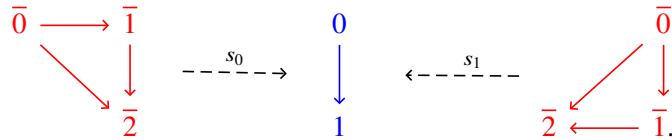
Convention 2.1.15. It will sometimes be useful to denote the terminal category by $[0]$, and to denote the maps $0 : * \rightarrow [1]$ and $1 : * \rightarrow [1]$ by $d_1 : [0] \rightarrow [1]$ and $d_0 : [0] \rightarrow [1]$, respectively. We will visualize the three categories $[0]$, $[1]$ and $[2]$ as follows:



The five face maps $d_0, d_1 : [0] \rightarrow [1]$ and $d_0, d_1, d_2 : [1] \rightarrow [2]$ and the relations between them may be displayed as follows:



The two degeneracy maps $s_0, s_1 : [2] \rightarrow [1]$ may be interpreted as the following ‘projections’:



Let us establish some of the relations implicit in this figure.

Lemma 2.1.16 (Simplicial identities). *There are vertex identifications*

$$\begin{aligned} d_2(0) = \bar{0} = d_1(0), & \quad d_2(1) = \bar{1} = d_0(0), & \quad d_0(1) = \bar{2} = d_1(1), \\ s_0(\bar{0}) = 0 = s_0(\bar{1}), & \quad s_0(\bar{2}) = 1, \quad s_1(\bar{0}) = 0, & \quad s_1(\bar{1}) = 1 = s_1(\bar{2}). \end{aligned}$$

There are identifications of morphisms

$$\begin{aligned} s_0 d_0 &= \text{id}_{[1]}, & s_0 d_1 &= \text{id}_{[1]}, & s_0 d_2 &= \text{const}_0, \\ s_1 d_0 &= \text{const}_1, & s_1 d_1 &= \text{id}_{[1]}, & s_1 d_2 &= \text{id}_{[1]}. \end{aligned}$$

These identifications may be chosen so that the morphism identifications are compatible with the vertex identifications.

The last statement means that if we evaluate one of the six morphism identifications at one of the two vertices 0 and 1, the resulting identification agrees with the one obtained by combining the twelve vertex identifications. For example, in the case of evaluating $s_0 d_0 = \text{id}_{[1]}$ at 0 it asserts that the following square of identifications commutes:

$$\begin{array}{ccc} s_0(d_0(0)) & \xrightarrow{=} & s_0(\bar{1}) \\ \downarrow = & & \downarrow = \\ \text{id}_{[1]}(0) & \xrightarrow{=} & 0. \end{array}$$

Proof. Recall that s_0 and s_1 are evaluation at the source and target. The defining pullback squares for $d_2 = \min$ and $d_0 = \max$ therefore give

$$s_0 d_2 = \text{const}_0, \quad s_1 d_2 = \text{id}_{[1]}, \quad s_0 d_0 = \text{id}_{[1]}, \quad s_1 d_0 = \text{const}_1.$$

The two remaining identifications follow from the definition of d_1 as the constant-arrow functor.

The vertex identifications for s_0 and s_1 are the specified source and target identifications of

$$\bar{0} = \text{id}_0, \quad \bar{1} = \text{can}, \quad \bar{2} = \text{id}_1.$$

At the vertices, these constructions give identifications

$$\begin{aligned} d_2(0) &= \bar{0} = d_1(0), \\ d_2(1) &= \bar{1} = d_0(0), \\ d_0(1) &= \bar{2} = d_1(1). \end{aligned}$$

We choose them with the source and target comparisons just specified. This is possible because the categories of morphisms $0 \rightarrow 0$, $0 \rightarrow 1$, and $1 \rightarrow 1$ in $[1]$ are contractible, by initiality of 0 and terminality of 1.

Each of these six vertex comparisons is chosen with its specified source and target identifications. Applying s_0 or s_1 therefore gives the corresponding commuting square of identifications, as required. $\square$

Definition 2.1.17 (Commutative triangle). Given a category C , we define a *commutative triangle in C* to be a functor $[2] \rightarrow C$. We define the category of commutative triangles in C as

$$\text{Triang}(C) \quad := \quad \text{Fun}([2], C).$$

The three functors $d_0, d_1, d_2: [1] \rightarrow [2]$ induce three restriction functors

$$d_0^*, d_1^*, d_2^*: \text{Triang}(C) \rightarrow \text{Ar}(C).$$

We will frequently denote commutative triangles by

$$\sigma = \begin{array}{ccc} & y & \\ f \nearrow & & \searrow g \\ x & \xrightarrow{h} & z, \end{array}$$

where $f = d_2^*(\sigma)$, $g = d_0^*(\sigma)$ and $h = d_1^*(\sigma)$. We use the identifications of Lemma 2.1.16 whenever we restrict a triangle to its edges and vertices. Thus the three edges have specified common vertices, and the simplicial identities respect those identifications. For families of triangles, these comparisons are obtained by uncurrying and restricting the evaluation functor; compatibility follows by applying evaluation to the comparisons above. We will often leave commutative triangles unnamed, and only label their three edges. We may sometimes denote in the same manner triples of morphisms (f, g, h) that do *not* (or not necessarily) come from a commutative triangle, which will then be mentioned explicitly.

Definition 2.1.18. Given morphisms $f: x \rightarrow y$ and $g: y \rightarrow z$ in C , a *composite* of f and g is a morphism $h: x \rightarrow z$ that comes equipped with a commutative triangle σ of the above form:

$$\begin{array}{ccc} & y & \\ f \nearrow & & \searrow g \\ x & \xrightarrow{h} & z. \end{array}$$

We define the *category of composites* $\text{Comp}_{f,g}(C)$ as the following pullback:

$$\begin{array}{ccc} \text{Comp}_{f,g}(C) & \longrightarrow & \text{Triang}(C) \\ \downarrow & \lrcorner & \downarrow (d_2^*, d_0^*) \\ * & \xrightarrow{(f,g)} & \text{Ar}(C) \times_C \text{Ar}(C). \end{array}$$

Here the right vertical map is the map induced by the following commutative diagram of restriction maps:

$$\begin{array}{ccc} \text{Triang}(C) & \xrightarrow{d_0^*} & \text{Ar}(C) \\ d_2^* \downarrow & & \downarrow \text{ev}_0 \\ \text{Ar}(C) & \xrightarrow{\text{ev}_1} & C. \end{array}$$

Example 2.1.19. The two functors $s_0, s_1: [2] \rightarrow [1]$ induce functors

$$s_0^*, s_1^*: \text{Ar}(C) \rightarrow \text{Triang}(C),$$

so that every morphism $f: x \rightarrow y$ in C gives rise to two commutative diagrams $s_0^*(f)$ and $s_1^*(f)$ in C . As a consequence of the simplicial identities from Lemma 2.1.16, these commutative triangles have the following form:

$$\begin{array}{ccc} & x & \\ \text{id}_x \nearrow & & \searrow f \\ x & \xrightarrow{f} & y \end{array} \quad \text{and} \quad \begin{array}{ccc} & y & \\ f \nearrow & & \searrow \text{id}_y \\ x & \xrightarrow{f} & y. \end{array}$$

In particular, these triangles exhibit f as a composite of id_x and f , as well as a composite of f and id_y .

2.1.2 Commutative squares

Throughout, C denotes a category.

Definition 2.1.20 (Commutative square). We define a *commutative square* in C to be a functor $[1] \times [1] \rightarrow C$. We define the category of commutative squares in C as

$$\text{Sq}(C) := \text{Fun}([1] \times [1], C).$$

We will often display commutative squares as follows:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ g \downarrow & & \downarrow h \\ z & \xrightarrow{k} & w. \end{array}$$

In ordinary category theory, a commutative square consists of morphisms f, g, h and k as above that satisfy the condition that $h \circ f = k \circ g$. We will now formulate a synthetic analogue of this.

Definition 2.1.21. We define two functors $j_0, j_1: [2] \rightarrow [1] \times [1]$ as

$$j_0 := (s_0, s_1) \quad \text{and} \quad j_1 := (s_1, s_0).$$

These two commutative triangles in $[1] \times [1]$ may be visualized as follows:

$$j_0 = \begin{array}{ccc} (0, 0) & \longrightarrow & (0, 1) \\ & \searrow & \downarrow \\ & & (1, 1), \end{array} \quad j_1 = \begin{array}{ccc} (0, 0) & & \\ \downarrow & \searrow & \\ (1, 0) & \longrightarrow & (1, 1). \end{array}$$

The simplicial relations $s_0 \circ d_1 = \text{id}_{[1]} = s_1 \circ d_1$ provide an identification making the following square commute:

$$\begin{array}{ccc} [1] & \xrightarrow{d_1} & [2] \\ d_1 \downarrow & & \downarrow j_0 \\ [2] & \xrightarrow{j_1} & [1] \times [1]. \end{array} \tag{2.1}$$

Axiom D (Commutative square axiom). The commutative square (2.1) is a pushout square: For every category C , the commutative square

$$\begin{array}{ccc} \text{Sq}(C) & \xrightarrow{j_1^*} & \text{Triang}(C) \\ j_0^* \downarrow & & \downarrow d_1^* \\ \text{Triang}(C) & \xrightarrow{d_1^*} & \text{Ar}(C) \end{array}$$

is a pullback square.

Construction 2.1.22. In a common categorical context Γ , consider terms $\Gamma \vdash \sigma, \tau : \text{Triang}(C)$ equipped with an identification $\Gamma \vdash \varphi : d_1^*(\sigma) = d_1^*(\tau)$. By Axiom D, these determine a term

$$\Gamma \vdash \square(\sigma, \tau, \varphi) : \text{Sq}(C)$$

with specified identifications $\sigma = j_0^*(\square(\sigma, \tau, \varphi))$ and $\tau = j_1^*(\square(\sigma, \tau, \varphi))$. In the absolute case it is the commutative square

$$\square(\sigma, \tau, \varphi) = \begin{array}{ccc} x & \xrightarrow{f} & y \\ g \downarrow & \searrow \sigma & \downarrow k \\ z & \xrightarrow{h} & w \end{array}$$

We will sometimes just write $\square(\sigma, \tau)$ instead of $\square(\sigma, \tau, \varphi)$ when the identification φ is left implicit.

Construction 2.1.23. We construct functors $p_0, p_2 : [1] \times [1] \rightarrow [2]$, pictorially represented by the following diagrams:

$$p_0 = \begin{array}{ccc} 0 & \longrightarrow & 1 \\ \parallel & \searrow & \downarrow \\ 0 & \longrightarrow & 2, \end{array} \quad p_2 = \begin{array}{ccc} 0 & \longrightarrow & 2 \\ \downarrow & \searrow & \parallel \\ 1 & \longrightarrow & 2. \end{array}$$

More precisely, $p_0 := \square(\text{id}_{[2]}, d_1 s_0)$ is the commutative square determined by the two commutative triangles $\sigma = \text{id}_{[2]}$ and $\tau = d_1 \circ s_0$, with identification

$$d_1^*(\sigma) = \text{id}_{[2]} \circ d_1 = d_1 = d_1 \circ \text{id}_{[1]} = d_1 \circ s_0 \circ d_1 = d_1^*(\tau).$$

Similarly, $p_2 := \square(d_1 s_1, \text{id}_{[2]})$ is the square determined by $\sigma = d_1 \circ s_1$ and $\tau = \text{id}_{[2]}$, with analogous identification between $d_1^*(\sigma)$ and $d_1^*(\tau)$.

By construction these two maps satisfy the following two relations:

$$p_0 \circ j_0 = \text{id}_{[2]}, \quad p_0 \circ j_1 = d_1 \circ s_0, \quad p_2 \circ j_0 = d_1 \circ s_1, \quad p_2 \circ j_1 = \text{id}_{[2]}.$$

Lemma 2.1.24. *We have the relations*

$$\square(s_0, s_1) = \text{pr}_1 \quad \text{and} \quad \square(s_1, s_0) = \text{pr}_2$$

of functors $[1] \times [1] \rightarrow [1]$.

Proof. It suffices to observe that these two projection maps have the correct restrictions to the triangles j_0 and j_1 . $\square$

Lemma 2.1.25. *We have*

$$\max = \square(s_1, s_1) \quad \text{and} \quad \min = \square(s_0, s_0).$$

Proof. It suffices to check that $\max$ and $\min$ restrict correctly along j_0 and j_1 , and this follows directly by unwinding the definitions of $\square(s_1, s_1)$ and $\square(s_0, s_0)$ from Construction 2.1.22. $\square$

Lemma 2.1.26. *We have relations*

$$\begin{aligned} j_0 p_0(x, y) &= (\min(x, y), y), & j_1 p_0(x, y) &= (y, \min(x, y)), \\ j_0 p_2(x, y) &= (y, \max(x, y)), & j_1 p_2(x, y) &= (\max(x, y), y). \end{aligned}$$

Proof. Since $p_0 := \square(\text{id}_{[2]}, d_1 s_0)$ and $p_2 := \square(d_1 s_1, \text{id}_{[2]})$, we have

$$\begin{aligned} s_0 p_0 &= \square(s_0, s_0 d_1 s_0) = \min, & s_1 p_0 &= \square(s_1, s_1 d_1 s_0) = \square(s_1, s_0) = \text{pr}_2 \\ s_0 p_2 &= \square(s_0 d_1 s_1, s_0) = \square(s_1, s_0) = \text{pr}_2, & s_1 p_2 &= \square(s_1 d_1 s_1, s_1) = \max, \end{aligned}$$

where we used Lemma 2.1.24. Using $j_0 = (s_0, s_1)$ and $j_1 = (s_1, s_0)$ we get the desired four relations. $\square$

Lemma 2.1.27. *The map $\langle 0, 1, 2 \rangle: * \sqcup * \sqcup * \rightarrow [2]^\simeq$ is an equivalence.*

Proof. Recall the functors $j_0 = (s_0, s_1): [2] \rightarrow [1] \times [1]$ and $p_0: [1] \times [1] \rightarrow [2]$ from Definition 2.1.21 and Construction 2.1.23. They satisfy $p_0 j_0 = \text{id}_{[2]}$. Choose an inverse to the preferred equivalence

$$\kappa := (\text{pr}_1^\simeq, \text{pr}_2^\simeq): ([1] \times [1])^\simeq \xrightarrow{\sim} ([1]^\simeq)^2$$

from Lemma 1.4.17, and put $\bar{j}_0 := \kappa \circ j_0^\simeq$ and $\bar{p}_0 := p_0^\simeq \circ \kappa^{-1}$. Let

$$a: * \sqcup * \sqcup * \longrightarrow (* \sqcup *)^2$$

send 0, 1, 2 to (0, 0), (0, 1), (1, 1), respectively, and let

$$b: (* \sqcup *)^2 \longrightarrow * \sqcup * \sqcup *$$

send (0, 0), (0, 1), (1, 0), (1, 1) to 0, 1, 0, 2, respectively. Then $b \circ a = \text{id}$. The defining identifications for groupoid-core functoriality and the displayed descriptions of j_0 and p_0 show that the following squares commute:

$$\begin{array}{ccc} * \sqcup * \sqcup * & \xrightarrow{a} & (* \sqcup *)^2 \\ \downarrow \langle 0, 1, 2 \rangle & & \downarrow \langle 0, 1 \rangle^2 \\ \simeq & \xrightarrow{\bar{j}_0} & ([1]^\simeq)^2 \end{array} \quad \text{and} \quad \begin{array}{ccc} (* \sqcup *)^2 & \xrightarrow{b} & * \sqcup * \sqcup * \\ \downarrow \langle 0, 1 \rangle^2 & & \downarrow \langle 0, 1, 2 \rangle \\ ([1]^\simeq)^2 & \xrightarrow{\bar{p}_0} & \simeq. \end{array}$$

The comparison for composition of groupoid-core functors, together with $p_0 j_0 = \text{id}_{[2]}$, gives $\bar{p}_0 \bar{j}_0 = \text{id}$. Thus this exhibits the map

$$\langle 0, 1, 2 \rangle: * \sqcup * \sqcup * \longrightarrow [2]^\simeq$$

as a retract of

$$\langle 0, 1 \rangle^2: (* \sqcup *)^2 \longrightarrow ([1]^\simeq)^2.$$

The latter map is an equivalence because it is the product of the equivalence in Axiom C.1 with itself. The claim follows from closure of equivalences under retracts (Lemma 1.3.9). $\square$

Lemma 2.1.28. *For categories C and D , the maps $C^\simeq \rightarrow (C \sqcup D)^\simeq$ and $D^\simeq \rightarrow (C \sqcup D)^\simeq$ induce an equivalence $C^\simeq \sqcup D^\simeq \xrightarrow{\sim} (C \sqcup D)^\simeq$.*

Proof. Let $i_0: * \rightarrow * \sqcup *$ and $i_1: * \rightarrow * \sqcup *$ denote the inclusions of the first and second point, respectively. By combining Lemma 1.6.4 and Axiom B.5', there are pullback squares

$$\begin{array}{ccc} C^\simeq & \longrightarrow & (C \sqcup D)^\simeq \\ \downarrow & \lrcorner & \downarrow \\ (*)^\simeq & \xrightarrow{i_0^\simeq} & (* \sqcup *)^\simeq \end{array} \quad \text{and} \quad \begin{array}{ccc} D^\simeq & \longrightarrow & (C \sqcup D)^\simeq \\ \downarrow & \lrcorner & \downarrow \\ (*)^\simeq & \xrightarrow{i_1^\simeq} & (* \sqcup *)^\simeq. \end{array}$$

It remains to show that the two maps $i_0^\simeq$ and $i_1^\simeq$ induce an equivalence

$$q: (*)^\simeq \sqcup (*)^\simeq \longrightarrow (* \sqcup *)^\simeq.$$

The defining identifications for groupoid-core functoriality give a commutative square

$$\begin{array}{ccc} (*)^\simeq \sqcup (*)^\simeq & \xrightarrow{q} & (* \sqcup *)^\simeq \\ \downarrow \iota_* \sqcup \iota_* & & \downarrow \iota_* \sqcup \iota_* \\ * \sqcup * & \xrightarrow{\text{id}} & * \sqcup *. \end{array}$$

The left vertical functor is an equivalence by Proposition 1.4.15, and the right vertical functor is an equivalence by Proposition 1.4.15, since $* \sqcup *$ is an anima. Hence q is an equivalence by 2-out-of-3. The two pullback squares above, descent for coproducts, and base change along q now identify $C^\simeq \sqcup D^\simeq$ with $(C \sqcup D)^\simeq$. $\square$

Remark 2.1.29. By Proposition 1.7.16 and Lemma 2.1.27, there are preferred equivalences

$$\text{Map}([1], [1]) \simeq \text{Fun}([1], [1])^\simeq \simeq [2]^\simeq \simeq * \sqcup * \sqcup *.$$

We may think of these three points as picking out the three morphisms $\text{id}_0: 0 \rightarrow 0$, $\text{id}_1: 1 \rightarrow 1$ and $\text{can}: 0 \rightarrow 1$ of $[1]$. In particular, we see that $[1]$ only has three morphisms.

2.2 The Segal axiom

In this section, we will introduce the *Segal axiom*, which will provide unique compositions of morphisms in categories.

Axiom E (Segal axiom). For any category C , the functor

$$(d_2^*, d_0^*): \text{Triang}(C) \rightarrow \text{Ar}(C) \times_C \text{Ar}(C)$$

is an equivalence.

In particular, for every pair of composable morphisms $f: x \rightarrow y$ and $g: y \rightarrow z$ in C the category $\text{Comp}_{f,g}(C)$ of composites of f and g is contractible, being defined as the fiber of the equivalence (d_2^*, d_0^*) .

Notation 2.2.1. In a common categorical context Γ , composable morphisms $\Gamma \vdash f: x \rightarrow y$ and $\Gamma \vdash g: y \rightarrow z$ determine a term

$$\Gamma \vdash g \circ f: x \rightarrow z,$$

which we call their unique composite. It comes with a chosen 2-simplex σ and specified identifications $d_2^*(\sigma) = f$, $d_0^*(\sigma) = g$, and $d_1^*(\sigma) = g \circ f$.

Observation 2.2.2 (Exercise 2.6.1). Let C_0 and C_1 be categories and consider morphisms $f_0: x_0 \rightarrow y_0$ and $g_0: y_0 \rightarrow z_0$ in C_0 and morphisms $f_1: x_1 \rightarrow y_1$ and $g_1: y_1 \rightarrow z_1$ in C_1 . Then the composite $(g_0, g_1) \circ (f_0, f_1): (x_0, x_1) \rightarrow (z_0, z_1)$ in $C_0 \times C_1$ is given by $(g_0 \circ f_0, g_1 \circ f_1)$.

For fixed objects $x, y, z : C$, the definition of $g \circ f$ works for expressions with values in $C(x, y)$ and $C(y, z)$ and a common source. Thus composition is functorial in f and g . We may make this functoriality more explicit as follows:

Construction 2.2.3 (Composition functors). The Segal axiom gives an equivalence

$$q_C = (d_2^*, d_0^*) : \text{Triang}(C) \longrightarrow \text{Ar}(C) \times_C \text{Ar}(C).$$

Here the identification joining the two short edges is induced by $d_2(1) = \bar{1} = d_0(0)$ from Lemma 2.1.16. The vertex identifications of that lemma make q_C a functor over C^3 : on triangles we evaluate at $\bar{0}, \bar{1}, \bar{2}$, and on composable pairs we take the source and target of the first edge and the target of the second.

Choose an inverse j_C over C^3 , together with an identification $\varepsilon_C : q_C j_C = \text{id}$ over C^3 . Composition is the functor

$$\text{Ar}(C) \times_C \text{Ar}(C) \xrightarrow{j_C} \text{Triang}(C) \xrightarrow{d_1^*} \text{Ar}(C).$$

Thus we first complete a composable pair (f, g) to a triangle, and then take its long edge. We denote this edge by $g \circ f$.

The vertex identifications of Lemma 2.1.16 also identify the endpoints of the long edge with the first and last vertices. Hence composition is a functor over $C \times C$. Passing to fibers over prescribed vertices x, y, z gives a functor

$$- \circ - : C(y, z) \times C(x, y) \longrightarrow C(x, z).$$

The source and target identifications of $g \circ f$ are obtained from these vertex comparisons followed by the corresponding components of ε_C . Different choices of the inverse give identified composition functors over $C \times C$.

Convention 2.2.4 (Composition of morphism expressions). For functors $f, g, h : C \rightarrow D$, consider morphism expressions $\alpha(x) : f(x) \rightarrow g(x)$ and $\beta(x) : g(x) \rightarrow h(x)$ in context $x : C$. Their endpoint identifications define a functor $C \rightarrow \text{Ar}(D) \times_D \text{Ar}(D)$. Composing with the composition functor gives a morphism expression

$$\beta(x) \circ \alpha(x) : f(x) \rightarrow h(x).$$

This construction also acts on identifications of morphism expressions that retain the endpoints.

Lemma 2.2.5. *For a functor $F : C \rightarrow D$ and morphisms $f : x \rightarrow y$ and $g : y \rightarrow z$ in C , there is an identification*

$$F(g \circ f) = F(g) \circ F(f)$$

in $D(F(x), F(z))$. This identification is natural in the pair (g, f) : it forms an identification over $D \times D$ between two functors $\text{Ar}(C) \times_C \text{Ar}(C) \rightarrow \text{Ar}(D)$.

Proof. Consider the chosen triangle in C expressing $g \circ f$ as a composite. Applying F to this triangle produces a triangle in D with short edges $F(f), F(g)$. By the Segal axiom, this triangle is identified with the chosen triangle in D expressing $F(g) \circ F(f)$ as a composite of $F(f)$ and $F(g)$, in a way compatible with the two short-edge comparison identifications. The relation $F(g \circ f) = F(g) \circ F(f)$ now follows by passing to the long edges of these identified triangles. On source and target, this

identification is the restriction of the two a priori identifications on the short edges, and so it is compatible with the vertex identifications.

For naturality, we apply this argument to terms in the categorical context $\text{Ar}(C) \times_C \text{Ar}(C)$. $\square$

Lemma 2.2.6 (Unitality). *For every morphism $f: x \rightarrow y$, there are identifications*

$$\text{id}_y \circ f = f = f \circ \text{id}_x$$

in $C(x, y)$, with its given endpoint identifications. They are natural in f : they define identifications of functors $\text{Ar}(C) \rightarrow \text{Ar}(C)$ over $C \times C$.

Proof. We use the degenerate triangle

$$s_0^*(f) = \begin{array}{ccc} & x & \\ \text{id}_x \nearrow & & \searrow f \\ x & \xrightarrow{f} & y, \end{array}$$

obtained by restricting f along s_0 . Its short edges are (id_x, f) and its long edge is f . These identifications respect the three vertices: after uncurrying, their compatibility is given by the vertex identifications of Lemma 2.1.16.

The chosen triangle $j_C(\text{id}_x, f)$ has the same short edges, with comparison given by ε_C . The Segal equivalence therefore identifies these two triangles, with this comparison on their short edges. Restriction to the long edge gives an identification

$$f \circ \text{id}_x = d_1^* j_C(\text{id}_x, f) = d_1^* s_0^*(f) = f.$$

This identification also respects the prescribed source and target: the interchange law allows us to compare endpoints before or after comparing the triangles. The short-edge identifications of the degenerate triangle then cancel with their inverses, leaving the vertex comparisons followed by the components of ε_C . These are exactly the source and target identifications used to define composition.

The same argument with s_1 gives $\text{id}_y \circ f = f$.

For naturality in f , we apply both constructions to the universal arrow on $\text{Ar}(C)$, working with term expressions in context $\text{Ar}(C)$. (In terms of actual functors, this translates to restriction and lifting through the Segal equivalence.) This results in the required identifications of functors over $C \times C$. $\square$

Proposition 2.2.7 (Associativity, cf. [RS17, Proposition 5.9]). *For composable morphisms $f: x \rightarrow y$, $g: y \rightarrow z$, and $h: z \rightarrow w$, there is an identification*

$$h \circ (g \circ f) = (h \circ g) \circ f$$

in $C(x, w)$. These identifications are natural in the composable triple: they form an identification over $C \times C$ between the two composition functors from $\text{Ar}(C) \times_C \text{Ar}(C) \times_C \text{Ar}(C)$ to $\text{Ar}(C)$.

Proof. As in the proof of unitality, we establish the desired identification (including the source and target compatibilities) for term expressions $f: x \rightarrow y$, $g: y \rightarrow z$ and $h: z \rightarrow w$ in an arbitrary categorical context. Taking this context to be $\text{Ar}(C) \times_C \text{Ar}(C) \times_C \text{Ar}(C)$ then gives the final claim.

Consider the commutative triangle presenting the composite of f and g , which has $g \circ f$ as its long edge. Consider also the commutative triangle presenting the composite of id_x and $g \circ f$. Using the unitality of composition from Lemma 2.2.6, we may identify its long edge with $g \circ f$ in $C(x, z)$. Using the commutative square axiom, we may glue these two triangles into a commutative square

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ \text{id}_x \downarrow & \searrow g \circ f & \downarrow g \\ x & \xrightarrow{g \circ f} & z. \end{array}$$

The pullback property in the commutative square axiom gives identifications of its two triangle restrictions with the chosen triangles, compatible with the specified identification of their long edges. Since this last identification lies in $C(x, z)$, the endpoint comparisons agree at the two corners on the diagonal. At the other two corners, compatibility follows from the corresponding triangle. To identify these with the corners of the displayed square, recall that $j_0 = (s_0, s_1)$ and $j_1 = (s_1, s_0)$. The simplicial identities identify their boundary edges with the coordinate edges of $[1] \times [1]$; for example, $j_0 d_2 = (0, \text{id}_{[1]})$ and $j_1 d_2 = (\text{id}_{[1]}, 0)$. The vertex compatibilities of Lemma 2.1.16 give the specified comparisons under both product projections, so the product universal property identifies the corner comparisons as well. Thus all four edges have their prescribed endpoints, including the identification of the left side with id_x .

In a similar fashion, gluing the triangles for (g, h) and $(h \circ g, \text{id}_w)$ along the unitality identification in $C(y, w)$ gives a commutative square

$$\begin{array}{ccc} y & \xrightarrow{h \circ g} & w \\ g \downarrow & \searrow h \circ g & \downarrow \text{id}_w \\ z & \xrightarrow{h} & w, \end{array}$$

again with the displayed endpoint identifications.

Under the equivalence $\text{Sq}(C) \simeq \text{Ar}(\text{Ar}(C))$, we may regard these commutative squares as morphisms $\text{id}_x \rightarrow g$ and $g \rightarrow \text{id}_w$ in $\text{Ar}(C)$. Their composition in $\text{Ar}(C)$ corresponds to a commutative square in C of the form

$$\begin{array}{ccc} x & \xrightarrow{(h \circ g) \circ f} & w \\ \text{id}_x \downarrow & & \downarrow \text{id}_w \\ x & \xrightarrow{h \circ (g \circ f)} & w. \end{array}$$

Here the top and bottom edges are identified with $(h \circ g) \circ f$ and $h \circ (g \circ f)$ by applying Lemma 2.2.5 to the two evaluation functors $\text{ev}_0, \text{ev}_1 : \text{Ar}(C) \rightarrow C$. That lemma also ensures that these identifications retain the prescribed endpoints.

The two triangle restrictions of this square identify its diagonal with the composites along the top and right, and along the left and bottom, respectively. Indeed, the Segal equivalence compares each triangle with the chosen triangle for its two short edges. By Lemma 2.2.6, these composites are identified with the top and bottom edges themselves in $C(x, w)$. We obtain the required identification

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

All constructions use restriction, the two axioms' universal properties, and composition in $\text{Ar}(C)$, so they apply to the universal composable triple as asserted at the start. $\square$

Observe that another way of formulating the Segal axiom, Axiom E, is that the commutative square

$$\begin{array}{ccc} \text{Triang}(C) & \xrightarrow{d_2^*} & \text{Ar}(C) \\ d_0^* \downarrow & & \downarrow \text{ev}_1 \\ \text{Ar}(C) & \xrightarrow{\text{ev}_0} & C \end{array}$$

is a pullback square. This has the following corollary:

Corollary 2.2.8. *For every category C , the commutative squares*

$$\begin{array}{ccc} \text{Triang}(C) & \xrightarrow{p_0^*} & \text{Sq}(C) \\ \text{ev}_0 \downarrow & & \downarrow \text{res} \\ C & \xrightarrow{(p_{[1]})^*} & \text{Fun}([1] \times \{0\}, C) \end{array} \quad \text{and} \quad \begin{array}{ccc} \text{Triang}(C) & \xrightarrow{p_2^*} & \text{Sq}(C) \\ \text{ev}_2 \downarrow & & \downarrow \text{res} \\ C & \xrightarrow{(p_{[1]})^*} & \text{Fun}([1] \times \{1\}, C) \end{array}$$

are pullback squares.

Proof. We will prove the claim for the left square and leave the analogous proof of the right square to the reader. Note that this square can be written as the outer square in the following commutative diagram:

$$\begin{array}{ccc} \text{Triang}(C) & \xrightarrow{p_0^*} & \text{Sq}(C) \\ d_1^* \downarrow & & \downarrow j_1^* \\ \text{Ar}(C) & \xrightarrow{s_0^*} & \text{Triang}(C) \\ \text{ev}_0 \downarrow & & \downarrow d_2^* \\ C & \xrightarrow{(p_{[1]})^*} & \text{Ar}(C), \end{array}$$

and so it will suffice that each of these two squares is a pullback square. The first one follows from applying the pasting law of pullback squares to the following commutative diagram:

$$\begin{array}{ccccc} \text{Triang}(C) & \xrightarrow{p_0^*} & \text{Sq}(C) & \xrightarrow{j_0^*} & \text{Triang}(C) \\ d_1^* \downarrow & \lrcorner & \downarrow j_1^* & \lrcorner & \downarrow d_1^* \\ \text{Ar}(C) & \xrightarrow{s_0^*} & \text{Triang}(C) & \xrightarrow{d_1^*} & \text{Ar}(C). \end{array}$$

Here the outer square is evidently a pullback, and the right square is a pullback by the commutative square axiom, Axiom D. The second one similarly follows from applying the pasting law of pullback squares to the following commutative diagram:

$$\begin{array}{ccccc} \text{Ar}(C) & \xrightarrow{s_0^*} & \text{Triang}(C) & \xrightarrow{d_0^*} & \text{Ar}(C) \\ \text{ev}_0 \downarrow & \lrcorner & d_2^* \downarrow & \lrcorner & \downarrow \text{ev}_0 \\ C & \xrightarrow{(p_{[1]})^*} & \text{Ar}(C) & \xrightarrow{\text{ev}_1} & C. \end{array}$$

Here the outer rectangle is again evidently a pullback square, and the right square is a pullback square by the Segal axiom, Axiom E. $\square$

Remark 2.2.9. Informally, the previous corollary says that commutative triangles in C may equivalently be encoded in the following three ways:

- (1) One might consider commutative triangles $[2] \rightarrow C$, which we will display as

$$\begin{array}{ccc} & y & \\ f \nearrow & & \searrow g \\ x & \xrightarrow{gf} & z. \end{array}$$

- (2) One might consider commutative squares $[1] \times [1] \rightarrow C$ whose restriction to $[1] \times \{0\}$ is constant, which we will display as

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ \parallel & & \downarrow g \\ x & \xrightarrow{gf} & z. \end{array}$$

- (3) One might consider commutative squares $[1] \times [1] \rightarrow C$ whose restriction to $[1] \times \{1\}$ is constant, which we will display as

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ gf \downarrow & & \downarrow g \\ z & \xlongequal{\quad} & z. \end{array}$$

We now explain how the usual naturality relation follows from our definition of natural transformations. Let $\alpha: f \rightarrow g$ be a natural transformation between functors $f, g: C \rightarrow D$. For a morphism $u: a \rightarrow b$ in C , precomposition of the interval diagram of $\alpha: C \times [1] \rightarrow D$ with $u \times \text{id}_{[1]}$ gives a commutative square

$$\begin{array}{ccc} f(a) & \xrightarrow{\alpha(a)} & g(a) \\ f(u) \downarrow & & \downarrow g(u) \\ f(b) & \xrightarrow{\alpha(b)} & g(b). \end{array}$$

Its two triangles have the same long edge. After taking composites, we write

$$\alpha(u): g(u) \circ \alpha(a) = \alpha(b) \circ f(u)$$

for the resulting comparison in $D(f(a), g(b))$. Its endpoint identifications are those supplied by Lemma 2.1.16 and the comparison with the chosen composites in Construction 2.2.3.

The same construction gives the following interchange law.

Lemma 2.2.10 (Interchange for natural transformations). *Let $F, G: C \rightarrow D$ and $u, v: B \rightarrow C$ be functors, and let $\alpha: F \rightarrow G$ and $\beta: u \rightarrow v$ be natural transformations. There is a commutative square in $\text{Fun}(B, D)$*

$$\begin{array}{ccc} Fu & \xrightarrow{\alpha u} & Gu \\ F\beta \downarrow & & \downarrow G\beta \\ Fv & \xrightarrow{\alpha v} & Gv. \end{array}$$

This gives an identification in $\text{Hom}_{\text{Fun}(B, D)}(Fu, Gv)$ of the form

$$(G\beta) \circ (\alpha u) = (\alpha v) \circ (F\beta).$$

Proof. Let $A: [1] \times C \rightarrow D$ and $H: [1] \times B \rightarrow C$ encode α and β , respectively. The composite

$$[1] \times [1] \times B \xrightarrow{\text{id}_{[1]} \times H} [1] \times C \xrightarrow{A} D$$

curries to a functor $[1] \times [1] \rightarrow \text{Fun}(B, D)$. Its four edges, with the endpoint identifications supplied by A and H , are precisely the four transformations displayed above. Restricting to the two triangles, using the restriction identifications fixed above, gives the two composites with a common long edge and the prescribed source/target. This gives the displayed identification. $\square$

Finally, the conversion of Construction 2.1.5 is compatible with composition of primitive identifications. For $\alpha: f = g$ and $\beta: g = h$, consider the constant triangle at f , with vertex identifications id_f , α , and $\beta \circ \alpha$. Its short edges represent α and β , using α to identify the second short edge with the constant interval at g . Its long edge represents $\beta \circ \alpha$. The Segal equivalence identifies this triangle with the chosen composite triangle, retaining the short-edge comparisons. Passing to the long edge therefore gives the required compatibility with the prescribed source/target, as in Construction 2.2.3.

2.3 The Rezk axiom

The notions of identity morphisms and composition of morphisms in categories, provided by the Segal axiom, lead to the notion of an *isomorphism* in a category:

Definition 2.3.1 (Isomorphisms). Consider a morphism $f: x \rightarrow y$ in a category C . We say that f is *invertible*, or that it is an *isomorphism*, if there are commutative triangles in C of the form

$$\begin{array}{ccc} y & \xrightarrow{g} & x \\ & \searrow \text{id}_y & \downarrow f \\ & & y \end{array} \quad \begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow \text{id}_x & \downarrow h \\ & & x. \end{array}$$

An invertible natural transformation is called a *natural isomorphism*. The *Rezk axiom* will relate invertible morphisms to the primitive identifications of objects from Definition 1.1.2. We start by defining the category $\text{Iso}(C)$ of isomorphisms in C .

Definition 2.3.2. Given a category C , we define the category $\text{Iso}(C)$ via the following pullback diagram:

$$\begin{array}{ccccc} \text{Iso}(C) & \xrightarrow{\quad} & \text{Triang}(C) \times_{d_0, \text{Ar}(C), d_2} \text{Triang}(C) & \xrightarrow{\quad} & \text{Ar}(C) \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \Delta \\ & & \text{Triang}(C) \times \text{Triang}(C) & \xrightarrow{d_0^* \times d_2^*} & \text{Ar}(C) \times \text{Ar}(C) \\ & & \downarrow d_1^* \times d_1^* & & \\ & & C \times C & \xrightarrow{(x,y) \mapsto (\text{id}_y, \text{id}_x)} & \text{Ar}(C) \times \text{Ar}(C). \end{array}$$

Explicitly, an absolute object of $\text{Iso}(C)$ is a commutative diagram in C of the form

$$\begin{array}{ccccc}
 & & x & & \\
 & g \nearrow & & \searrow \text{id}_x & \\
 y & & & & x. \\
 & \searrow \text{id}_y & & \nearrow h & \\
 & & y & &
 \end{array}$$

We denote the composite functor at the top of the diagram by

$$\pi_{\text{Iso}}: \text{Iso}(C) \rightarrow \text{Ar}(C).$$

Notice that by definition a morphism $f \in \text{Ar}(C)$ is invertible in C if and only if it lifts to an absolute object of $\text{Iso}(C)$. We may therefore think of the fiber of π_{Iso} over f as the category of ‘proofs that f is invertible’. The following lemma shows that there is at most one such proof:

Lemma 2.3.3. *For a category C , the map $\pi_{\text{Iso}}: \text{Iso}(C) \rightarrow \text{Ar}(C)$ is an embedding.*

Proof. We use the unit and associativity comparisons over prescribed vertices established in Lemma 2.2.6 and Proposition 2.2.7. We need to show that the diagonal

$$\Delta: \text{Iso}(C) \rightarrow \text{Iso}(C) \times_{\text{Ar}(C)} \text{Iso}(C)$$

is an equivalence, or equivalently that the first projection

$$\text{pr}_1: \text{Iso}(C) \times_{\text{Ar}(C)} \text{Iso}(C) \rightarrow \text{Iso}(C)$$

is an equivalence. To this end, consider the two functors

$$\begin{aligned}
 \text{Ar}(C) \times_{\text{ev}_1, C, \text{ev}_0} \text{Iso}(C) &\rightarrow \text{Ar}(C) \times_{\text{ev}_1, C, \text{ev}_1} \text{Iso}(C), & (g, f) &\mapsto (f \circ g, f) \\
 \text{Iso}(C) \times_{\text{ev}_1, C, \text{ev}_0} \text{Ar}(C) &\rightarrow \text{Iso}(C) \times_{\text{ev}_0, C, \text{ev}_0} \text{Ar}(C), & (f, h) &\mapsto (f, h \circ f).
 \end{aligned}$$

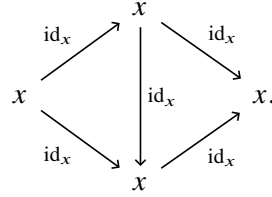
We claim that these functors are equivalences. For the first functor, note that for $f \in \text{Iso}(C)$ we may functorially extract both a left inverse f_l with an identification $f_l \circ f = \text{id}_{s(f)}$ as well as a right inverse f_r with an identification $f \circ f_r = \text{id}_{t(f)}$. Using associativity of composition, a left inverse to the first functor is then given by $(g', f) \mapsto (f_l \circ g', f)$, while a right inverse is given by $(g', f) \mapsto (f_r \circ g', f)$. One argues similarly for the second functor.

The claim now follows from the observation that the projection pr_1 is the left vertical map in the following pullback square, where the right vertical map is the fiber product over $\text{Iso}(C)$ of the two equivalences just constructed, hence an equivalence itself:

$$\begin{array}{ccc}
 \text{Iso}(C) \times_{\text{Ar}(C)} \text{Iso}(C) & \longrightarrow & \text{Ar}(C) \times_{\text{ev}_1, C, \text{ev}_0} \text{Iso}(C) \times_{\text{ev}_1, C, \text{ev}_0} \text{Ar}(C) \\
 \text{pr}_1 \downarrow & \lrcorner & \downarrow (g, f, h) \mapsto (f \circ g, f, h \circ f) \\
 \text{Iso}(C) & \xrightarrow{f \mapsto (\text{id}_y, f, \text{id}_x)} & \text{Ar}(C) \times_{\text{ev}_1, C, \text{ev}_1} \text{Iso}(C) \times_{\text{ev}_0, C, \text{ev}_0} \text{Ar}(C).
 \end{array}
 \quad \square$$

As one expects, the identity morphism of any term $\Gamma \vdash x : C$ is invertible in the same context:

Construction 2.3.4. We construct a functor $i: C \rightarrow \text{Iso}(C)$ lifting the functor $1: C \rightarrow \text{Ar}(C)$ given by $x \mapsto \text{id}_x$. Informally, i sends x to the following commutative diagram in C :



More formally, restriction along the map $p_{[2]}: [2] \rightarrow [0]$ induces a functor $p_{[2]}^*: C \rightarrow \text{Triang}(C)$, giving a functor $(p_{[2]}^*, p_{[2]}^*): C \rightarrow \text{Triang}(C) \times_{\text{Ar}(C)} \text{Triang}(C)$. Since the two relevant edges of this diagram are both the identity on x , this functor factors through $\text{Iso}(C)$, providing the desired functor $i: C \rightarrow \text{Iso}(C)$.

By construction the following diagram commutes:

$$\begin{array}{ccccc}
 C & \xrightarrow{i} & \text{Iso}(C) & \xrightarrow{(\text{ev}_0, \text{ev}_1)} & C \times C \\
 & \searrow x \mapsto \text{id}_x & \downarrow \pi_{\text{Iso}} & \nearrow (\text{ev}_0, \text{ev}_1) & \\
 & & \text{Ar}(C) & &
 \end{array}$$

Axiom F (Rezk axiom). For any category C , the functor $i: C \rightarrow \text{Iso}(C)$ is an equivalence.

Together with representability of identifications, the Rezk axiom identifies the anima of identifications between two objects with the category of isomorphisms between them:

Corollary 2.3.5. *For every category C and absolute objects x, y of C , there is an equivalence*

$$(x = y) \simeq * \times_{C \times C} \text{Iso}(C),$$

where the maps defining the fiber product are $(x, y): * \rightarrow C \times C$ and $(\text{ev}_0, \text{ev}_1): \text{Iso}(C) \rightarrow C \times C$.

Proof. By Axiom B.7, the anima $(x = y)$ is the pullback of $x, y: * \rightarrow C$. Applying Lemma 1.6.13, we obtain a pullback square

$$\begin{array}{ccc}
 (x = y) & \xrightarrow{\quad} & C \\
 \downarrow & \lrcorner & \downarrow \Delta \\
 * & \xrightarrow{(x, y)} & C \times C.
 \end{array}$$

The Rezk axiom identifies C with $\text{Iso}(C)$ through the equivalence i , and $(\text{ev}_0, \text{ev}_1)i = \Delta$. Replacing C by $\text{Iso}(C)$ therefore identifies this pullback with the required fiber. $\square$

Corollary 2.3.6. *For functors $f, g: C \rightarrow D$, there is an equivalence*

$$(f = g) \simeq * \times_{\text{Fun}(C, D) \times \text{Fun}(C, D)} \text{Iso}(\text{Fun}(C, D)),$$

where the maps defining the fiber product are (f, g) and $(\text{ev}_0, \text{ev}_1)$. Thus identifications of functors are equivalently natural isomorphisms of functors.

Proof. By Definition 1.7.3, currying identifies the anima ($f = g$) with the anima of identifications between the corresponding absolute objects of $\text{Fun}(C, D)$. Applying Corollary 2.3.5 to these objects gives the asserted equivalence. $\square$

Corollary 2.3.7. *For a category C , the functor $p_{[1]}^*: C \rightarrow \text{Ar}(C)$ is an embedding.*

Proof. This functor factors as $C \xrightarrow{i} \text{Iso}(C) \xrightarrow{\pi_{\text{Iso}}} \text{Ar}(C)$. The first functor is an equivalence by the Rezk axiom, while the second one is an embedding by Lemma 2.3.3. The claim follows. $\square$

2.4 Groupoids

A category is called a *groupoid* if all its morphisms are invertible. In the synthetic setting, we need to phrase this condition for arbitrary terms of the arrow category, as follows.

Definition 2.4.1 (Groupoid). We say that a category C is a groupoid if in an arbitrary categorical context, every morphism expression $f: x \rightarrow y$ in C is an isomorphism. Explicitly: for every category Γ and every functor $f: \Gamma \rightarrow \text{Ar}(C)$, there exists a dashed arrow $\tilde{f}$ making the following diagram commute:

$$\begin{array}{ccc} & \text{Iso}(C) & \\ \tilde{f} \nearrow & \downarrow \pi_{\text{Iso}} & \\ \Gamma & \xrightarrow{f} & \text{Ar}(C). \end{array}$$

The following result provides various alternative formulations for a category to be a groupoid:

Proposition 2.4.2. *Let X be a category. Then the following conditions are equivalent:*

- (1) *The category X is a groupoid;*
- (2) *The functor $\pi_{\text{Iso}}: \text{Iso}(X) \rightarrow \text{Ar}(X)$ admits a section, in the sense of Definition 1.3.2;*
- (3) *The functor $p_{[1]}^*: X \rightarrow \text{Ar}(X)$, induced by the map $p_{[1]}: [1] \rightarrow *$, admits a section;*
- (4) *The functor $p_{[1]}^*: X \rightarrow \text{Ar}(X)$ is an equivalence.*

Proof. To see that (1) implies (2), take $\Gamma = \text{Ar}(X)$ and let $f = \text{id}_{\text{Ar}(X)}$, so that $s = \tilde{f}$ provides a section. Conversely, if $s: \text{Ar}(X) \rightarrow \text{Iso}(X)$ is a section of π_{Iso} , a lift $\tilde{f}$ of f may be given as $\tilde{f} := s \circ f$.

The fact that (2) and (3) are equivalent is immediate from the following commutative diagram:

$$\begin{array}{ccc} & X & \\ i \swarrow & & \searrow (p_{[1]})^* \\ \text{Iso}(X) & \xrightarrow{\pi_{\text{Iso}}} & \text{Ar}(X), \end{array}$$

using that the map i is an equivalence by Axiom F. Finally, note that the functor $(p_{[1]})^*$ always admits a retraction given by $\text{ev}_0: \text{Ar}(X) \rightarrow X$, and thus the equivalence between (3) and (4) is an instance of Lemma 1.3.3. $\square$

Convention 2.4.3. From now on we will mostly take (4) as our working definition of groupoids.

Example 2.4.4. The terminal category $*$ is a groupoid: there is a commutative triangle

$$\begin{array}{ccc} & \text{Ar}(*) & \\ (p_{[1]})^* \nearrow & & \searrow \\ * & \xrightarrow{\quad} & *, \end{array}$$

and since the bottom and right diagonal maps are equivalences (see Lemma 1.7.9) the claim follows by 2-out-of-3.

Example 2.4.5. If X and Y are groupoids, then their product $X \times Y$ is a groupoid. This is an immediate consequence of Lemma 1.7.10.

Lemma 2.4.6. *Let $C \rightarrow D$ be an equivalence of categories. Then C is a groupoid if and only if D is a groupoid.*

Proof. This follows from the following commutative diagram:

$$\begin{array}{ccc} C & \xrightarrow{\sim} & D \\ (p_{[1]})^* \downarrow & & \downarrow (p_{[1]})^* \\ \text{Ar}(C) & \xrightarrow{\sim} & \text{Ar}(D). \end{array}$$

Here the bottom functor is again an equivalence by the functoriality of functor categories, Construction 1.7.7. $\square$

Lemma 2.4.7. *Let X be a groupoid. Then for every category C , the functor category $\text{Fun}(C, X)$ is a groupoid.*

Proof. This follows from the fact that the map $\text{Fun}(C, X) \rightarrow \text{Ar}(\text{Fun}(C, X))$ is equivalent to the map $\text{Fun}(C, X) \rightarrow \text{Fun}(C, \text{Ar}(X))$ induced by the equivalence $X \xrightarrow{\sim} \text{Ar}(X)$. $\square$

Lemma 2.4.8. *Groupoids are closed under retracts: if X is a groupoid and Y is a retract of X , then Y is also a groupoid.*

Proof. If Y is a retract of X , then the functor $Y \rightarrow \text{Ar}(Y)$ is a retract of the functor $X \rightarrow \text{Ar}(X)$. In particular, if the latter is an equivalence then so is the former by Lemma 1.3.9. $\square$

Lemma 2.4.9. *Consider functors $f: X \rightarrow C$ and $g: Y \rightarrow C$ and assume that X and Y are groupoids. Then the pullback $X \times_C Y$ is a groupoid.*

Proof. The functor $X \times_C Y \rightarrow \text{Ar}(X \times_C Y)$ factors as

$$X \times_C Y \xrightarrow{\sim} X \times_{\text{Ar}(C)} Y \xrightarrow{\sim} \text{Ar}(X) \times_{\text{Ar}(C)} \text{Ar}(Y) \simeq \text{Ar}(X \times_C Y).$$

Here the second functor is an equivalence since X and Y are groupoids. The last equivalence is Lemma 1.7.11. Finally, we claim that the first functor is a pullback of the diagonal of the embedding

$i: C \rightarrow \text{Ar}(C)$ from Corollary 2.3.7 and hence is an equivalence. To see this, consider the following commutative diagram:

$$\begin{array}{ccccc}
 X \times_C Y & \longrightarrow & X \times_{\text{Ar}(C)} Y & \longrightarrow & X \times Y \\
 \downarrow & & \downarrow & & \downarrow \\
 C & \xrightarrow[\sim]{\Delta_i} & C \times_{\text{Ar}(C)} C & \longrightarrow & C \times C \\
 & & \downarrow & \lrcorner & \downarrow i \times i \\
 & & \text{Ar}(C) & \xrightarrow{\Delta} & \text{Ar}(C) \times \text{Ar}(C).
 \end{array}$$

The right bottom square, the right outer rectangle and the top outer rectangle are pullback squares by Lemma 1.6.13. It thus follows from Lemma 1.6.11 that also the top left square is a pullback square. $\square$

Lemma 2.4.10. *Consider a pullback square*

$$\begin{array}{ccc}
 X' & \longrightarrow & X \\
 f' \downarrow & \lrcorner & \downarrow f \\
 Y' & \longrightarrow & Y
 \end{array}$$

of categories. If X, Y and Y' are groupoids, then so is X' .

Proof. This is a special case of Lemma 2.4.9. $\square$

Lemma 2.4.11. *The empty category $\emptyset$ is a groupoid.*

Proof. We need to show that the functor $\emptyset \rightarrow \text{Ar}(\emptyset)$ is an equivalence, or equivalently that the map $\text{ev}_0: \text{Ar}(\emptyset) \rightarrow \emptyset$ is an equivalence. But this is immediate from Axiom B.4'. $\square$

2.5 Recognizing animae

Animae enter the foundational language primitively as homotopy-coherent collections, whereas the groupoid condition is a property of a category expressed by invertibility of its morphisms. The two notions enter the theory independently. The following axiom asserts that they determine the same class of categories, allowing us to pass between collection-oriented anima language and structural groupoid arguments.

Axiom G (Groupoid recognition axiom). A category is an anima if and only if it is a groupoid.

Proposition 2.5.1. *For any anima X and any category A , the canonical functor*

$$\iota_{A,X}: \text{Map}(A, X) \xrightarrow{\sim} \text{Fun}(A, X)$$

is an equivalence.

Proof. By the groupoid recognition axiom and Lemma 2.4.7, the category $\text{Fun}(A, X)$ is an anima. The claim follows from Propositions 1.4.15 and 1.7.16. $\square$

Lemma 2.5.2. *For functors $X \rightarrow C$ and $Y \rightarrow C$ with X and Y animae, the pullback $X \times_C Y$ is an anima.*

Proof. This follows from Lemma 2.4.9 and the groupoid recognition axiom. $\square$

Corollary 2.5.3. *Let C be a category. Then the functor $\iota_C : C^\simeq \rightarrow C$ is an embedding.*

Proof. We need to show that the diagonal $C^\simeq \rightarrow C^\simeq \times_C C^\simeq$ is an equivalence. Its source is an anima, and its target is an anima by Lemma 2.5.2. By Proposition 1.4.11, it therefore suffices to show that for every anima Z the induced map

$$\text{Map}(Z, C^\simeq) \rightarrow \text{Map}(Z, C^\simeq \times_C C^\simeq) \simeq \text{Map}(Z, C^\simeq) \times_{\text{Map}(Z, C)} \text{Map}(Z, C^\simeq)$$

is an equivalence, i.e. we need to show that the map $\text{Map}(Z, C^\simeq) \rightarrow \text{Map}(Z, C)$ is an embedding. But this map is even an equivalence by Corollary 1.4.13. $\square$

Lemma 2.5.4. *For any two categories C and D , the canonical map*

$$\text{Ar}(C) \sqcup \text{Ar}(D) \rightarrow \text{Ar}(C \sqcup D)$$

is an equivalence.

Proof. When $C = * = D$, we have $\text{Ar}(*) \simeq *$ and the statement says that the map $* \sqcup * \rightarrow \text{Ar}(* \sqcup *)$ is an equivalence. Since $* \sqcup *$ is an anima, it is a groupoid by the recognition axiom. The claim follows, since for a groupoid X the functor $X \rightarrow \text{Ar}(X)$ is an equivalence. For arbitrary C and D , it follows from Lemma 1.7.11 that there are pullback squares

$$\begin{array}{ccccc} \text{Ar}(C) & \longrightarrow & \text{Ar}(C \sqcup D) & \longleftarrow & \text{Ar}(D) \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ \text{Ar}(*) & \longrightarrow & \text{Ar}(* \sqcup *) & \longleftarrow & \text{Ar}(*) \end{array}$$

The claim thus follows from Lemma 1.6.16. $\square$

Corollary 2.5.5. *Let X and Y be groupoids. Then their disjoint union $X \sqcup Y$ is a groupoid.*

Proof. The map $X \sqcup Y \rightarrow \text{Ar}(X \sqcup Y)$ factors as

$$X \sqcup Y \rightarrow \text{Ar}(X) \sqcup \text{Ar}(Y) \rightarrow \text{Ar}(X \sqcup Y),$$

where the first map is the disjoint union of the two equivalences $X \xrightarrow{\sim} \text{Ar}(X)$ and $Y \xrightarrow{\sim} \text{Ar}(Y)$, while the second is the equivalence from Lemma 2.5.4. $\square$

Corollary 2.5.6. *The empty category $\emptyset$ is an anima. For animae X and Y , the coproduct $X \sqcup Y$ is an anima.*

Proof. This follows from Lemma 2.4.11 and Corollary 2.5.5 and the groupoid recognition axiom. $\square$

2.6 Exercises

Exercise 2.6.1. Let C_0 and C_1 be categories and consider morphisms $f_0: x_0 \rightarrow y_0$ and $g_0: y_0 \rightarrow z_0$ in C_0 and morphisms $f_1: x_1 \rightarrow y_1$ and $g_1: y_1 \rightarrow z_1$ in C_1 . Show that the composite $(g_0, g_1) \circ (f_0, f_1): (x_0, x_1) \rightarrow (z_0, z_1)$ in $C_0 \times C_1$ is given by $(g_0 \circ f_0, g_1 \circ f_1)$.

Exercise 2.6.2. Let $f: x \rightarrow y$ be a morphism in a category C . Prove that the following assertions are equivalent:

- (i) f is an isomorphism;
- (ii) for any object c of C , f induces an equivalence of the form

$$f \circ -: C(c, x) \xrightarrow{\sim} C(c, y);$$

- (iii) for any object c of C , f induces an equivalence of the form

$$- \circ f: C(y, c) \xrightarrow{\sim} C(x, c).$$

3 Constructions of categories

In this chapter, we introduce various fundamental categorical constructions in the setting of synthetic category theory: subcategories and full subcategories, localizations, geometric realizations, exponentiable functors, joins, and slices.

3.1 Subcategories

In ordinary category theory, if one is given a collection M of morphisms in a category C which is closed under composition, then one may form the subcategory of C whose morphisms are precisely the morphisms in M (and the objects in the subcategory are precisely the objects whose identity morphism is in M). In this section, we discuss the analogous situation in the context of synthetic category theory.

We start by defining the notion of a subcategory. For this, we need the following auxiliary construction:

Construction 3.1.1. For categories C , D and E , we construct a map

$$\mathrm{Map}(C, D) \rightarrow \mathrm{Map}(\mathrm{Map}(E, C), \mathrm{Map}(E, D))$$

by currying the internal composition functor from Proposition 1.4.8:

$$\mu_{E,C,D}: \mathrm{Map}(C, D) \times \mathrm{Map}(E, C) \longrightarrow \mathrm{Map}(E, D).$$

For an absolute functor $f: C \rightarrow D$, the specialization of the resulting map at f is identified with the postcomposition functor $f_*: \mathrm{Map}(E, C) \rightarrow \mathrm{Map}(E, D)$ constructed in Construction 1.4.3. This follows by uncurrying and comparing the two double-evaluation functors. Here we use the convention from Construction 1.4.1 of passing back and forth between functors and their representing points in the mapping anima.

Definition 3.1.2 (Subcategory). A functor $f: A \rightarrow C$ is called a *subcategory* if the following two conditions are satisfied:

- (1) The map $f_*: \mathrm{Map}([1], A) \rightarrow \mathrm{Map}([1], C)$ is an embedding;
- (2) For every category D , the induced square

$$\begin{array}{ccc} \mathrm{Map}(D, A) & \xrightarrow{f_*} & \mathrm{Map}(D, C) \\ \downarrow & & \downarrow \\ \mathrm{Map}(\mathrm{Map}([1], D), \mathrm{Map}([1], A)) & \xrightarrow{(f_*)_*} & \mathrm{Map}(\mathrm{Map}([1], D), \mathrm{Map}([1], C)) \end{array}$$

is a pullback square. Here the two vertical maps are given by Construction 3.1.1.

In other words, if $f: A \rightarrow C$ is a subcategory, then a given functor $D \rightarrow C$ factors through A if and only if the associated map on morphisms $\text{Map}([1], D) \rightarrow \text{Map}([1], C)$ factors through $\text{Map}([1], A)$.

Remark 3.1.3. The above notion of subcategory corresponds to what is known as a *replete subcategory* in the higher category theory literature. There is no sensible notion of non-replete subcategories in the synthetic setting.

Example 3.1.4. The identity functor $\text{id}_C: C \rightarrow C$ and the empty category $\emptyset \rightarrow C$ are both seen to be subcategories.

Lemma 3.1.5. *Every subcategory $f: A \rightarrow C$ is an embedding.*

Proof. By Proposition 1.6.20, it will suffice to show that the induced map $\text{Map}(D, A) \rightarrow \text{Map}(D, C)$ is an embedding for every category D . But this follows from its defining property, as it is the base change of the map $(f_*)^*$ induced by the map $f_*: \text{Map}([1], A) \rightarrow \text{Map}([1], C)$ which was assumed to be an embedding. $\square$

We would now like to construct subcategories of C out of the data of a collection of morphisms in C closed under composition. Let us start by defining this notion.

Definition 3.1.6 (Collection of morphisms). Let C be a category. A *collection of morphisms in C* is an anima M equipped with an embedding

$$m: M \hookrightarrow \text{Map}([1], C).$$

We say that M is *closed under composition* if the following two conditions are satisfied:

- If a morphism $f: x \rightarrow y$ in C is in M , then so are id_x and id_y . More concretely, there are commutative diagrams of the following form:

$$\begin{array}{ccccc} M & \xleftarrow{\quad\quad\quad} & M & \xrightarrow{\quad\quad\quad} & M \\ m \downarrow & & \downarrow m & & \downarrow m \\ \text{Map}([1], C) & \xleftarrow{f \mapsto \text{id}_x} & \text{Map}([1], C) & \xrightarrow{f \mapsto \text{id}_y} & \text{Map}([1], C). \end{array}$$

- For maps $f: x \rightarrow y$ and $g: y \rightarrow z$ in M , also the composition of f and g is in M . More concretely, there exists a dotted arrow in the following diagram:

$$\begin{array}{ccc} M \times_{\text{ev}_1, C^\simeq, \text{ev}_0} M & \xrightarrow{\quad\quad\quad} & M \\ \downarrow & & \downarrow m \\ \text{Map}([1], C) \times_{\text{ev}_1, C^\simeq, \text{ev}_0} \text{Map}([1], C) & \xleftarrow{\sim} \text{Map}([2], C) \xrightarrow{d_1^*} & \text{Map}([1], C). \end{array}$$

Convention 3.1.7. Let $m: M \hookrightarrow \text{Map}([1], C)$ be a collection of morphisms in C . We say that a morphism $f: x \rightarrow y$ in C is *contained in M* , or simply *is in M* , if the corresponding absolute object of $\text{Map}([1], C)$ factors through m . Since m is an embedding, such a factorization, if it exists, is unique, so being contained in M is a property. Thus, in phrases such as “a morphism in M ”, the word “morphism” refers to a morphism in C contained in the subanima $M \hookrightarrow \text{Map}([1], C)$, not to a morphism of the anima M .

Example 3.1.8. Taking $M = \text{Map}([1], C)$, it is clear that the identity map $M \rightarrow \text{Map}([1], C)$ is a collection of morphisms in C closed under composition.

Example 3.1.9. Let $A \rightarrow C$ be a subcategory. Then the map $M_A := \text{Map}([1], A) \rightarrow \text{Map}([1], C)$ is a collection of morphisms in C closed under composition. This is immediate from the fact that the functor $A \rightarrow C$ preserves compositions and identities.

Also the collection of isomorphisms in C determines a collection of morphisms closed under composition:

Lemma 3.1.10. *For a category C , the map $\pi_{\text{Iso}}^\sim : \text{Iso}(C)^\sim \rightarrow \text{Map}([1], C)$ defines a collection of morphisms in C closed under composition.*

Proof. We saw in Lemma 2.3.3 that the functor π_{Iso} is an embedding, hence $\pi_{\text{Iso}}^\sim$ is an embedding by Corollary 1.6.21. The identity-isomorphism construction shows that the source and target identities of every isomorphism again lie in $\text{Iso}(C)^\sim$. Given composable isomorphisms f and g , choose their functorially specified left and right inverses. Associativity and unitality exhibit $f^{-1} \circ g^{-1}$ as a two-sided inverse of $g \circ f$. This construction is functorial in (f, g) and supplies the dotted composition map in Definition 3.1.6. Thus $\text{Iso}(C)^\sim$ is closed under composition. $\square$

We may now state the subcategory axiom:

Axiom H (Subcategory axiom). Let C be a category and let $M \hookrightarrow \text{Map}([1], C)$ be a collection of morphisms in C closed under composition. Then there exists a category $\langle M \rangle_C$ equipped with a functor $i_M : \langle M \rangle_C \rightarrow C$ satisfying the following two conditions:

- (1) The functor i_M sends morphisms in $\langle M \rangle_C$ into M : the first projection pr_1 in the following pullback square is an equivalence:

$$\begin{array}{ccc} \text{Map}([1], \langle M \rangle_C) \times_{\text{Map}([1], C)} M & \xrightarrow{\text{pr}_2} & M \\ \text{pr}_1 \downarrow \simeq & \lrcorner & \downarrow m \\ \text{Map}([1], \langle M \rangle_C) & \xrightarrow{(i_M)_*} & \text{Map}([1], C). \end{array}$$

Equivalently: the map $(i_M)_* : \text{Map}([1], \langle M \rangle_C) \rightarrow \text{Map}([1], C)$ factors through the map $m : M \rightarrow \text{Map}([1], C)$.

- (2) The functor i_M is universal with this property: for any other category D , the following commutative square is a pullback square:

$$\begin{array}{ccc} \text{Map}(D, \langle M \rangle_C) & \xrightarrow{(i_M)_*} & \text{Map}(D, C) \\ \downarrow & \lrcorner & \downarrow \\ \text{Map}(\text{Map}([1], D), M) & \xleftarrow{m_*} & \text{Map}(\text{Map}([1], D), \text{Map}([1], C)). \end{array}$$

Here the vertical maps use Construction 3.1.1 and the map from (1).

Corollary 3.1.11. *In the situation of Axiom H, the functor $\text{Map}([1], \langle M \rangle_C) \rightarrow M$ is an equivalence.*

Proof. Applying the universal property of $\langle M \rangle_C$ to $D = [1]$, and using the equivalence $\text{Map}([1], [1]) \simeq \{\text{id}_0, \text{id}_1, \text{can}\}$ from Remark 2.1.29, we obtain a pullback square

$$\begin{array}{ccc} \text{Map}([1], \langle M \rangle_C) & \longrightarrow & \text{Map}([1], C) \\ \downarrow & \lrcorner & \downarrow \begin{smallmatrix} (f: x \rightarrow y) \\ \downarrow \\ (\text{id}_x, \text{id}_y, f) \end{smallmatrix} \\ M^3 & \hookrightarrow & \text{Map}([1], C)^3. \end{array}$$

The claim now follows from the fact that the map $M \hookrightarrow \text{Map}([1], C)$ is an embedding and that M is closed under composition, so that f lies in M if and only if each of $\text{id}_x, \text{id}_y, f$ lie in M . $\square$

Corollary 3.1.12. *In the situation of Axiom H, the functor $i_M: \langle M \rangle_C \rightarrow C$ exhibits $\langle M \rangle_C$ as a subcategory of C . In particular, i_M is an embedding.*

Proof. By Corollary 3.1.11, the map $\text{Map}([1], \langle M \rangle_C) \rightarrow \text{Map}([1], C)$ agrees with the embedding $M \hookrightarrow \text{Map}([1], C)$, giving condition (1) of Definition 3.1.2. Condition (2) then follows immediately from the universal property of $\langle M \rangle_C$. $\square$

Lemma 3.1.13. *Let $f: A \rightarrow C$ be a subcategory and let $M_A := \text{Map}([1], A) \hookrightarrow \text{Map}([1], C)$ denote the induced collection of morphisms in C . Then the map $i_{M_A}: \langle M_A \rangle_C \rightarrow C$ factors through an equivalence $\langle M_A \rangle_C \xrightarrow{\sim} A$.*

Proof. Since the functor $\langle M_A \rangle_C \rightarrow C$ induces the inclusion $\text{Map}([1], A) \hookrightarrow \text{Map}([1], C)$, the fact that A is a subcategory of C implies that this functor factors through A . It further follows that the induced map $\text{Map}(D, \langle M_A \rangle_C) \rightarrow \text{Map}(D, A)$ is an equivalence for every category D , since both sides sit in the same pullback square. The claim thus follows from Proposition 1.4.11. $\square$

In particular, this shows:

Corollary 3.1.14. *When $M = \text{Map}([1], C)$ is the collection of all morphisms, the map $i_M: \langle M \rangle_C \rightarrow C$ is an equivalence.* $\square$

We will now explain the functoriality of the construction of the subcategory $\langle M \rangle_C$ in the pair (C, M) .

Definition 3.1.15. Let C and D be categories, and assume that they come equipped with collections of morphisms $m: M \hookrightarrow \text{Map}([1], C)$ and $n: N \hookrightarrow \text{Map}([1], D)$. A functor $f: C \rightarrow D$ is said to *send morphisms in M to morphisms in N* if there exists a map $\tilde{f}: M \rightarrow N$ making the following diagram commute:

$$\begin{array}{ccc} M & \xrightarrow{\tilde{f}} & N \\ m \downarrow & & \downarrow n \\ \text{Map}([1], C) & \xrightarrow{f_*} & \text{Map}([1], D). \end{array}$$

Since the map n is an embedding, the map $\tilde{f}: M \rightarrow N$ is unique if it exists.

Construction 3.1.16. Let C and D be categories equipped with collections of morphisms M and N closed under composition, and let $f: C \rightarrow D$ be a functor which sends morphisms in M to morphisms in N . We construct a functor $f|: \langle M \rangle_C \rightarrow \langle N \rangle_D$ making the following diagram commute:

$$\begin{array}{ccc} \langle M \rangle_C & \xrightarrow{f|} & \langle N \rangle_D \\ i_M \downarrow & & \downarrow i_N \\ C & \xrightarrow{f} & D. \end{array}$$

By the defining property of $\langle N \rangle_D$, it suffices to show that the map $f \circ i_M: \langle M \rangle_C \rightarrow D$ sends all morphisms to morphisms in N . But this follows from the following commutative diagram:

$$\begin{array}{ccccc} \text{Map}([1], \langle M \rangle_C) & \xrightarrow{\cong} & M & \xrightarrow{\tilde{f}} & N \\ & \searrow (i_M)_* & \downarrow m & & \downarrow n \\ & & \text{Map}([1], C) & \xrightarrow{f_*} & \text{Map}([1], D). \end{array}$$

Remark 3.1.17 (Exercise 3.8.3). The construction $f \mapsto f|$ satisfies the following properties:

- (1) The map $f|$ is uniquely determined by f .
- (2) Given an identification $f = f'$, we obtain an induced identification $f| = f'|$.
- (3) The construction is compatible with identities and composition through identifications

$$(\text{id}_C)| = \text{id}_{\langle M \rangle_C} \quad \text{and} \quad (g \circ f)| = g| \circ f|.$$

- (4) If the maps f and $\tilde{f}$ are equivalences, then also $f|: \langle M \rangle_C \rightarrow \langle N \rangle_D$ is an equivalence.

3.2 Full subcategories

As a consequence of the subcategory axiom, we may in particular form *full* subcategories.

Construction 3.2.1. Taking $E = *$ in Construction 3.1.1, and using $C^\simeq = \text{Map}(*, C)$ and $D^\simeq = \text{Map}(*, D)$, gives a map

$$\text{Map}(C, D) \longrightarrow \text{Map}(C^\simeq, D^\simeq).$$

Its specialization at $f: C \rightarrow D$ is the functor $f^\simeq: C^\simeq \rightarrow D^\simeq$ from Construction 1.4.16.

Definition 3.2.2 (Full subcategory). A functor $f: A \rightarrow C$ is called a *full subcategory* if the induced map $f^\simeq: A^\simeq \rightarrow C^\simeq$ is an embedding and for every category D , the commutative square

$$\begin{array}{ccc} \text{Map}(D, A) & \xrightarrow{f_*} & \text{Map}(D, C) \\ \downarrow & & \downarrow \\ \text{Map}(D^\simeq, A^\simeq) & \xrightarrow{(f^\simeq)_*} & \text{Map}(D^\simeq, C^\simeq) \end{array}$$

is a pullback square.

More informally, this says that a functor $D \rightarrow C$ factors through A if and only if the map $D^\simeq \rightarrow C^\simeq$ on objects factors through the embedding $A^\simeq \rightarrow C^\simeq$.

Corollary 3.2.3. *Every full subcategory is an embedding.*

Proof. For every category D , the map $\text{Map}(D, A) \rightarrow \text{Map}(D, C)$ is the base change of the embedding

$$(f^\simeq)_* : \text{Map}(D^\simeq, A^\simeq) \longrightarrow \text{Map}(D^\simeq, C^\simeq).$$

It is therefore an embedding. The claim follows from Proposition 1.6.20. $\square$

Definition 3.2.4 (Collection of objects). Consider a category C . A *collection of objects* is a pair (Γ, j) consisting of an anima Γ equipped with an embedding $j : \Gamma \hookrightarrow C^\simeq$. In this case, we define an anima M_Γ via the following pullback:

$$\begin{array}{ccc} M_\Gamma & \xhookrightarrow{m} & \text{Map}([1], C) \\ \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ \Gamma \times \Gamma & \xhookrightarrow{j \times j} & C^\simeq \times C^\simeq. \end{array}$$

The map m is an embedding, as it is a base change of the embedding $j \times j$. The resulting collection M_Γ is closed under composition: the identity of an object represented by Γ again has both endpoints in Γ , and the composite of two composable morphisms with endpoints represented by Γ has the same property. This allows us to make the following definition:

Definition 3.2.5. Given an embedding $j : \Gamma \hookrightarrow C^\simeq$, we define

$$\langle \Gamma \subseteq C \rangle := \langle M_\Gamma \rangle_C,$$

and refer to it as the *full subcategory of C spanned by Γ* .

Lemma 3.2.6. *There is a preferred equivalence $\langle \Gamma \subseteq C \rangle^\simeq \xrightarrow{\sim} \Gamma$ fitting in a commutative triangle*

$$\begin{array}{ccc} \langle \Gamma \subseteq C \rangle^\simeq & \xrightarrow{\sim} & \Gamma \\ & \searrow & \swarrow j \\ & C^\simeq. & \end{array}$$

Proof. Given a category D , consider the following two commutative squares:

$$\begin{array}{ccc} \text{Map}(D, \langle \Gamma \subseteq C \rangle) & \xrightarrow{\quad} & \text{Map}(D, C) \\ \downarrow & \lrcorner & \downarrow \\ \text{Map}(\text{Map}([1], D), M_\Gamma) & \xrightarrow{\quad} & \text{Map}(\text{Map}([1], D), \text{Map}([1], C)) \\ \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1)_* \\ \text{Map}(\text{Map}([1], D), \Gamma \times \Gamma) & \xrightarrow{\quad} & \text{Map}(\text{Map}([1], D), C^\simeq \times C^\simeq). \end{array}$$

The top square is a pullback by definition of $\langle \Gamma \subseteq C \rangle$, and the bottom square is a pullback by definition of M_Γ . Taking $D = *$ we thus obtain a pullback square of the form

$$\begin{array}{ccc} \langle \Gamma \subseteq C \rangle^\simeq & \xrightarrow{\quad} & C^\simeq \\ \downarrow & \lrcorner & \downarrow \Delta \\ \Gamma \times \Gamma & \xrightarrow{j \times j} & C^\simeq \times C^\simeq. \end{array}$$

Since j is an embedding, this implies that $\langle \Gamma \subseteq C \rangle^\simeq \simeq \Gamma$, cf. Lemma 1.6.22. It is immediate from the construction of this equivalence that it lives over $C^\simeq$. $\square$

Proposition 3.2.7. *The inclusion $\tilde{j}: \langle \Gamma \subseteq C \rangle \rightarrow C$ is a full subcategory, in the sense of Definition 3.2.2.*

Proof. We showed in Lemma 3.2.6 that the induced functor $\tilde{j}^\simeq: \langle \Gamma \subseteq C \rangle^\simeq \rightarrow C^\simeq$ agrees, under the preferred equivalence $\langle \Gamma \subseteq C \rangle^\simeq \xrightarrow{\sim} \Gamma$, with $j: \Gamma \rightarrow C^\simeq$, hence is an embedding by assumption. We thus need to show that for every category D , the top square in the following commutative diagram is a pullback square:

$$\begin{array}{ccc}
\text{Map}(D, \langle \Gamma \subseteq C \rangle) & \xrightarrow{\quad} & \text{Map}(D, C) \\
\downarrow & & \downarrow \\
\text{Map}(D^\simeq, \langle \Gamma \subseteq C \rangle^\simeq) & \xrightarrow{\quad} & \text{Map}(D^\simeq, C^\simeq) \\
\downarrow & & \downarrow \\
\text{Map}(D^\simeq \times D^\simeq, \langle \Gamma \subseteq C \rangle^\simeq \times \langle \Gamma \subseteq C \rangle^\simeq) & & \text{Map}(D^\simeq \times D^\simeq, C^\simeq \times C^\simeq) \\
\downarrow (\text{ev}_0, \text{ev}_1)^* & & \downarrow (\text{ev}_0, \text{ev}_1)^* \\
\text{Map}(\text{Map}([1], D), \langle \Gamma \subseteq C \rangle^\simeq \times \langle \Gamma \subseteq C \rangle^\simeq) & \xrightarrow{\quad} & \text{Map}(\text{Map}([1], D), C^\simeq \times C^\simeq).
\end{array}$$

The bottom two horizontal maps are embeddings since they are induced by the embedding $\langle \Gamma \subseteq C \rangle^\simeq \rightarrow C^\simeq$. The outer rectangle agrees with the outer pullback square in the proof of Lemma 3.2.6, hence is a pullback square itself. It then follows from Lemma 1.6.28 that the top square is also a pullback square. $\square$

Corollary 3.2.8. *For any category D , there is a preferred pullback*

$$\begin{array}{ccc}
\text{Map}(D, \langle \Gamma \subseteq C \rangle) & \xrightarrow{\quad} & \text{Map}(D, C) \\
\downarrow & \lrcorner & \downarrow \\
\text{Map}(D^\simeq, \Gamma) & \xrightarrow{j_*} & \text{Map}(D^\simeq, C^\simeq).
\end{array}$$

Proof. This is immediate from Proposition 3.2.7 and Lemma 3.2.6. $\square$

Lemma 3.2.9. *The inclusion $\langle C^\simeq \subseteq C \rangle \hookrightarrow C$ is an equivalence.*

Proof. Since the map $j: C^\simeq \rightarrow C^\simeq$ is the identity, it follows from Corollary 3.2.8 that for every category D the induced map

$$\text{Map}(D, \langle C^\simeq \subseteq C \rangle) \rightarrow \text{Map}(D, C)$$

is an equivalence. The claim thus follows from Proposition 1.4.11. $\square$

Lemma 3.2.10. *Let C and D be categories, and let $j: \Gamma \hookrightarrow C^\simeq$ and $k: \Lambda \hookrightarrow D^\simeq$ be embeddings. Then there is a preferred equivalence*

$$\langle \Gamma \times \Lambda \subseteq C \times D \rangle \xrightarrow{\sim} \langle \Gamma \subseteq C \rangle \times \langle \Lambda \subseteq D \rangle.$$

Proof. This is an exercise in unwinding definitions and is left to the reader: see Exercise 3.8.2. $\square$

Lemma 3.2.11. Consider a functor $f: B \rightarrow C$. Given an embedding $\Gamma \hookrightarrow C^\approx$, we define $\Lambda \hookrightarrow B^\approx$ via the pullback

$$\begin{array}{ccc} \Lambda & \hookrightarrow & B^\approx \\ \downarrow & \lrcorner & \downarrow f^\approx \\ \Gamma & \hookrightarrow & C^\approx. \end{array}$$

Then the square

$$\begin{array}{ccc} \langle \Lambda \subseteq B \rangle & \longrightarrow & B \\ \downarrow & & \downarrow f \\ \langle \Gamma \subseteq C \rangle & \longrightarrow & C \end{array}$$

is a pullback square.

Proof. By Proposition 1.6.3, it will suffice to check that this square becomes a pullback square after applying the functor $\text{Map}(D, -)$ for every category D . To this end, consider the following commutative diagram:

$$\begin{array}{ccccc} & & \text{Map}(D, B) & \xrightarrow{\quad} & \text{Map}(D^\approx, B^\approx) \\ & \nearrow & \downarrow & \nearrow & \downarrow \\ \text{Map}(D, \langle \Lambda \subseteq B \rangle) & \xrightarrow{\quad} & \text{Map}(D^\approx, \Lambda) & & \\ \downarrow & & \downarrow & & \downarrow \\ & \nearrow & \text{Map}(D, C) & \xrightarrow{\quad} & \text{Map}(D^\approx, C^\approx) \\ \text{Map}(D, \langle \Gamma \subseteq C \rangle) & \xrightarrow{\quad} & \text{Map}(D^\approx, \Gamma) & & \end{array}$$

Here the top and bottom face of the cube are pullback squares by Corollary 3.2.8, while the right face is a pullback square since $\text{Map}(D^\approx, -)$ preserves pullback squares by Proposition 1.6.3. It follows from the pasting law of pullback squares that also the left face of the cube is a pullback square, finishing the proof. $\square$

Lemma 3.2.12. Let $f: B \rightarrow C$ be a functor exhibiting B as a full subcategory of C . Then f is an equivalence if and only if the induced map $f^\approx: B^\approx \rightarrow C^\approx$ is an equivalence.

Proof. The forward implication follows by functoriality of groupoid cores. Conversely, suppose that $f^\approx$ is an equivalence. The defining pullback square for a full subcategory then shows that $\text{Map}(D, B) \rightarrow \text{Map}(D, C)$ is an equivalence for every category D . Hence f is an equivalence by Proposition 1.4.11. $\square$

3.3 Localizations

In ordinary category theory, if one is given a category C equipped with a collection of morphisms W in C , one may form the *localization* of C at W : the initial functor $C \rightarrow C[W^{-1}]$ out of C which sends all morphisms in W to isomorphisms. Our goal in this section is to define localizations in the context of synthetic category theory.

Construction 3.3.1. Let C be a category equipped with a collection of morphisms $w: W \hookrightarrow \text{Map}([1], C)$, in the sense of Definition 3.1.6. We will construct for every category D a full subcategory

$$\text{Fun}^W(C, D) \subseteq \text{Fun}(C, D)$$

In order to do this, we first define the anima $\text{Map}^W(C, D)$ via the following pullback square:

$$\begin{array}{ccc} \text{Map}^W(C, D) & \xrightarrow{\lambda_{C,D}^W} & \text{Map}(C, D) \\ \downarrow & \lrcorner & \downarrow \\ & \text{Map}(\text{Map}([1], C), \text{Map}([1], D)) & \\ & \downarrow w^* & \\ \text{Map}(W, \text{Iso}(D)^{\simeq}) & \xrightarrow{(\pi_{\text{Iso}})^*} & \text{Map}(W, \text{Map}([1], D)), \end{array}$$

where the top right vertical map is the map from Construction 3.1.1. The bottom map is induced by the embedding $\pi_{\text{Iso}}^{\simeq}$ from Lemma 2.3.3 and hence is an embedding. It follows that also $\lambda_{C,D}^W$ is an embedding. We can now define

$$\text{Fun}^W(C, D) := \langle \text{Map}^W(C, D) \subseteq \text{Fun}(C, D) \rangle.$$

In particular, the map $\lambda_{C,D}^W$ extends to a functor $\tilde{\lambda}_{C,D}^W: \text{Fun}^W(C, D) \rightarrow \text{Fun}(C, D)$. We may think of the category $\text{Fun}^W(C, D)$ as the full subcategory of $\text{Fun}(C, D)$ spanned by those functors sending the morphisms in W to isomorphisms in D .

Remark 3.3.2. The construction of $\text{Fun}^W(C, D)$ is functorial in the pair (C, W) : if C' is another category equipped with a collection of morphisms $w': W' \hookrightarrow \text{Map}([1], C')$ and $f: C \rightarrow C'$ is a functor which sends morphisms in W to morphisms in W' (in the sense of Definition 3.1.15), then for every category D there is an induced square

$$\begin{array}{ccc} \text{Map}^{W'}(C', D) & \xrightarrow{f^*} & \text{Map}^W(C, D) \\ \lambda_{C',D}^{W'} \downarrow & & \downarrow \lambda_{C,D}^W \\ \text{Map}(C', D) & \xrightarrow{f^*} & \text{Map}(C, D), \end{array}$$

which in turn induces a commutative diagram

$$\begin{array}{ccc} \text{Fun}^{W'}(C', D) & \xrightarrow{f^*} & \text{Fun}^W(C, D) \\ \tilde{\lambda}_{C',D}^{W'} \downarrow & & \downarrow \tilde{\lambda}_{C,D}^W \\ \text{Fun}(C', D) & \xrightarrow{f^*} & \text{Fun}(C, D) \end{array}$$

via Construction 3.1.16.

Before moving on to the localization axiom, we will establish some basic results that express that the categories $\text{Fun}^W(C, D)$ behave as we expect. As a first result, we show that every functor $C \rightarrow D$ lies in $\text{Fun}^W(C, D)$ whenever D is a groupoid:

Lemma 3.3.3. *Let C be a category equipped with a collection of morphisms $W \hookrightarrow \text{Map}([1], C)$, and let Γ be a groupoid. Then the functor*

$$\text{Fun}^W(C, \Gamma) \rightarrow \text{Fun}(C, \Gamma)$$

is an equivalence.

Proof. The functor $\text{Iso}(\Gamma)^\simeq \rightarrow \text{Map}([1], \Gamma)$ is an equivalence, since Γ is a groupoid. It follows from its defining property that also the map $\text{Map}^W(C, \Gamma) \rightarrow \text{Map}(C, \Gamma)$ is an equivalence, and hence both sides define the same full subcategory of $\text{Fun}(C, \Gamma)$. Since the full subcategory determined by $\text{Map}(C, \Gamma)$ is just $\text{Fun}(C, \Gamma)$ by Lemma 3.2.9, this finishes the proof. $\square$

The following lemma expresses the fact that any functor $C \rightarrow D$ takes isomorphisms in C to isomorphisms in D :

Proposition 3.3.4. *In the case of $W := \text{Iso}(C)^\simeq \hookrightarrow \text{Map}([1], C)$, the forgetful functor*

$$\text{Fun}^W(C, D) \rightarrow \text{Fun}(C, D)$$

is an equivalence.

Proof. Since the comparison map is the embedding of a full subcategory by construction, it suffices to prove that it induces an equivalence on objects. In other words, it suffices to check that the induced map

$$\text{Map}^W(C, D) \rightarrow \text{Map}(C, D)$$

is an equivalence. Since this is an embedding of animae, it suffices to check that any functor of the form $f: C \rightarrow D$ has the property that the induced map

$$f_*: \text{Map}([1], C) \rightarrow \text{Map}([1], D)$$

sends isomorphisms of C to isomorphisms of D . For this, since there is a canonical equivalence $C^\simeq \simeq \text{Iso}(C)^\simeq$ given by the rule $x \mapsto \text{id}_x$, it suffices to check that $f(\text{id}_x) = \text{id}_{f(x)}$ holds for any object x of C . This is the identity comparison for the functor f . $\square$

Construction 3.3.5. Let $l: C \rightarrow L$ be a functor which sends all morphisms in W to isomorphisms in L , in the sense of Definition 3.1.15. Then we see that for every category D the induced map $l^*: \text{Fun}(L, D) \rightarrow \text{Fun}(C, D)$ factors through $\text{Fun}^W(C, D)$, using the following composite:

$$\text{Fun}(L, D) \simeq \text{Fun}^{\text{Iso}(L)^\simeq}(L, D) \xrightarrow{l^*} \text{Fun}^W(C, D).$$

Here the map l^* is obtained via Remark 3.3.2.

Definition 3.3.6 (Localization). Given a collection of morphisms W in C , we say that a functor $l: C \rightarrow L$ is a *localization of C at W* if it sends all morphisms in W to isomorphisms in L and is universal with respect to this property: for every category D , the functor

$$l^*: \text{Fun}(L, D) \rightarrow \text{Fun}^W(C, D)$$

is an equivalence. A functor $l: C \rightarrow L$ is called a *localization functor* if it is a localization of C at some collection of morphisms W .

If a localization exists, it is unique; see Exercise 3.8.4. We will frequently denote it by $C[W^{-1}]$, and refer to the functor $l: C \rightarrow C[W^{-1}]$ as the *localization functor*.

Axiom I (Localization axiom). For every collection of morphisms $w: W \hookrightarrow \text{Map}([1], C)$ in C there exists a localization $l: C \rightarrow C[W^{-1}]$ of C at W .

Proposition 3.3.7. *Given a collection of morphisms W in a category C as well as a functor $\lambda: C \rightarrow L$ sending all morphisms in W to isomorphisms in L , assume that, for every category D , the induced map*

$$\lambda^*: \text{Map}(L, D) \rightarrow \text{Map}^W(C, D)$$

is an equivalence. Then λ exhibits L as a localization of C at W .

Proof. Let $l: C \rightarrow C[W^{-1}]$ be a localization of C at W . Then the induced map

$$l^*: \text{Map}(C[W^{-1}], D) \rightarrow \text{Map}^W(C, D)$$

is an equivalence because it is the map on groupoid cores induced by the equivalence $l^*: \text{Fun}(C[W^{-1}], D) \xrightarrow{\sim} \text{Fun}^W(C, D)$. Since l inverts W , the assumption on L implies that there exists a functor $u: L \rightarrow C[W^{-1}]$ equipped with an identification $u \circ \lambda = l$. To see that u is an equivalence, consider for every category D the commutative triangle

$$\begin{array}{ccc} \text{Map}(C[W^{-1}], D) & \xrightarrow{u^*} & \text{Map}(L, D) \\ & \searrow l^* \quad \swarrow \lambda^* & \\ & \text{Map}^W(C, D). & \end{array}$$

By Proposition 1.4.11, it suffices to show that u^* is an equivalence for every category D . This is immediate by the 2-out-of-3 property as both diagonal maps are equivalences.

Thus u is an equivalence compatible with the maps out of C , and the universal property of l transfers to λ . $\square$

Remark 3.3.8. The previous proposition and the localization axiom give a pushout square

$$\begin{array}{ccc} [1] \times W & \xrightarrow{\tilde{w}} & C \\ \text{pr}_2 \downarrow & \lrcorner & \downarrow \\ W & \longrightarrow & C[W^{-1}], \end{array}$$

where the map $\tilde{w}$ is obtained by uncurrying the composite $W \xrightarrow{w} \text{Map}([1], C) \rightarrow \text{Fun}([1], C)$; see Exercise 3.8.5.

Construction 3.3.9. The construction $C \mapsto C[W^{-1}]$ is functorial in the pair (C, W) : if D is another category equipped with a collection of morphisms W' and $f: C \rightarrow D$ is a functor which sends W

to W' , the composite $C \rightarrow D \rightarrow D[W'^{-1}]$ sends all morphisms in W to isomorphisms, and hence induces a canonical dashed functor making the square commute:

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ \downarrow \iota & & \downarrow \iota \\ C[W^{-1}] & \dashrightarrow & D[W'^{-1}]. \end{array}$$

3.4 Geometric realizations

The localization of a category C at *all* its morphisms deserves special notation and terminology:

Definition 3.4.1 (Geometric realization). In the case where $W := \text{Map}([1], C)$ consists of all morphisms in C , we write

$$\|C\| := C[W^{-1}],$$

and we denote the localization functor by $\lambda_C: C \rightarrow \|C\|$. We refer to $\|C\|$ as the *geometric realization* of C .

In other words, the geometric realization is characterized by the property that for every category D , the functor

$$\lambda_C^*: \text{Fun}(\|C\|, D) \rightarrow \text{Fun}^{\text{Map}([1], C)}(C, D)$$

is an equivalence.

By Construction 3.3.9, the assignment $C \mapsto \|C\|$ is functorial in C : every functor $f: C \rightarrow D$ induces a functor

$$\|f\|: \|C\| \rightarrow \|D\|.$$

Lemma 3.4.2. *Assume that X is a groupoid. Then the localization functor $\lambda_X: X \rightarrow \|X\|$ is an equivalence.*

Proof. Using that $\text{Iso}(X) \simeq \text{Fun}([1], X)$, we obtain by Proposition 3.3.4 equivalences

$$\text{Fun}^{\text{Map}([1], X)}(X, D) \simeq \text{Fun}^{\text{Iso}(X) \simeq}(X, D) \simeq \text{Fun}(X, D),$$

which shows that the identity map $X \rightarrow X$ exhibits X as a localization of X at all its morphisms. This gives the claim. $\square$

Lemma 3.4.3. *If C is a category and Γ is a groupoid, then precomposition with the localization functor $\lambda_C: C \rightarrow \|C\|$ defines an equivalence*

$$\lambda_C^*: \text{Fun}(\|C\|, \Gamma) \xrightarrow{\sim} \text{Fun}(C, \Gamma)$$

Proof. We may simply compose the equivalence $\text{Fun}(\|C\|, \Gamma) \xrightarrow{\sim} \text{Fun}^{\text{Map}([1], C)}(C, \Gamma)$ with the equivalence $\text{Fun}^{\text{Map}([1], C)}(C, \Gamma) \xrightarrow{\sim} \text{Fun}(C, \Gamma)$ from Lemma 3.3.3. $\square$

Since we have obtained $\|C\|$ from C by inverting all its morphisms, one may expect it to be a groupoid. While this is indeed the case, this question is somewhat more subtle than it may seem at first, since a priori the localization may create new morphisms that do not lift to C itself. This problem is closely related to a variety of other statements we may expect to hold. Let us first formulate these and prove that they are equivalent; we will see afterwards that they are in fact satisfied.

Proposition 3.4.4. *The following conditions are equivalent:*

- (1) *For every category C , the category $\|C\|$ is a groupoid;*
- (2) *The geometric realization $\|[1]\|$ of the walking morphism $[1]$ is contractible;*
- (3) *For every category C , the functor $p_{[1]}^* : C \rightarrow \text{Fun}([1], C)$ is a full subcategory of $\text{Fun}([1], C)$ (in the sense of Definition 3.2.2);*
- (4) *For every category C , the functor $\pi_{\text{Iso}} : \text{Iso}(C) \rightarrow \text{Fun}([1], C)$ is a full subcategory;*
- (5) *Every category C for which the map $C^\simeq \simeq \text{Map}(*, C) \xrightarrow{p_{[1]}^*} \text{Map}([1], C)$ is an equivalence is a groupoid;*
- (6) *For every category C , the subcategory $\langle \text{Iso}(C)^\simeq \rangle_C \subseteq C$ spanned by all the isomorphisms in C is a groupoid;*
- (7) *For every category C , the inclusion $C^\simeq \hookrightarrow C$ induces an equivalence $C^\simeq \xrightarrow{\sim} \langle \text{Iso}(C)^\simeq \rangle_C$ between the groupoid core of C and the subcategory of C spanned by all its isomorphisms.*

Proof. We start by showing that (1) implies (2). By (1), the category $\|[1]\|$ is a groupoid, so it suffices by Proposition 1.4.11 to show that for every groupoid Γ the induced map

$$\text{Map}(*, \Gamma) \rightarrow \text{Map}(\|[1]\|, \Gamma)$$

is an equivalence. By Lemma 3.4.3, this map corresponds under the canonical equivalences to the map

$$p_{[1]}^* : \Gamma \rightarrow \text{Map}([1], \Gamma).$$

But this map is an equivalence since Γ is a groupoid, finishing the proof.

For (2) $\implies$ (3), the equivalence $\|[1]\| \xrightarrow{\sim} *$ and the universal property of geometric realization show that precomposition with $p_{[1]} : [1] \rightarrow *$ induces an equivalence

$$C \xrightarrow{\sim} \text{Fun}^{\text{Map}([1], [1])}([1], C),$$

where the right-hand side is a full subcategory of $\text{Fun}([1], C)$ by construction. Thus $p_{[1]}^*$ exhibits C as that full subcategory.

The Rezk equivalence $C \xrightarrow{\sim} \text{Iso}(C)$ identifies $p_{[1]}^*$ with the composite $C \rightarrow \text{Iso}(C) \xrightarrow{\pi_{\text{Iso}}} \text{Fun}([1], C)$. Hence (3) and (4) are equivalent.

For (3) $\implies$ (5), recall that C is a groupoid if and only if the map $C \rightarrow \text{Fun}([1], C)$ is an equivalence. Since this functor is a full subcategory by (3), it follows from Lemma 3.2.12 that we may check this on objects, giving the criterion from (5).

We will now deduce (6) from (5). To show $\langle \text{Iso}(C)^\simeq \rangle_C$ is a groupoid, it suffices by (5) to show that the map $(\langle \text{Iso}(C)^\simeq \rangle_C)^\simeq \rightarrow \text{Map}([1], \langle \text{Iso}(C)^\simeq \rangle_C)$ is an equivalence. To this end, consider the following commutative diagram, where the two horizontal maps are induced by the inclusion $i: \langle \text{Iso}(C)^\simeq \rangle_C \hookrightarrow C$:

$$\begin{array}{ccc} (\langle \text{Iso}(C)^\simeq \rangle_C)^\simeq & \xrightarrow{i^\simeq} & C^\simeq \\ p_{[1]}^* \downarrow & & \downarrow p_{[1]}^* \\ \text{Map}([1], \langle \text{Iso}(C)^\simeq \rangle_C) & \xrightarrow{\simeq} \text{Iso}(C)^\simeq & \xrightarrow{\pi_{\text{Iso}}^\simeq} \text{Map}([1], C). \end{array}$$

By the 2-out-of-3 property it remains to show that the top map is an equivalence. To construct an inverse, we consider the inclusion $\iota_C: C^\simeq \rightarrow C$. It sends all morphisms to isomorphisms, so induces a functor $C^\simeq \rightarrow \langle \text{Iso}(C)^\simeq \rangle_C$. Since the source is a groupoid, it factors through a map $\varphi: C^\simeq \rightarrow (\langle \text{Iso}(C)^\simeq \rangle_C)^\simeq$. To check the relations $\varphi \circ i^\simeq = \text{id}$ and $i^\simeq \circ \varphi = \text{id}$, we may do so after composing with the embedding $\iota_C: C^\simeq \hookrightarrow C$ from Corollary 2.5.3 and the embeddings $(\langle \text{Iso}(C)^\simeq \rangle_C)^\simeq \hookrightarrow \langle \text{Iso}(C)^\simeq \rangle_C$ and $\langle \text{Iso}(C)^\simeq \rangle_C \hookrightarrow C$ (Lemma 3.1.5) after which the claim becomes a tautology. This establishes (6).

Assume (6). The functor $C^\simeq \rightarrow \langle \text{Iso}(C)^\simeq \rangle_C$ constructed above is then a functor between groupoids. Its map on groupoid cores is inverse to the equivalence $i^\simeq$ constructed in the preceding paragraph. Since the groupoid-core inclusions of its source and target are equivalences by Axiom G and Proposition 1.4.15, the functor itself is an equivalence by 2-out-of-3. This proves (7).

Finally, we show that (7) implies (1). The localization functor $\lambda_C: C \rightarrow \|C\|$ sends all morphisms of C to isomorphisms, hence it factors through the subcategory of $\|C\|$ spanned by its isomorphisms. By (7) this subcategory is equivalent to the groupoid core of $\|C\|$. The resulting functor $C \rightarrow \|C\|^\simeq$ still inverts all morphisms in C , hence induces a functor $\|C\| \rightarrow \|C\|^\simeq$. By uniqueness of the localization factorization, the composite $\|C\| \rightarrow \|C\|^\simeq \hookrightarrow \|C\|$ is identified with the identity. Thus the embedding $\|C\|^\simeq \hookrightarrow \|C\|$ admits a section and is an equivalence by Lemma 1.6.24. It follows that $\|C\|$ is a groupoid. $\square$

Proposition 3.4.5. *All of the equivalent conditions from Proposition 3.4.4 are satisfied.*

Proof. It will suffice to show that the map $\|[1]\| \rightarrow *$ is an equivalence. By Proposition 1.4.11, it will suffice to show that for any category D , the induced functor $D^\simeq \simeq \text{Map}(*, D) \rightarrow \text{Map}(\|[1]\|, D) \simeq \text{Map}^{\text{Map}([1], [1])}([1], D)$ is an equivalence. Recall from Remark 2.1.29 that the groupoid $\text{Map}([1], [1])$ is a disjoint union of three points, corresponding to the identity on 0, the identity on 1, and the canonical map from 0 to 1 in $[1]$. Spelling out the definition of $\text{Map}^{\text{Map}([1], [1])}([1], D)$, we obtain a pullback square

$$\begin{array}{ccc} \text{Map}^{\text{Map}([1], [1])}([1], D) & \longrightarrow & \text{Map}([1], D) \\ \downarrow & \lrcorner & \downarrow (f: x \rightarrow y) \\ D^{\simeq 3} & \longrightarrow & \text{Map}([1], D)^3. \end{array}$$

The bottom map is the third power of the embedding $p_{[1]}^*: D^\simeq \rightarrow \text{Map}([1], D)$. Hence the pullback consists of those morphisms $f: x \rightarrow y$ for which id_x , id_y , and f lie in the image of $p_{[1]}^*$. The first two conditions are automatic, while the third says precisely that f is an isomorphism. The Rezk equivalence $D^\simeq \xrightarrow{\sim} \text{Iso}(D)^\simeq$ therefore identifies the pullback with $D^\simeq$, compatibly with the displayed map from $D^\simeq$. This proves the claim. $\square$

Proposition 3.4.6 (Objectwise criterion for natural isomorphisms). *Let $f, g: C \rightarrow D$ be functors and let $\alpha: f \rightarrow g$ be a natural transformation. Then α is a natural isomorphism if and only if $\alpha_x: f(x) \rightarrow g(x)$ is an isomorphism for every object x of C .*

Proof. By Propositions 3.4.4 and 3.4.5, the functor $\text{Iso}(D) \rightarrow \text{Fun}([1], D)$ is a full subcategory. Unwinding, this means we have a pullback square

$$\begin{array}{ccc} \text{Map}(C, \text{Iso}(D)) & \longrightarrow & \text{Map}(C, \text{Fun}([1], D)) \\ \downarrow & \lrcorner & \downarrow \\ \text{Map}(C^\simeq, \text{Iso}(D)^\simeq) & \longrightarrow & \text{Map}(C^\simeq, \text{Map}([1], D)). \end{array}$$

The transformation α determines a term of the upper right corner, and the right-hand map restricts it to its components. If every component is invertible, applying this hypothesis to the universal object in the anima $C^\simeq$ gives a lift to the lower left corner. The pullback property then supplies a lift $C \rightarrow \text{Iso}(D)$ of the functor encoding α . It follows from Lemma 1.7.11 that $\text{Fun}(C, \text{Iso}(D)) \simeq \text{Iso}(\text{Fun}(C, D))$, so that this lift provides the inverse data for α in $\text{Fun}(C, D)$, showing that α is a natural isomorphism. Conversely, evaluation at each object sends an inverse of α to an inverse of its component. $\square$

Corollary 3.4.7. *Consider a pushout square of categories*

$$\begin{array}{ccc} C' & \longrightarrow & C \\ \downarrow & & \downarrow \\ D' & \longrightarrow & D. \end{array}$$

Then the induced square

$$\begin{array}{ccc} \|C'\| & \longrightarrow & \|C\| \\ \downarrow & & \downarrow \\ \|D'\| & \longrightarrow & \|D\| \end{array}$$

is again a pushout square.

Proof. By Definition 1.8.1, it will suffice to show that the square becomes a pullback square after applying $\text{Map}(-, E)$ for every category E . By Corollary 1.4.13 and Lemma 3.4.3, there are equivalences

$$\text{Map}(\|C\|, E) \simeq \text{Map}(\|C\|, E^\simeq) \simeq \text{Map}(C, E^\simeq),$$

and similarly for C replaced with C' , D and D' . The claim now follows from the assumption on the original square. $\square$

3.5 Exponentiable functors

For a category T , we may consider categories C equipped with a functor $f: C \rightarrow T$, which we will refer to as *categories over T* (to be discussed in more detail in Chapter 4 (outside selection)). Given

a functor $p : S \rightarrow T$, we obtain a notion of *base change along p* :

$$\begin{array}{ccc} \{\text{categories over } T\} & \rightarrow & \{\text{categories over } S\} \\ f : C \rightarrow T & \mapsto & p^*(f) : C \times_T S \rightarrow S. \end{array}$$

This assignment admits a left adjoint, given by postcomposition with p :

$$\begin{array}{ccc} \{\text{categories over } S\} & \rightarrow & \{\text{categories over } T\} \\ g : D \rightarrow S & \mapsto & p \circ g : D \rightarrow T. \end{array}$$

However, in general this assignment will not admit a *right* adjoint. Let us illustrate this with the following simple counterexample:

Example 3.5.1. Consider the functor $p = d_1 : [1] \rightarrow [2]$ given by $d_1(0) = 0$ and $d_1(1) = 2$. We will argue that pullback along p does not preserve pushouts of categories, and hence cannot admit a right adjoint. To see this, consider the following commutative square:

$$\begin{array}{ccc} [0] & \xrightarrow{d_2} & [1] \\ d_0 \downarrow & & \downarrow d_0 \\ [1] & \xrightarrow{d_2} & [2]. \end{array} \quad (3.1)$$

We see as a consequence of the Segal axiom Axiom E that this is a pushout square of categories. If we pull back this square along the functor $d_1 : [1] \rightarrow [2]$, we obtain the commutative square

$$\begin{array}{ccc} \emptyset & \longrightarrow & [0] \\ \downarrow & & \downarrow d_0 \\ [0] & \xrightarrow{d_1} & [1]. \end{array} \quad (3.2)$$

But this is no longer a pushout square of categories: if it were, then it would remain a pushout square after passing to geometric realization by Corollary 3.4.7. But since $\|[1]\| \xrightarrow{\sim} *$ by part (2) of Proposition 3.4.4, this square takes the form

$$\begin{array}{ccc} \emptyset & \longrightarrow & * \\ \downarrow & & \parallel \\ * & \xlongequal{\quad} & *, \end{array}$$

and this is only a pushout square if $* \sqcup * \xrightarrow{\sim} *$. Consequently, in every model in which $* \sqcup *$ is not equivalent to $*$, the functor d_1 is not exponentiable. This supplies the intended semantic counterexample. The non-equivalence $* \sqcup * \not\cong *$ is not a consequence of the axioms imposed so far, so we do not claim internally that d_1 fails to be exponentiable in every model.

As the previous example indicates, we cannot expect the existence of a right adjoint to p^* in full generality. There is, however, a large class of functors, the *exponentiable functors*, for which this right adjoint does exist. This class includes cartesian and cocartesian fibrations, to be introduced in Chapter 5 (outside selection).

In this section, we will give a precise definition of this class of functors. We start with some auxiliary definitions and constructions.

Definition 3.5.2. Let S be a category and let $f: C \rightarrow S$ and $g: D \rightarrow S$ be categories over S . We define the category $\text{Fun}_S(C, D)$ of *functors* $C \rightarrow D$ over S as the following pullback:

$$\begin{array}{ccc} \text{Fun}_S(C, D) & \longrightarrow & \text{Fun}(C, D) \\ \downarrow & \lrcorner & \downarrow g_* \\ * & \xrightarrow{f} & \text{Fun}(C, S). \end{array}$$

Note that objects of $\text{Fun}_S(C, D)$ are pairs (h, α) where $h: C \rightarrow D$ is a functor and α is an identification $f = g \circ h$. In other words, objects are commutative triangles of the form

$$\begin{array}{ccc} C & \xrightarrow{h} & D \\ & \searrow f & \swarrow g \\ & S. & \end{array}$$

We write $\text{Map}_S(C, D)$ for the groupoid core of $\text{Fun}_S(C, D)$, which fits into a pullback square of the form

$$\begin{array}{ccc} \text{Map}_S(C, D) & \longrightarrow & \text{Map}(C, D) \\ \downarrow & \lrcorner & \downarrow g_* \\ * & \xrightarrow{f} & \text{Map}(C, S). \end{array}$$

Construction 3.5.3. Let $p: S \rightarrow T$ be a functor. For categories C and D over T , we obtain an induced functor

$$p^*: \text{Fun}_T(C, D) \rightarrow \text{Fun}_S(C \times_T S, D \times_T S)$$

by noting that the composite $\text{Fun}_T(C, D) \rightarrow \text{Fun}(C, D) \xrightarrow{p^*} \text{Fun}(C \times_T S, D \times_T S)$ lands in the correct fiber.

Definition 3.5.4 (Dependent product). Consider a functor $p: S \rightarrow T$ and let $f: C \rightarrow S$ be a category over S . A *dependent product of C along p* is a category $g: D \rightarrow T$ over T equipped with a functor $\varepsilon: D \times_T S \rightarrow C$ over S which satisfies the property that for any other category $E \rightarrow T$ over T the composite functor

$$\text{Map}_T(E, D) \xrightarrow{p^*} \text{Map}_S(E \times_T S, D \times_T S) \xrightarrow{\varepsilon \circ -} \text{Map}_S(E \times_T S, C)$$

is an equivalence.

If a dependent product of C along p exists, it is unique; see Exercise 3.8.6. We will denote it by $p_*(f): p_*(C) \rightarrow T$.

Terminology 3.5.5. Assume that C admits a dependent product $p_*(C)$ along p . Given a functor $\varphi: D \rightarrow p_*(C)$ over T , we obtain a functor

$$\varphi^u := \varepsilon \circ p^*(\varphi): D \times_T S \rightarrow C$$

over S that we will refer to as the *relative uncurrying of φ* . Conversely, there is for every functor $\psi: D \times_T S \rightarrow C$ over S a *relatively curried* functor $\psi_c: D \rightarrow p_*(C)$, satisfying

$$(\varphi^u)_c = \varphi \quad \text{and} \quad (\psi_c)^u = \psi.$$

In this way, we may think of the equivalence in Definition 3.5.4 as a *relative* version of currying/uncurrying.

Construction 3.5.6. Consider a pullback square

$$\begin{array}{ccc} S' & \xrightarrow{h} & S \\ p' \downarrow & \lrcorner & \downarrow p \\ T' & \xrightarrow{g} & T \end{array}$$

of categories, and let $f: C \rightarrow S$ be a category over S . Assume that:

- There exists a dependent product $p_*(C) \rightarrow T$ of C along p ;
- There exists a dependent product $p'_*(C \times_S S') \rightarrow T'$ of $C \times_S S'$ along p' .

We will construct a functor

$$\mathrm{BC}_*: p_*(C) \times_T T' \rightarrow p'_*(C \times_S S')$$

over T' called the *Beck-Chevalley map*. By relative uncurrying, we may equivalently define BC_* by specifying a map $(p_*(C) \times_T T') \times_{T'} S' \rightarrow C \times_S S'$, which we take to be the map

$$(p_*(C) \times_T T') \times_{T'} S' \simeq (p_*(C) \times_T S) \times_S S' \xrightarrow{\varepsilon_{S S'}} C \times_S S'.$$

We are now ready to introduce the class of exponentiable functors.

Definition 3.5.7. Let $p: S \rightarrow T$ be a functor. We say that p is *exponentiable* if the following two conditions are satisfied:

- (1) For every pullback square

$$\begin{array}{ccc} S' & \xrightarrow{h} & S \\ p' \downarrow & \lrcorner & \downarrow p \\ T' & \xrightarrow{g} & T \end{array}$$

and every category C' over S' , there exists a dependent product $p'_*(C') \rightarrow T'$ of C' along p' .

- (2) For every pullback diagram

$$\begin{array}{ccccc} S'' & \xrightarrow{h'} & S' & \xrightarrow{h} & S \\ p'' \downarrow & \lrcorner & p' \downarrow & \lrcorner & \downarrow p \\ T'' & \xrightarrow{g'} & T' & \xrightarrow{g} & T \end{array}$$

and every $f': C' \rightarrow S'$, the Beck-Chevalley map $\mathrm{BC}_*: p'_*(C') \times_{T'} T'' \rightarrow p''_*(C' \times_{S'} S'')$ is an equivalence.

Remark 3.5.8. Note that exponentiable functors are by definition closed under base change: if $p: S \rightarrow T$ is exponentiable, then so is any base change $p': S' \rightarrow T'$ of p .

Construction 3.5.9. Let $p: S \rightarrow T$ be an exponentiable functor, and write $\varepsilon_C: p_*(C) \times_T S \rightarrow C$ and $\varepsilon_D: p_*(D) \times_T S \rightarrow D$ for the evaluation functors. A functor $\varphi: C \rightarrow D$ over S determines a composite

$$p_*(C) \times_T S \xrightarrow{\varepsilon_C} C \xrightarrow{\varphi} D$$

over S . Its relative currying is a functor $p_*(\varphi): p_*(C) \rightarrow p_*(D)$ over T . The uniqueness part of relative currying supplies identifications

$$p_*(\text{id}_C) = \text{id}_{p_*(C)} \quad \text{and} \quad p_*(\psi \circ \varphi) = p_*(\psi) \circ p_*(\varphi).$$

Thus dependent products are functorial in categories over S .

Lemma 3.5.10. *Let $p: S \rightarrow T$ be an exponentiable functor. Then the assignment $f \mapsto p_*(f)$ preserves pullbacks: given functors $C_1 \rightarrow C_3 \leftarrow C_2$ over S , there exists a preferred equivalence*

$$p_*(C_1 \times_{C_3} C_2) \xrightarrow{\sim} p_*(C_1) \times_{p_*(C_3)} p_*(C_2)$$

over T .

Proof. Functoriality of dependent products applied to the two projections constructs the displayed comparison functor. For every category E over T , the universal properties of dependent products and pullbacks give equivalences

$$\begin{aligned} \text{Map}_T(E, p_*(C_1 \times_{C_3} C_2)) &\simeq \text{Map}_S(E \times_T S, C_1 \times_{C_3} C_2) \\ &\simeq \text{Map}_S(E \times_T S, C_1) \times_{\text{Map}_S(E \times_T S, C_3)} \text{Map}_S(E \times_T S, C_2) \\ &\simeq \text{Map}_T(E, p_*(C_1)) \times_{\text{Map}_T(E, p_*(C_3))} \text{Map}_T(E, p_*(C_2)) \\ &\simeq \text{Map}_T(E, p_*(C_1) \times_{p_*(C_3)} p_*(C_2)). \end{aligned}$$

These equivalences are induced by the comparison functor and are natural in E . Taking E to be either side of the comparison constructs an inverse over T and identifications between the two composites and the respective identity functors, exactly as in the proof of Proposition 1.4.11. Hence the comparison is an equivalence. $\square$

Corollary 3.5.11. *Let $p: S \rightarrow T$ be an exponentiable functor. If $f: C \rightarrow S$ is an embedding, then also $p_*(f): p_*(C) \rightarrow T$ is an embedding.*

Proof. The identity functor of T exhibits $T \rightarrow T$ as the dependent product of $S \rightarrow S$ along p , so there is a preferred equivalence $p_*(S) \xrightarrow{\sim} T$. Since f is an embedding, its diagonal square is a pullback. Applying Lemma 3.5.10 shows that the diagonal square of $p_*(f): p_*(C) \rightarrow p_*(S) \simeq T$ is a pullback. Hence $p_*(f)$ is an embedding. $\square$

The terminology ‘exponentiable’ comes from the fact that it allows us to construct certain ‘exponential objects’ relative to a base category S :

Definition 3.5.12. Let $p: C \rightarrow S$ and $q: D \rightarrow S$ be two functors and assume that p is an exponentiable functor. We define the *relative functor category* $\underline{\text{Fun}}_S(C, D) \rightarrow S$ as

$$\begin{array}{ccc} \underline{\text{Fun}}_S(C, D) & & \\ \downarrow & & \\ S & & \end{array} \quad := \quad p_* \left(\begin{array}{ccc} C \times_S D & & \\ \downarrow p^*(q) & & \\ C & & \end{array} \right).$$

It comes equipped with an evaluation map

$$\text{ev}: \underline{\text{Fun}}_S(C, D) \times_S C \rightarrow D$$

over S satisfying the property that for every other functor $E \rightarrow S$ the composite functor

$$\mathrm{Map}_S(E, \underline{\mathrm{Fun}}_S(C, D)) \xrightarrow{- \times_S C} \mathrm{Map}_S(E \times_S C, \underline{\mathrm{Fun}}_S(C, D) \times_S C) \xrightarrow{\mathrm{ev}_0} \mathrm{Map}_S(E \times_S C, D).$$

is an equivalence. In other words, $\underline{\mathrm{Fun}}_S(C, D)$ is an ‘internal hom object’ or ‘exponential object’ for categories over S .

Remark 3.5.13. Note that, for a category T , the data of a functor $T \rightarrow \underline{\mathrm{Fun}}_S(C, D)$ is equivalent to the data of a pair (t, f) , where $t: T \rightarrow S$ is a functor and $f: T \times_S C \rightarrow T \times_S D$ is a functor over T .

Observation 3.5.14. For any functor $t: T \rightarrow S$ and categories C and D over S such that $C \rightarrow S$ is exponentiable, there is a pullback square

$$\begin{array}{ccc} \underline{\mathrm{Fun}}_T(T \times_S C, T \times_S D) & \longrightarrow & \underline{\mathrm{Fun}}_S(C, D) \\ \downarrow & \lrcorner & \downarrow \\ T & \xrightarrow{t} & S. \end{array}$$

In particular, the fiber of $\underline{\mathrm{Fun}}_S(C, D) \rightarrow S$ over an object s of S is the functor category $\mathrm{Fun}(C_s, D_s)$, where C_s and D_s are the fibers of C and D over s , respectively.

Proposition 3.5.15. *For every category S , the terminal functor $p_S: S \rightarrow *$ is exponentiable. Moreover, if $p: C \rightarrow S$ is exponentiable and $q: D \rightarrow S$ is any functor, there is a canonical equivalence*

$$\mathrm{Fun}_S(C, D) \xrightarrow{\sim} (p_S)_* \underline{\mathrm{Fun}}_S(C, D).$$

Proof. Let T be a category. The base change of p_S along $T \rightarrow *$ is the projection $\mathrm{pr}_T: T \times S \rightarrow T$. Given a category $r: E \rightarrow T \times S$, let $\delta_T: T \rightarrow \mathrm{Fun}(S, T \times S)$ be the currying of $\mathrm{id}_{T \times S}$. Define $\Pi_S(E) \rightarrow T$ by the pullback square

$$\begin{array}{ccc} \Pi_S(E) & \longrightarrow & \mathrm{Fun}(S, E) \\ \downarrow & \lrcorner & \downarrow r_* \\ T & \xrightarrow{\delta_T} & \mathrm{Fun}(S, T \times S). \end{array}$$

Uncurrying the upper horizontal functor gives an evaluation functor $\Pi_S(E) \times S \rightarrow E$ over $T \times S$. For every category $K \rightarrow T$, the pullback property and currying identify

$$\mathrm{Map}_T(K, \Pi_S(E)) \xrightarrow{\sim} \mathrm{Map}_{T \times S}(K \times S, E).$$

Thus $\Pi_S(E)$ is a dependent product of E along pr_T . If $T' \rightarrow T$ is a functor, the pullback defining $\Pi_S(E)$ base changes to the analogous pullback over T' , since functor categories preserve pullbacks. The resulting Beck–Chevalley map is therefore an equivalence. This proves that p_S is exponentiable.

Now let $p: C \rightarrow S$ and $q: D \rightarrow S$ be as in the statement. For every category K , the universal properties of the dependent product and the relative functor category give equivalences

$$\begin{aligned} \mathrm{Map}(K, (p_S)_* \underline{\mathrm{Fun}}_S(C, D)) &\simeq \mathrm{Map}_S(K \times S, \underline{\mathrm{Fun}}_S(C, D)) \\ &\simeq \mathrm{Map}_S(K \times C, D) \\ &\simeq \mathrm{Map}(K, \mathrm{Fun}_S(C, D)). \end{aligned}$$

These equivalences are induced by evaluation and are natural in K . Hence the canonical functor $\mathrm{Fun}_S(C, D) \rightarrow (p_S)_* \underline{\mathrm{Fun}}_S(C, D)$ is an equivalence by Proposition 1.4.11. $\square$

Corollary 3.5.16. *Let $p: C \rightarrow S$ be exponentiable and let $q: D \rightarrow S$ be any functor. Then there is a canonical equivalence*

$$\mathrm{Map}_S(C, D) \xrightarrow{\sim} ((p_S)_* \underline{\mathrm{Fun}}_S(C, D))^\simeq.$$

Proof. The definition of $\mathrm{Map}_S(C, D)$ and preservation of pullbacks by groupoid cores give $\mathrm{Map}_S(C, D) \xrightarrow{\sim} \mathrm{Fun}_S(C, D)^\simeq$. Now apply the preceding proposition. $\square$

3.6 Joins

A final category constructor we introduce is the *join* $C \star D$ of two categories. This construction admits two universal properties:

- There is a mapping-in universal property: the join comes equipped with a functor to $[1]$, and a map living over $[1]$ from a category T into the join is the same as a pair of maps $T_0 \rightarrow C$ and $T_1 \rightarrow D$, where the sources are the fibres of $T \rightarrow [1]$.
- There is a mapping-out universal property: the join is obtained from the product $[1] \times C \times D$ by collapsing $\{0\} \times C \times D$ onto C and $\{1\} \times C \times D$ onto D .

In order to obtain joins in arbitrary categorical contexts, we will in fact introduce versions of the join over an arbitrary base category Γ .

Using the join construction we define $C^\flat := C \star *$ and $C^\sharp := * \star C$. We may then inductively define the categories $[n]$ by $[0] = *$ and $[n+1] := [n]^\flat$.

Two universal properties

We start with the universal property of mapping out of $C \star D$:

Axiom J.1 (Join axiom). Let Γ be an anima and let $p: C \rightarrow \Gamma$ and $q: D \rightarrow \Gamma$ be functors. There is a category $C \star_\Gamma D$ over Γ and a pushout square

$$\begin{array}{ccc} (\{0\} \sqcup \{1\}) \times (C \times_\Gamma D) & \longrightarrow & [1] \times (C \times_\Gamma D) \\ \downarrow & & \downarrow \\ C \sqcup D & \xrightarrow{i_{C,D}} & C \star_\Gamma D, \end{array}$$

over Γ . Here the top map is induced by the inclusion $\{0\} \sqcup \{1\} \rightarrow [1]$, and the left vertical map is given by

$$(\{0\} \sqcup \{1\}) \times (C \times_\Gamma D) \xrightarrow{\sim} (C \times_\Gamma D) \sqcup (C \times_\Gamma D) \xrightarrow{\mathrm{pr}_C \sqcup \mathrm{pr}_D} C \sqcup D.$$

The displayed maps to $\Gamma \times [1]$ induce a height functor

$$h_{C,D}: C \star_\Gamma D \rightarrow \Gamma \times [1]$$

whose composite with the projection to Γ is the structural functor of the relative join.

For $\Gamma = *$, we write $C \star D$ for $C \star_* D$.

Remark 3.6.1. In other words, for every third category E , the commutative square

$$\begin{array}{ccc} \text{Fun}(C \star D, E) & \longrightarrow & \text{Fun}(C \times [1] \times D, E) \\ \downarrow & & \downarrow \\ \text{Fun}(C \sqcup D, E) & \longrightarrow & \text{Fun}(C \times \partial[1] \times D, E) \end{array}$$

is a pullback square.

Example 3.6.2. For $C = D = [0]$, we see that $[0] \star [0] \xrightarrow{\sim} [1]$.

Example 3.6.3. By Corollary 2.2.8, the commutative squares

$$\begin{array}{ccc} \{0\} \times [1] & \longrightarrow & [1] \times [1] \\ \downarrow & & \downarrow p_0 \\ * & \xrightarrow{0} & [2] \end{array} \quad \text{and} \quad \begin{array}{ccc} [1] \times \{0\} & \longrightarrow & [1] \times [1] \\ \downarrow & & \downarrow p_2 \\ * & \xrightarrow{2} & [2] \end{array}$$

are pushout squares. It follows that there are equivalences

$$[0] \star [1] \xrightarrow{\sim} [2] \xleftarrow{\sim} [1] \star [0].$$

Notation 3.6.4. For any functor $g: C \star D \rightarrow E$, we denote by

$$g_C: C \rightarrow E \quad \text{and} \quad g_D: D \rightarrow E$$

the composites of g with the inclusions $C \rightarrow C \star D$ and $D \rightarrow C \star D$, respectively.

Construction 3.6.5 (Functoriality of joins). Consider functors $f: C \rightarrow C'$ and $g: D \rightarrow D'$. Then the functors $f \times \text{id}_{[1]} \times g: C \times [1] \times D \rightarrow C' \times [1] \times D'$ and $f \sqcup g: C \sqcup D \rightarrow C' \sqcup D'$ induce a functor $f \star g: C \star D \rightarrow C' \star D'$. The universal property supplies identifications

$$\text{id}_C \star \text{id}_D = \text{id}_{C \star D} \quad \text{and} \quad (f' \star g') \circ (f \star g) = (f' \circ f) \star (g' \circ g).$$

Example 3.6.6. Taking $f = p_C: C \rightarrow [0]$ and $g = p_D: D \rightarrow [0]$, we obtain a map $p_C \star p_D: C \star D \rightarrow [0] \star [0] = [1]$ fitting in a commutative diagram as follows:

$$\begin{array}{ccc} C \sqcup D & \xrightarrow{i_{C,D}} & C \star D \\ \downarrow f \sqcup g & & \downarrow p_C \star p_D \\ \partial[1] = [0] \sqcup [0] & \hookrightarrow & [1]. \end{array}$$

We now come to the second universal property of $C \star D$:

Axiom J.2 (Join as dependent product). Let Γ be a category and let $C \rightarrow \Gamma$ and $D \rightarrow \Gamma$ be functors. There is a category $C \star_{\Gamma} D$ over Γ and a height functor $h_{C,D}: C \star_{\Gamma} D \rightarrow \Gamma \times [1]$. When Γ is an anima, we require these to agree with the category and height functor supplied by Axiom J.1. Write

$$j_{\Gamma} := \text{id}_{\Gamma} \times j: \Gamma \times \partial[1] \longrightarrow \Gamma \times [1]$$

for the weakening of the boundary inclusion. Then the canonical functor

$$C \sqcup D \longrightarrow (C \star_{\Gamma} D) \times_{\Gamma \times [1]} (\Gamma \times \partial[1])$$

is an equivalence, and its inverse exhibits the height functor $h_{C,D}: C \star_{\Gamma} D \rightarrow \Gamma \times [1]$ as a dependent product of the category $C \sqcup D$ over $\Gamma \times \partial[1]$ along j_{Γ} , in the sense of Definition 3.5.4.

Definition 3.6.7. Let $p: C \rightarrow S$ and $q: D \rightarrow S$ be functors. The category $C \star_S D \rightarrow S$ supplied by Axiom J.2 is called the *relative join* of C and D over S .

Remark 3.6.8. Let us spell out the resulting universal property more explicitly. Consider a category E over $\Gamma \times [1]$ and denote by E_0 and E_1 its restrictions along the two sections $\Gamma \rightarrow \Gamma \times [1]$. Then Axiom J.2 gives an equivalence

$$\text{Map}_{\Gamma \times [1]}(E, C \star_{\Gamma} D) \simeq \text{Map}_{\Gamma \times \partial[1]}(E \times_{\Gamma \times [1]} (\Gamma \times \partial[1]), C \sqcup D) \simeq \text{Map}_{\Gamma}(E_0, C) \times \text{Map}_{\Gamma}(E_1, D).$$

In other words, giving a functor $\varphi: E \rightarrow C \star_{\Gamma} D$ over $\Gamma \times [1]$ amounts to giving functors $\varphi_0: E_0 \rightarrow C$ and $\varphi_1: E_1 \rightarrow D$ over Γ .

Proposition 3.6.9. Let A and B be two categories. The canonical embedding $A \sqcup B \hookrightarrow A \star B$ induces an equivalence

$$A^{\simeq} \sqcup B^{\simeq} \xrightarrow{\sim} (A \star B)^{\simeq}.$$

Proof. Let $h: A \star B \rightarrow [1]$ be the height functor. By Axiom J.2, there is a pullback square

$$\begin{array}{ccc} A \sqcup B & \hookrightarrow & A \star B \\ \downarrow & & \downarrow h \\ \partial[1] & \hookrightarrow & \cdot \end{array}$$

The canonical equivalence $\partial[1] \xrightarrow{\sim} [1]^{\simeq}$ from Axiom C.1 identifies this with a pullback square

$$\begin{array}{ccc} A \sqcup B & \hookrightarrow & A \star B \\ \downarrow & & \downarrow h \\ [1]^{\simeq} & \hookrightarrow & [1] \end{array}$$

up to the specified comparison. The square formed by the two groupoid-core inclusions and h therefore induces a functor

$$u: (A \star B)^{\simeq} \longrightarrow (A \sqcup B)^{\simeq} \xrightarrow{\sim} A^{\simeq} \sqcup B^{\simeq}.$$

The functor u is an embedding: after composing with the groupoid-core inclusion of $A \sqcup B$ and then with $A \sqcup B \hookrightarrow A \star B$, it agrees with the groupoid-core inclusion of $A \star B$. Since these inclusions are embeddings, so is u by Lemma 1.6.26.

Applying groupoid cores to $A \sqcup B \hookrightarrow A \star B$ and using Lemma 2.1.28 gives a section

$$A^\simeq \sqcup B^\simeq \longrightarrow (A \star B)^\simeq$$

of u . Hence u is an equivalence by Lemma 1.6.24. $\square$

Unitality and associativity

We will now show that the join operation $(C, D) \mapsto C \star D$ is unital and associative.

Lemma 3.6.10 (Unitality of joins). *For every category C , the maps $C \rightarrow \emptyset \star C$ and $C \rightarrow C \star \emptyset$ are equivalences.*

Proof. By definition the category $\emptyset \star C$ sits in a pushout square

$$\begin{array}{ccc} \emptyset & \longrightarrow & \emptyset \\ \downarrow & & \downarrow \\ C & \longrightarrow & \emptyset \star C \end{array}$$

and hence the bottom map is an equivalence. A similar argument applies to $C \star \emptyset$. $\square$

Construction 3.6.11 (Associativity of joins). Given categories C , D and E , we construct an equivalence

$$a_{C,D,E} : C \star (D \star E) \xrightarrow{\sim} (C \star D) \star E.$$

Applying the mapping-out property of joins twice shows that a functor from either parenthesization to a category F consists of the same data: functors from C , D , and E to F ; families of connecting morphisms from C to D , from D to E , and from C to E ; and a commutative triangle relating these three families over $C \times D \times E$. The two pullback descriptions are identified using the equivalences

$$[0] \star [1] \xrightarrow{\sim} [2] \xleftarrow{\sim} [1] \star [0]$$

from Example 3.6.3.

The universal property therefore gives a comparison $a_{C,D,E} : C \star (D \star E) \rightarrow (C \star D) \star E$ which is the identity on C , D , and E . For every category F , precomposition with $a_{C,D,E}$ identifies the two pullback descriptions above and is consequently an equivalence of functor categories. Passing to groupoid cores and applying Proposition 1.4.11 shows that $a_{C,D,E}$ is an equivalence.

No triangle or pentagon compatibility among the chosen equivalences is asserted here. Such a statement would require additional specified comparison data, which is not part of the present naive language.

Definition 3.6.12. Let C be a category. We define its *left cone* $C^\triangleleft$ and its *right cone* $C^\triangleright$ as follows:

$$C^\triangleleft := [0] \star C, \quad \text{and} \quad C^\triangleright := C \star [0].$$

3.7 Slice categories

The join operation can be used to describe functors into slice categories.

Construction 3.7.1 (Slice category). Let $p_C : C \rightarrow \Gamma$ and $p_Y : Y \rightarrow \Gamma$ be categories over Γ , let $\psi : Y \rightarrow C$ be a functor over Γ , and assume that p_Y is exponentiable. Relative currying of ψ gives a section $\tilde{\psi} : \Gamma \rightarrow \underline{\text{Fun}}_\Gamma(Y, C)$. Relative currying of the constant-family functor $(\text{pr}_C, \text{pr}_C) : C \times_\Gamma Y \rightarrow C \times_\Gamma C$ gives a functor $p_Y^* : C \rightarrow \underline{\text{Fun}}_\Gamma(Y, C)$ over Γ . We define the *slice* $C_{/\psi} \rightarrow \Gamma$ by the pullback square of categories over Γ

$$\begin{array}{ccc} C_{/\psi} & \xrightarrow{\quad} & \underline{\text{Fun}}_\Gamma([1]_\Gamma \times_\Gamma Y, C) \\ \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ C & \xrightarrow{(p_Y^*, \tilde{\psi} \circ p_C)} & \underline{\text{Fun}}_\Gamma(Y, C) \times_\Gamma \underline{\text{Fun}}_\Gamma(Y, C). \end{array}$$

Here $[1]_\Gamma := \Gamma \times [1] \rightarrow \Gamma$. The relative functor category in the upper right is constructed by taking the dependent product along p_Y of the arrow category of $C \times_\Gamma Y$ over Y ; thus its existence uses only the assumed exponentiability of p_Y .

For $\Gamma = *$, the terminal functor $Y \rightarrow *$ is exponentiable by Proposition 3.5.15. Thus the absolute slice exists without an additional exponentiability hypothesis.

Remark 3.7.2. We will mostly be interested in the case $Y = \Gamma$ and $p_Y = \text{id}_\Gamma$. For a section $x : \Gamma \rightarrow C$, we write $C_{/x}$ for the resulting slice over Γ .

Lemma 3.7.3. *For every category X and every functor $\psi : Y \rightarrow C$, there is a pullback square of the form*

$$\begin{array}{ccc} \text{Fun}(X, C_{/\psi}) & \xrightarrow{\quad} & \text{Fun}(X \star Y, C) \\ \downarrow & \lrcorner & \downarrow \\ \text{Fun}(X, C) & \xrightarrow{1 \times \psi} & \text{Fun}(X, C) \times \text{Fun}(Y, C). \end{array}$$

Proof. Since the assignment $C \mapsto \text{Fun}(X, C)$ preserves pullbacks, applying it to the defining pullback square of $C_{/\psi}$ gives

$$\begin{array}{ccc} \text{Fun}(X, C_{/\psi}) & \xrightarrow{\quad} & \text{Fun}(X \times [1] \times Y, C) \\ \downarrow & \lrcorner & \downarrow (\text{ev}_0, \text{ev}_1) \\ \text{Fun}(X, C) & \xrightarrow{1 \times \psi} & \text{Fun}(X, C) \times \text{Fun}(Y, C) \xrightarrow{\text{pr}_X^* \times \text{pr}_Y^*} \text{Fun}(X \times Y, C) \times \text{Fun}(X \times Y, C). \end{array}$$

Because of the factorization of the bottom map, we may compute this pullback by first pulling back along the map $\text{pr}_X^* \times \text{pr}_Y^*$, giving a pullback square

$$\begin{array}{ccc} \text{Fun}(X, C_{/\psi}) & \xrightarrow{\quad} & \text{Fun}(X \sqcup Y, C) \times_{\text{Fun}(X \times \partial[1] \times Y, C)} \text{Fun}(X \times [1] \times Y, C) \\ \downarrow & \lrcorner & \downarrow \\ \text{Fun}(X, C) & \xrightarrow{1 \times \psi} & \text{Fun}(X, C) \times \text{Fun}(Y, C) \end{array}$$

$\text{Fun}(X, C) = \text{Fun}(X, C) \times * \xrightarrow{1 \times \psi} \text{Fun}(X, C) \times \text{Fun}(Y, C).$

Since the upper right corner is equivalent to $\text{Fun}(X \star Y, C)$ by Remark 3.6.1, this finishes the proof. $\square$

Remark 3.7.4. Informally speaking, the previous lemma says that a functor $X \rightarrow C/\psi$ into the slice consists of a map $g: X \star Y \rightarrow C$ fitting in a commutative diagram as follows:

$$\begin{array}{ccc} & Y & \\ \swarrow & & \searrow \psi \\ X \star Y & \xrightarrow{g} & C. \end{array}$$

Completely analogously we obtain a dual slice construction.

Construction 3.7.5. Let $p_C: C \rightarrow \Gamma$ and $p_X: X \rightarrow \Gamma$ be categories over Γ , let $\varphi: X \rightarrow C$ be a functor over Γ , and assume that p_X is exponentiable. Let $\bar{\varphi}: \Gamma \rightarrow \underline{\text{Fun}}_\Gamma(X, C)$ be its relative currying. Define the coslice $C_{\varphi/} \rightarrow \Gamma$ by the pullback square of categories over Γ

$$\begin{array}{ccc} C_{\varphi/} & \xrightarrow{\quad} & \underline{\text{Fun}}_\Gamma([1]_\Gamma \times_\Gamma X, C) \\ \downarrow & \lrcorner & \downarrow \\ C & \xrightarrow{(\bar{\varphi} \circ p_C, p_X^*)} & \underline{\text{Fun}}_\Gamma(X, C) \times_\Gamma \underline{\text{Fun}}_\Gamma(X, C). \end{array}$$

3.8 Exercises Chapter 3

Exercise 3.8.1. Consider a category Γ , a category D and a category $q: E \rightarrow \Gamma$ over Γ . Put $D_\Gamma := D \times \Gamma \rightarrow \Gamma$. Define a category $\underline{\text{Fun}}_\Gamma(D_\Gamma, E)$ over Γ by the pullback square

$$\begin{array}{ccc} \underline{\text{Fun}}_\Gamma(D_\Gamma, E) & \xrightarrow{\quad} & \text{Fun}(D, E) \\ \downarrow & \lrcorner & \downarrow q_* \\ \Gamma & \xrightarrow{\delta_\Gamma} & \text{Fun}(D, \Gamma), \end{array}$$

where δ_Γ is the currying of the projection $\Gamma \times D \rightarrow \Gamma$. Show that this category has the universal property of the relative functor category from D_Γ to E over Γ .

Exercise 3.8.2. Prove Lemma 3.2.10: show that for embeddings $\Gamma \hookrightarrow C^\approx$ and $\Lambda \hookrightarrow D^\approx$ the projections $C \times D \rightarrow C$ and $C \times D \rightarrow D$ induce an equivalence

$$\langle \Gamma \times \Lambda \subseteq C \times D \rangle \xrightarrow{\sim} \langle \Gamma \subseteq C \rangle \times \langle \Lambda \subseteq D \rangle.$$

Exercise 3.8.3. Prove the assertions from Remark 3.1.17: given a functor $f: C \rightarrow D$ sending a collection of morphisms M in C to a collection of morphisms N in D , we have:

- (1) The map $f|: \langle M \rangle_C \rightarrow \langle N \rangle_D$ is uniquely determined by f .
- (2) Given an identification $f = f'$, we obtain an induced identification $f| = f'|$.
- (3) The construction is compatible with identities and composition: we have identifications $(\text{id}_C)| = \text{id}_{\langle M \rangle_C}$ and $(g \circ f)| = g| \circ f|$ if $g: D \rightarrow E$ is another functor preserving a collection of morphisms.

(4) If the functors $f: C \rightarrow D$ and $\tilde{f}: M \rightarrow N$ are equivalences, then also $f|: \langle M \rangle_C \rightarrow \langle N \rangle_D$ is an equivalence.

Exercise 3.8.4. Let W be a collection of morphisms in a category C , and let $l: C \rightarrow L$ and $l': C \rightarrow L'$ be two localizations of C at W . Construct an equivalence $L \xrightarrow{\sim} L'$.

Exercise 3.8.5. Let W be a collection of morphisms in a category C , and let $C \rightarrow C[W^{-1}]$ be a localization of C at W . Construct a pushout square

$$\begin{array}{ccc} [1] \times W & \xrightarrow{\tilde{w}} & C \\ \text{pr}_2 \downarrow & \ulcorner & \downarrow \\ W & \longrightarrow & C[W^{-1}], \end{array}$$

where the map $\tilde{w}$ is obtained by uncurrying the composite $W \xrightarrow{w} \text{Map}([1], C) \rightarrow \text{Fun}([1], C)$.

Exercise 3.8.6. Let $p: S \rightarrow T$ be a functor, and let $f: C \rightarrow S$ be a category over S . Let $g: D \rightarrow T$ and $g': D' \rightarrow T$ be categories over T that are both a dependent product of C along p . Show that there is an equivalence $D \xrightarrow{\sim} D'$ of categories over T (in the sense of Definition 4.3.3 (outside selection)).

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